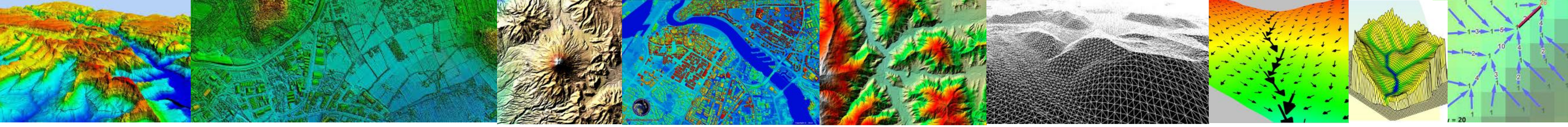
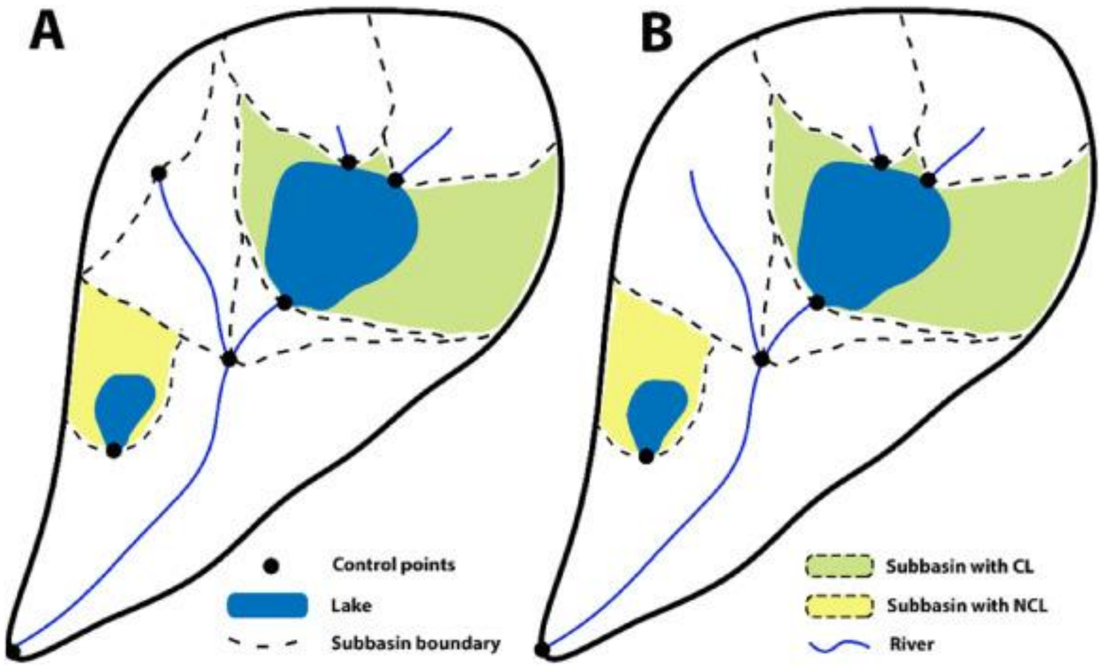


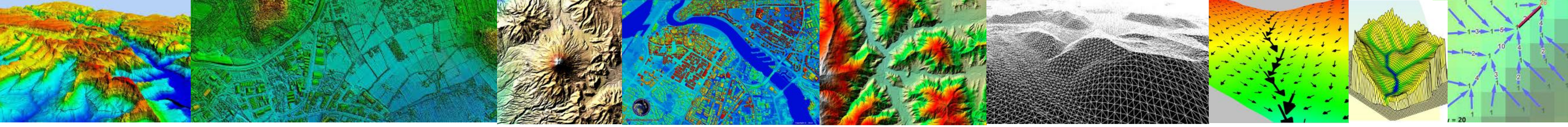
Curso de Hidrología

Unidad 6. Cuenca y red de drenaje

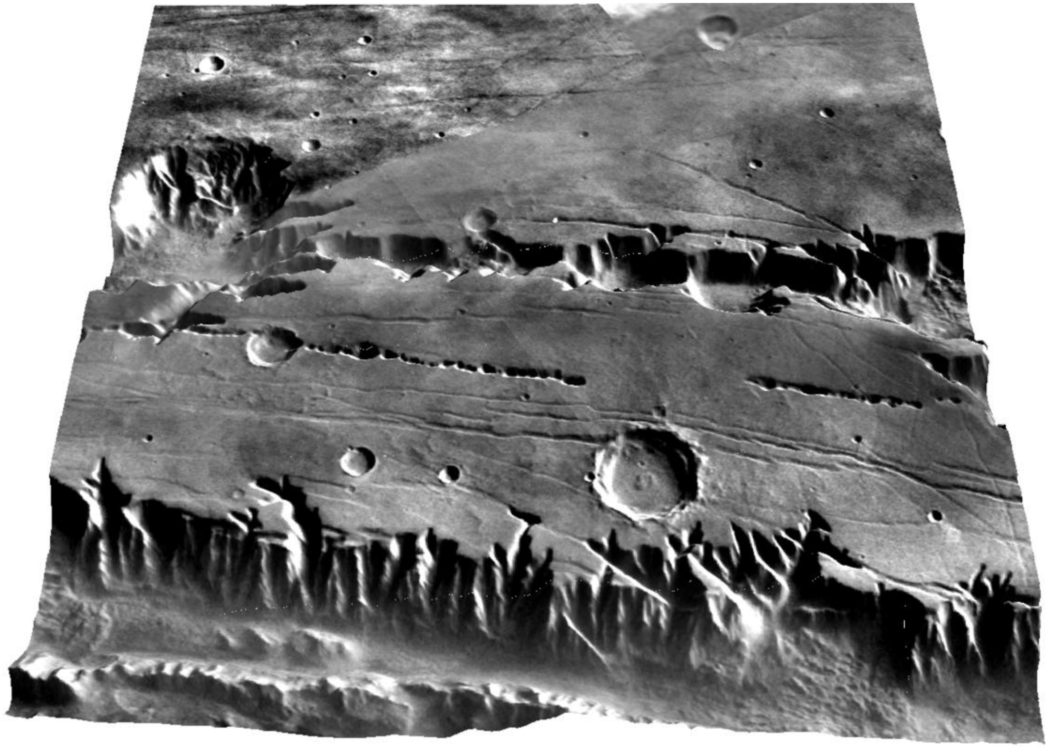
Objetivos

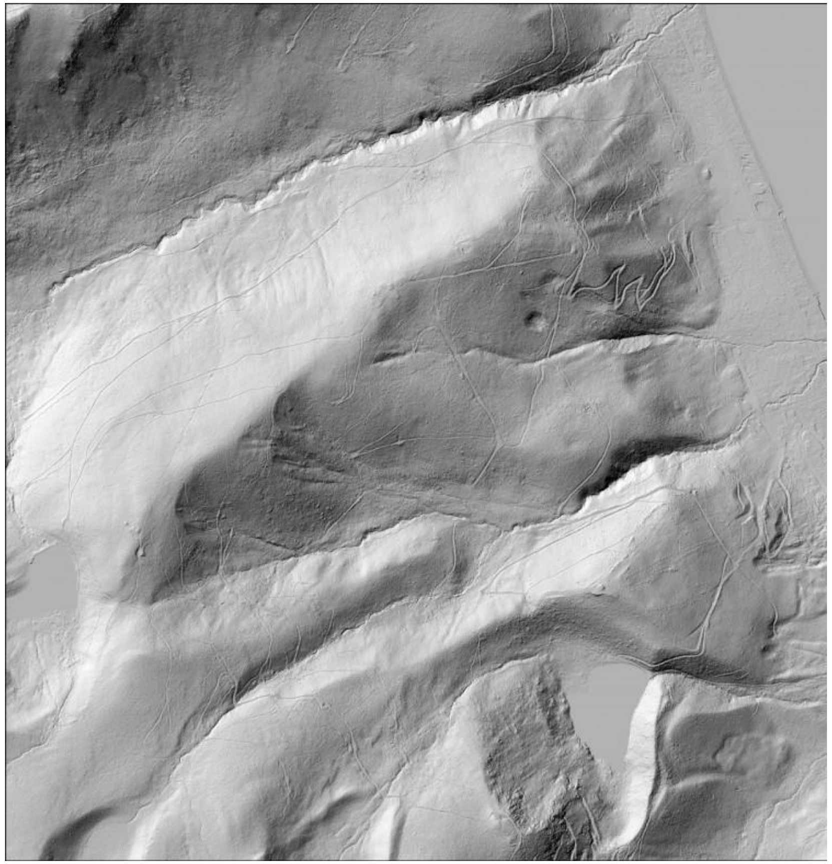
- Comprender los fundamentos de los MDE.
- Preprocesamiento y posprocesamiento de un MDE.
- Análisis de algoritmos de dirección y acumulación de flujo
- Determinar la red de drenaje y umbreles
- Delimitar cuencas y subcuencas de forma automática
- Derivar variables topográficas clave a partir del los MDEs

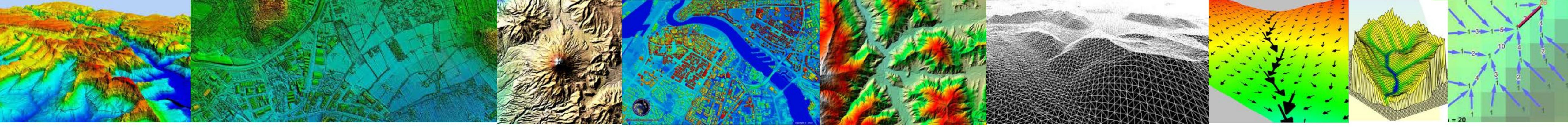




¿Cómo delimitar una cuenca y sus subcuencas, y como calcular su red de drenaje ?

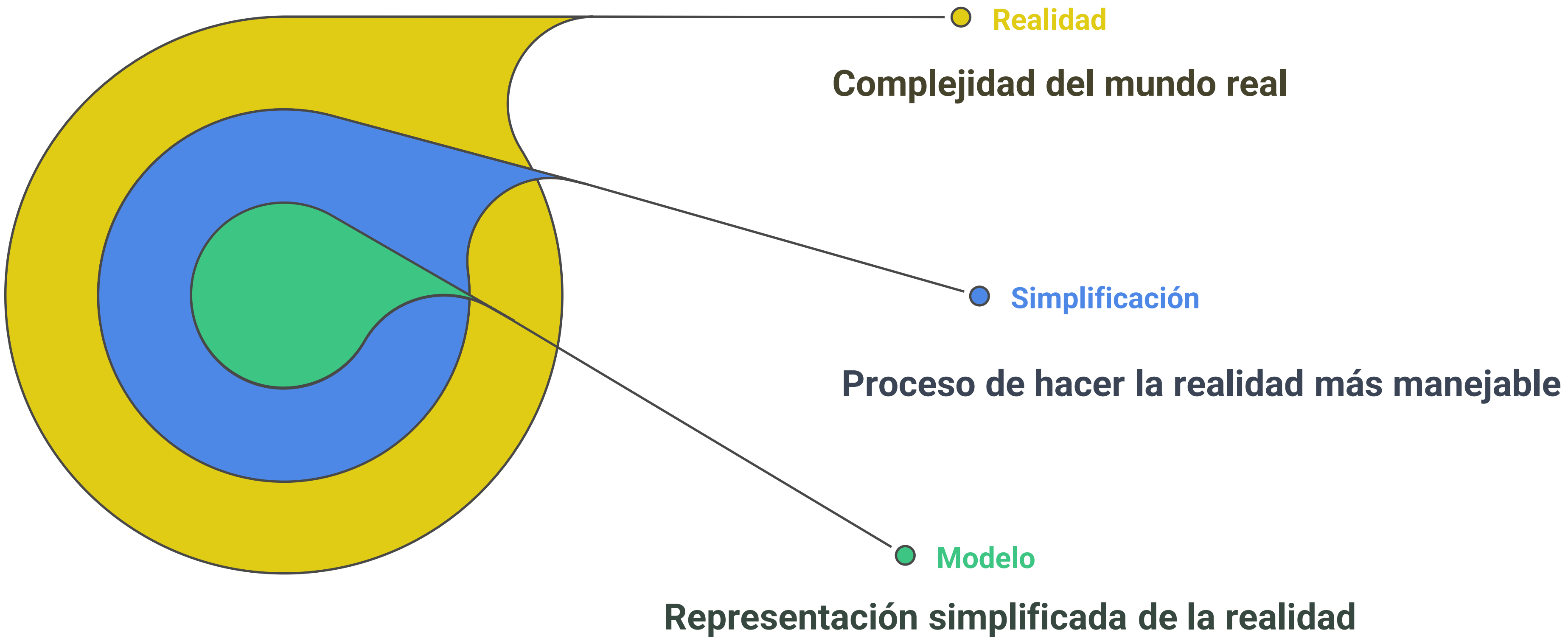


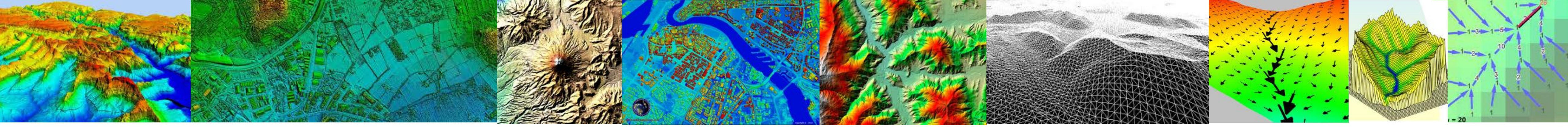




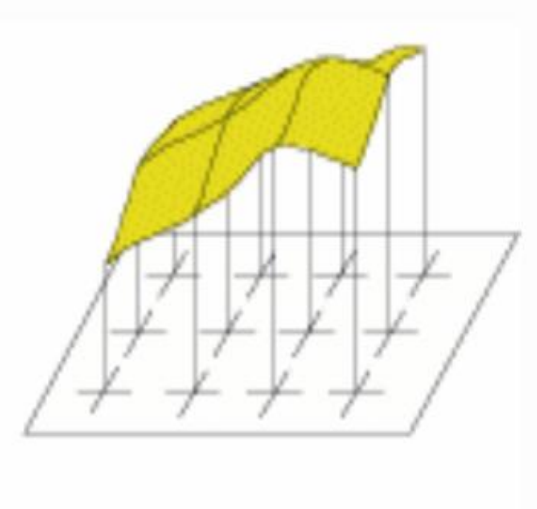
¿Qué es un modelo?

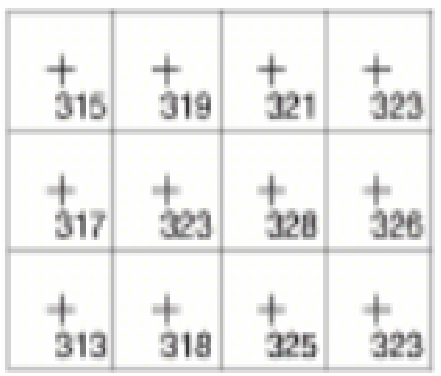
Un modelo es una herramienta diseñada para representar una versión simplificada de la realidad.

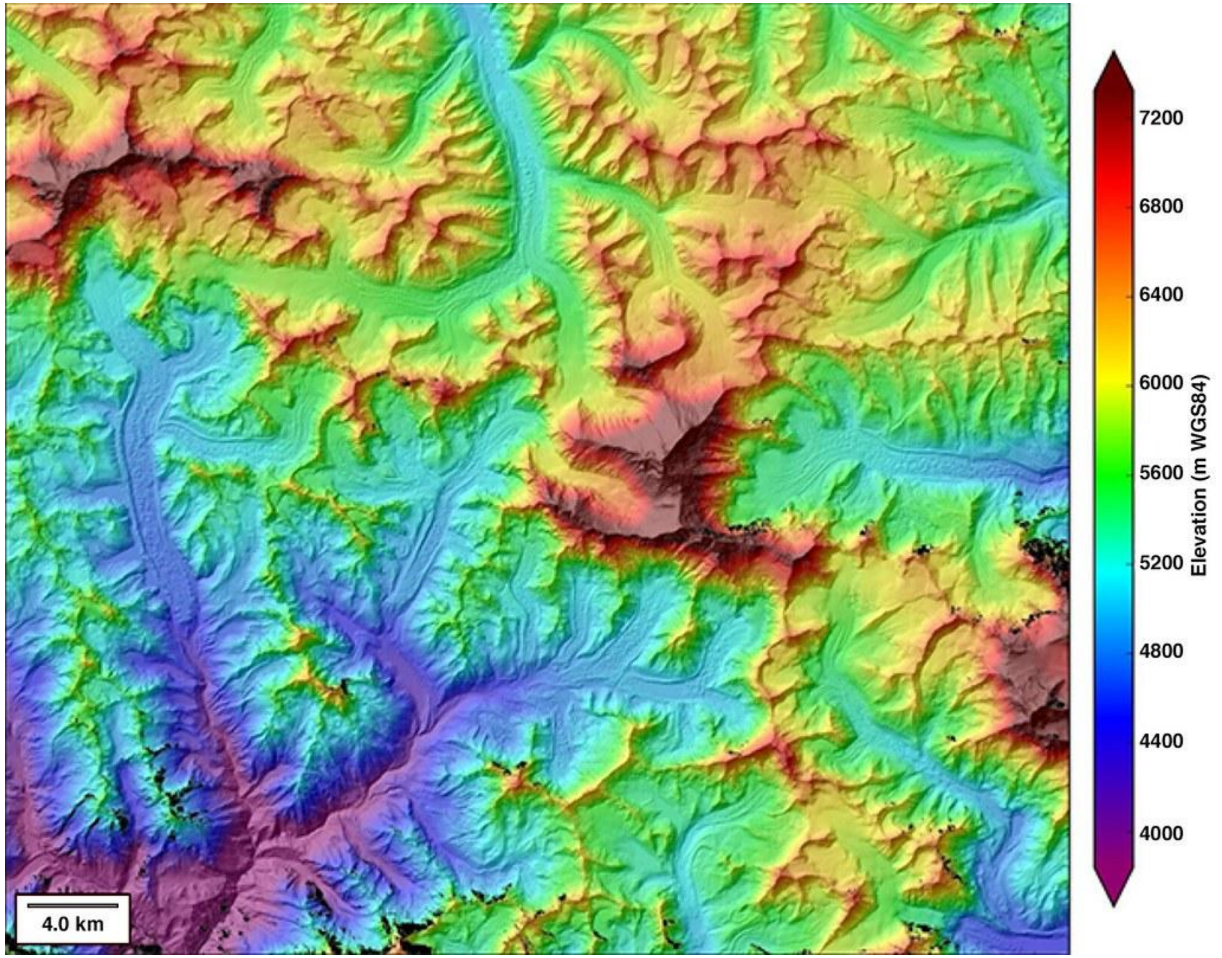


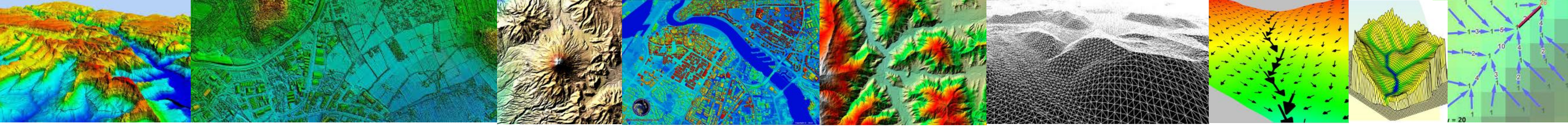


Los modelos digitales de elevación son un conjunto ordenado de números en una matriz que representa la distribución espacial de las elevaciones del terreno (Moore et al., 1991).

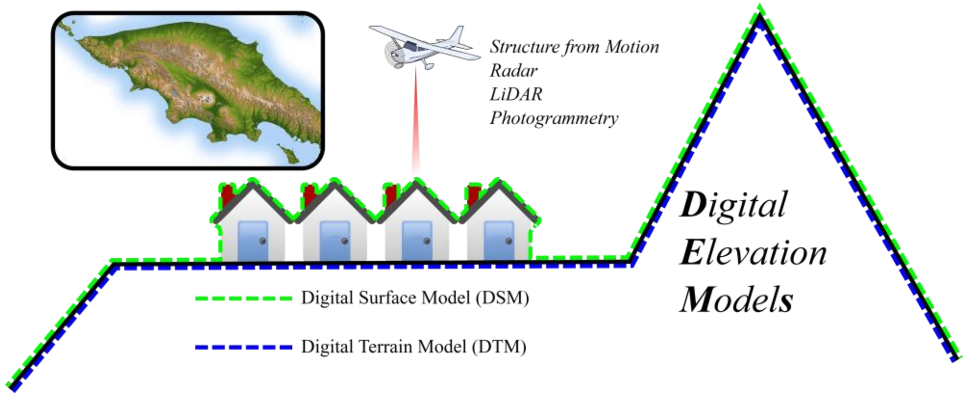


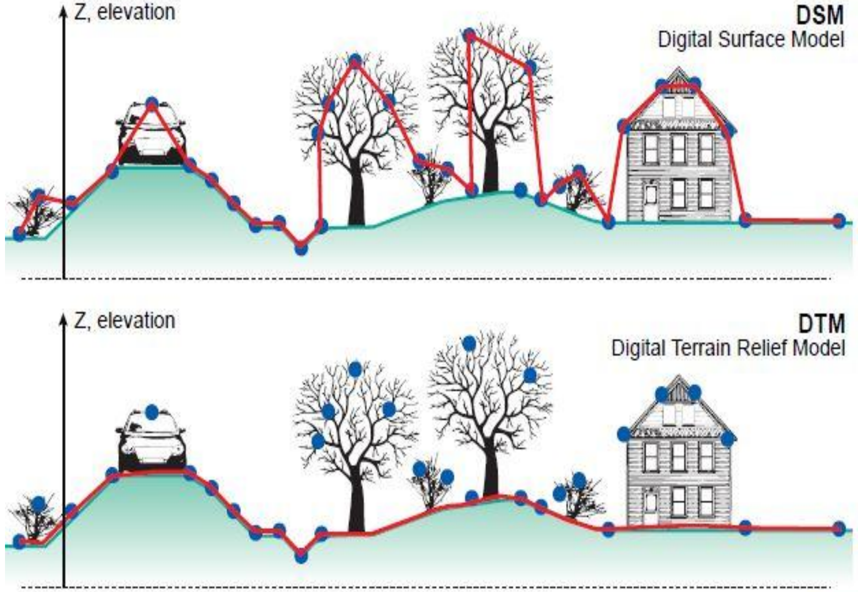


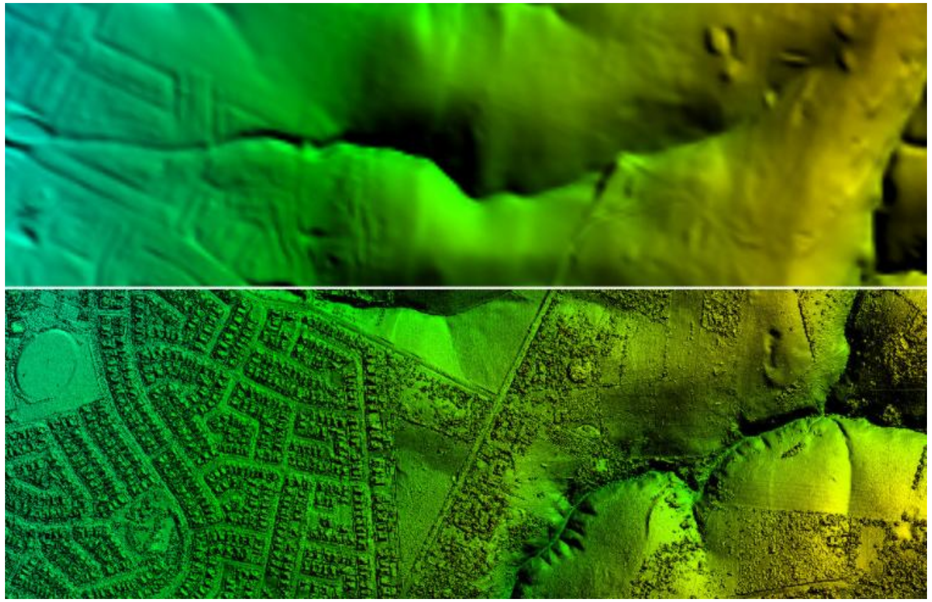


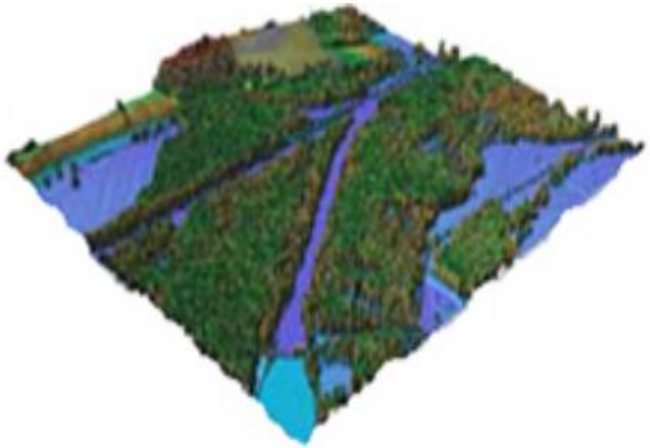


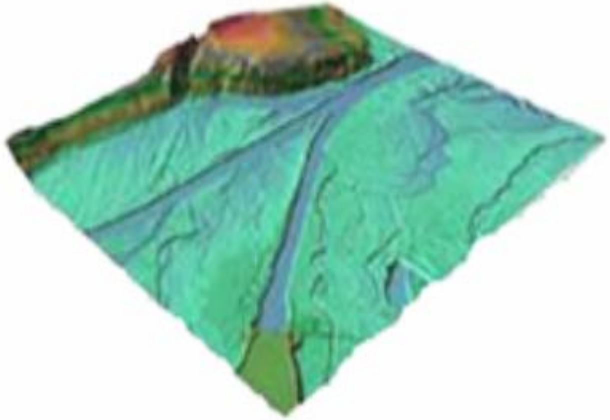
MDE es una representación de las elevaciones sobre un terreno, incluyendo las plantas y los edificios. El MDT no toma en cuenta los objetos que se encuentran sobre el terreno, como las plantas y los edificios.

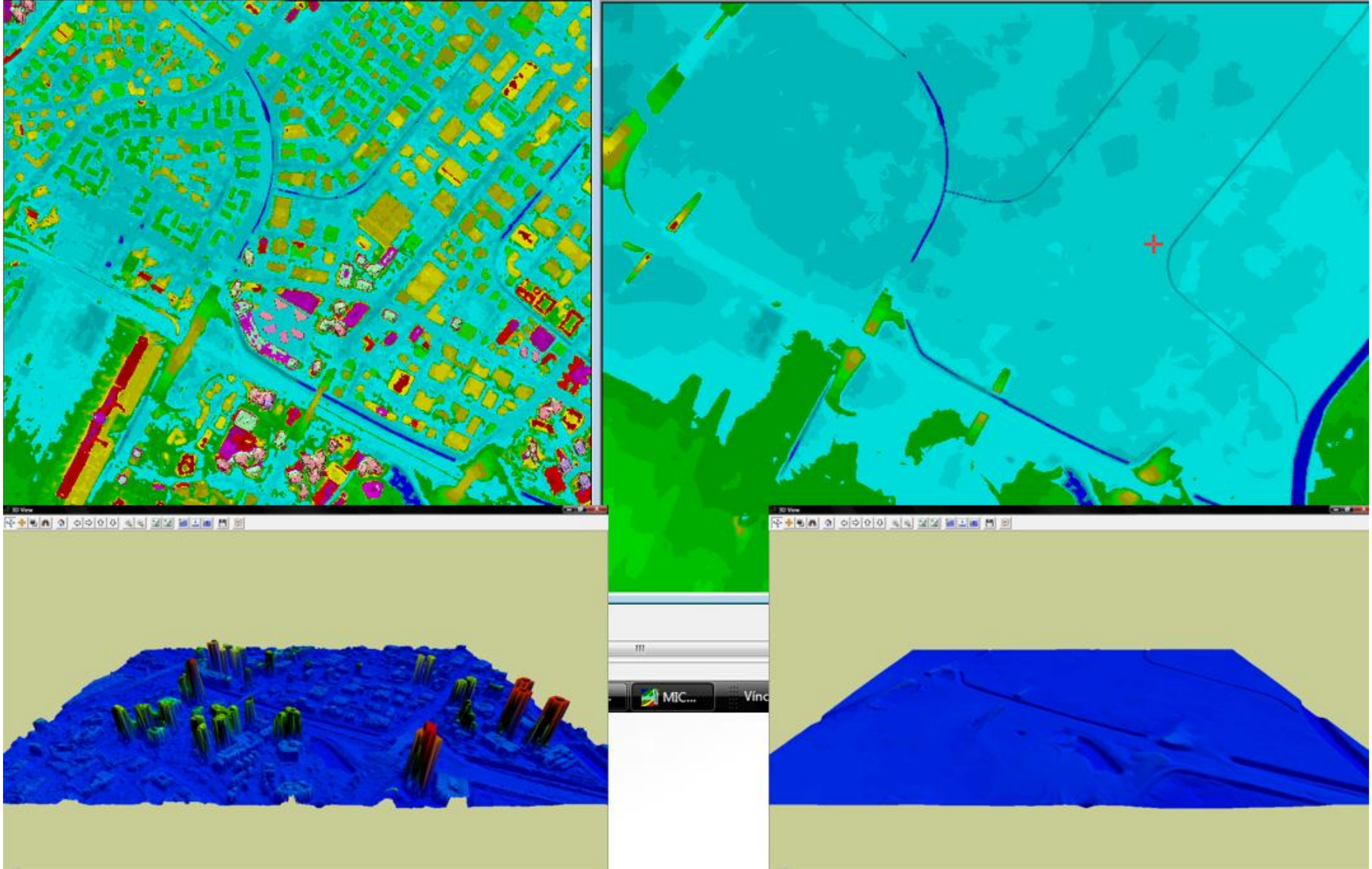


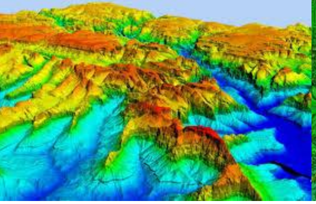


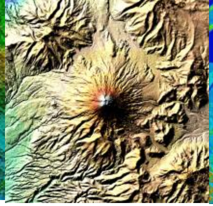


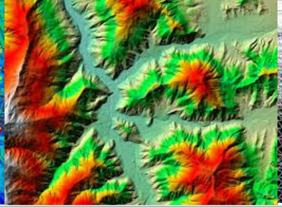
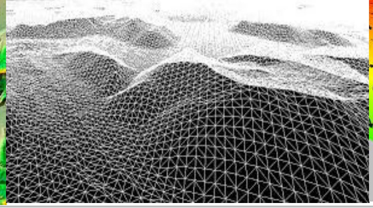
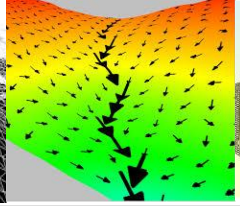
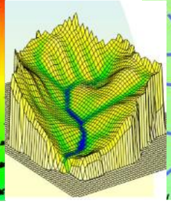
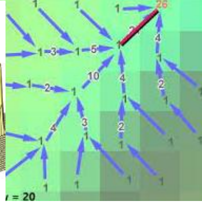


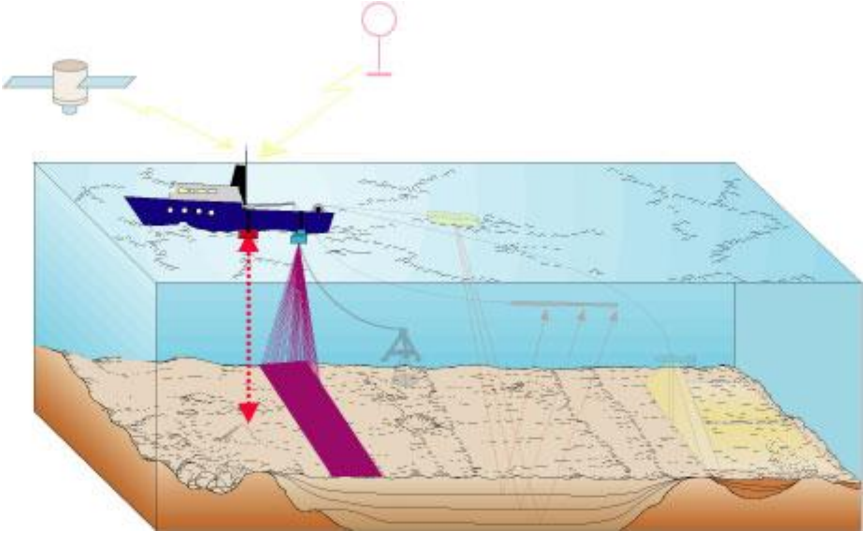
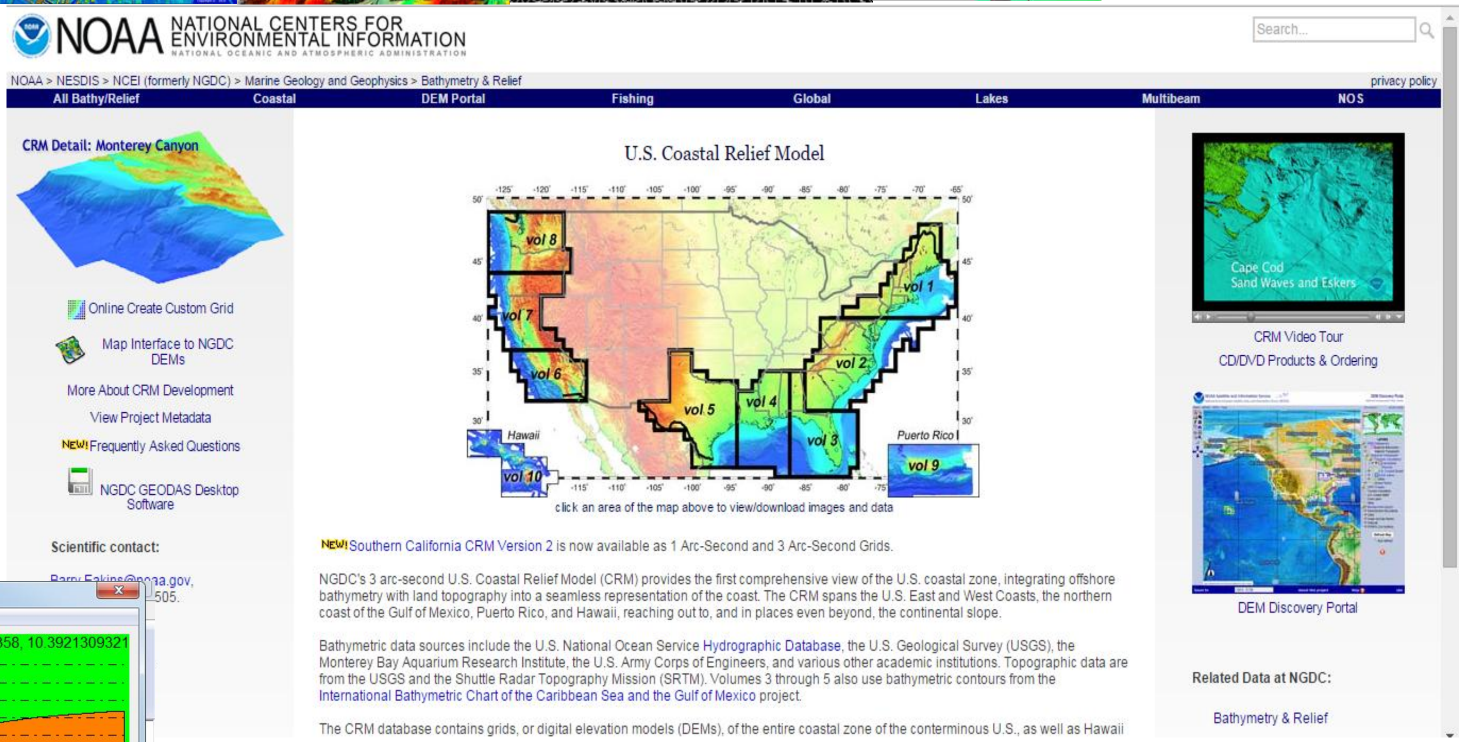



MIC...

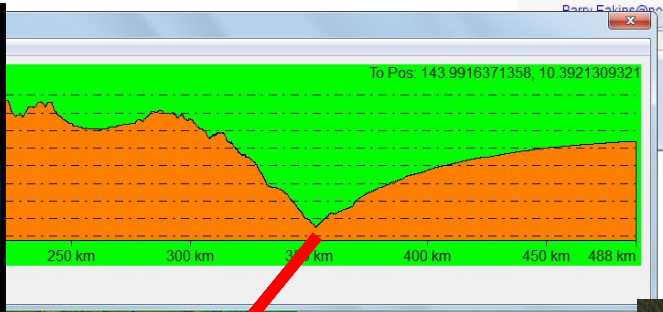


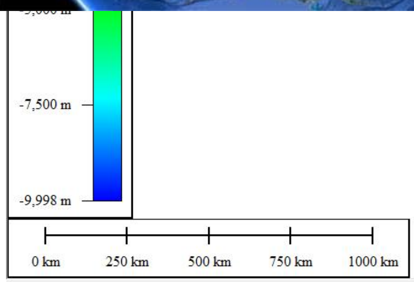


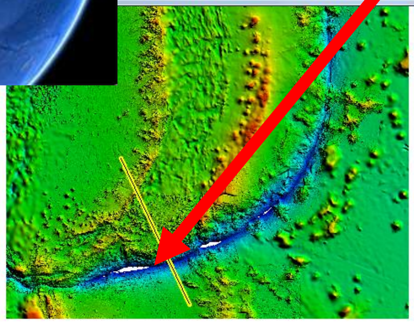






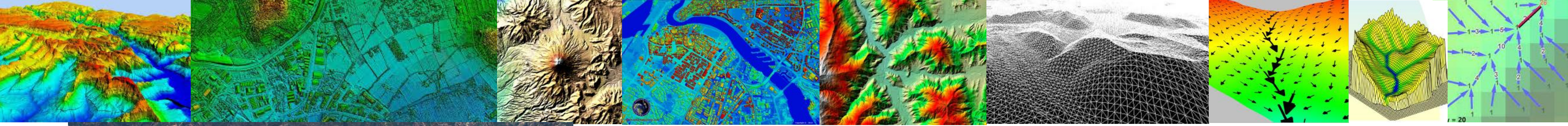



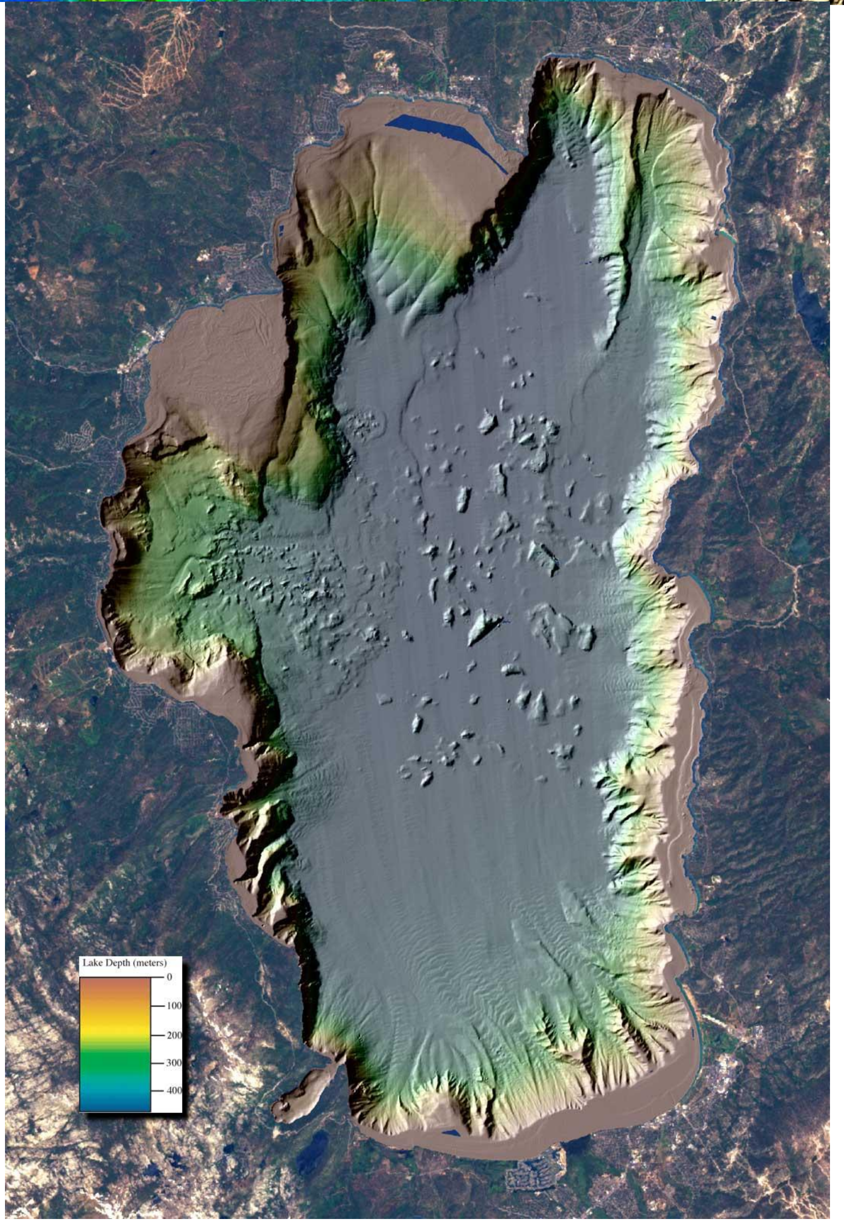
Batimetría: Islas Marianas

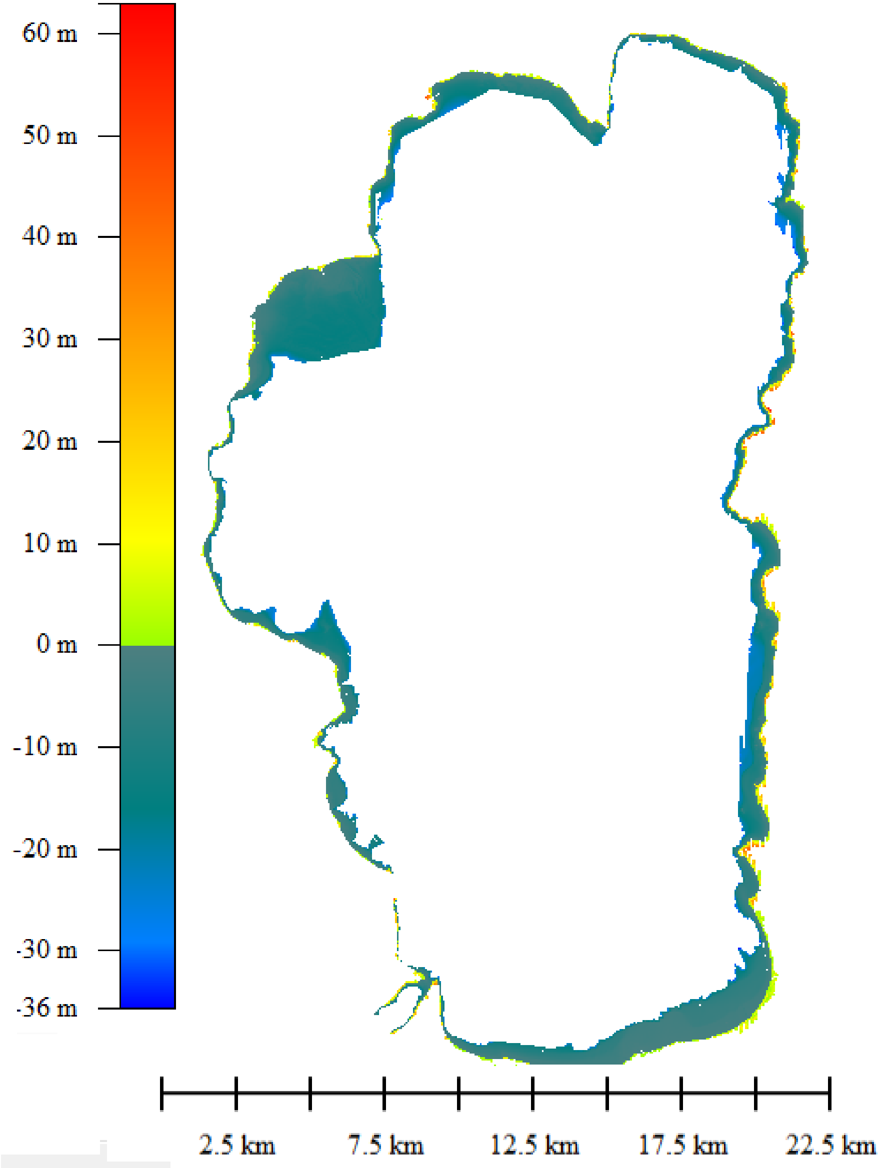


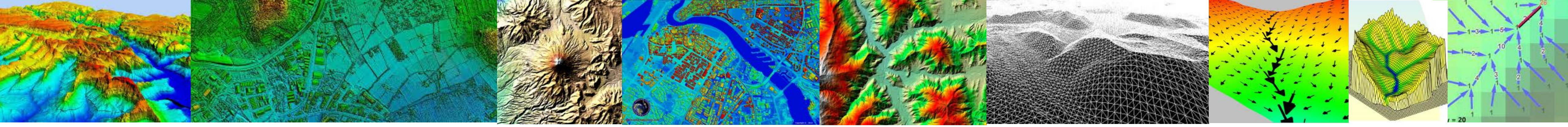




Acoustic Backscatter of Lake Tahoe

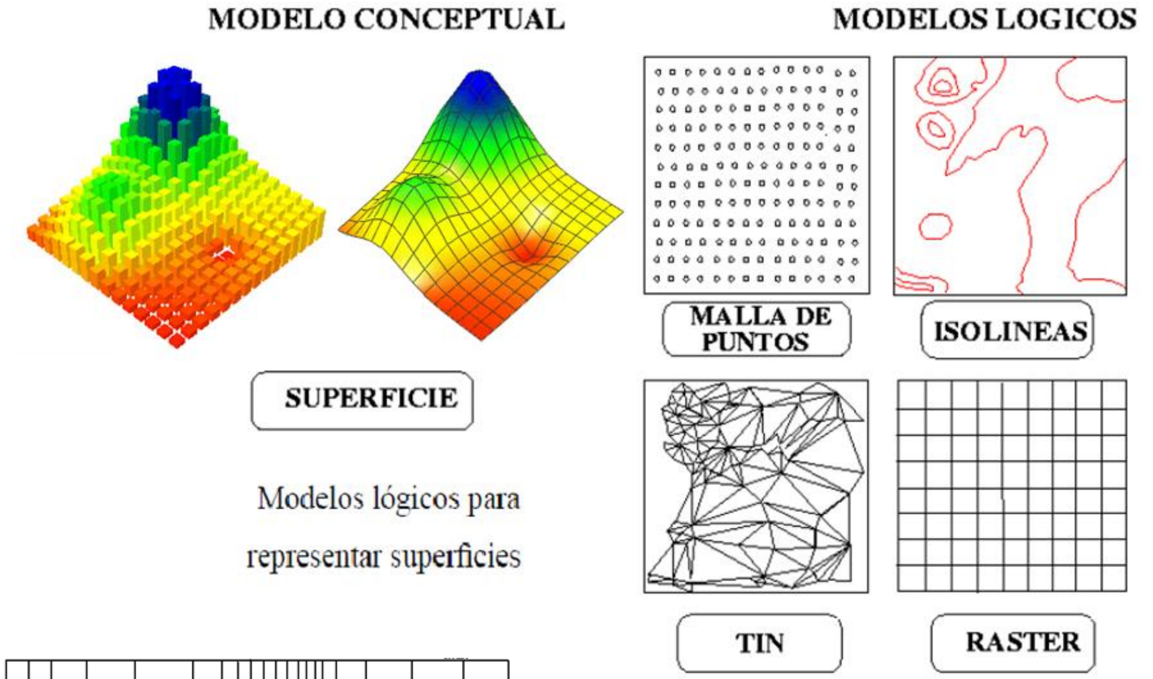


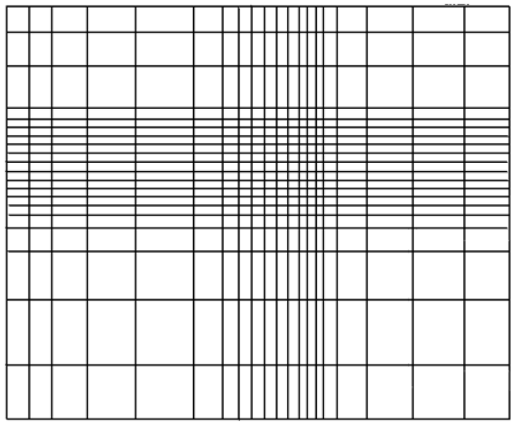


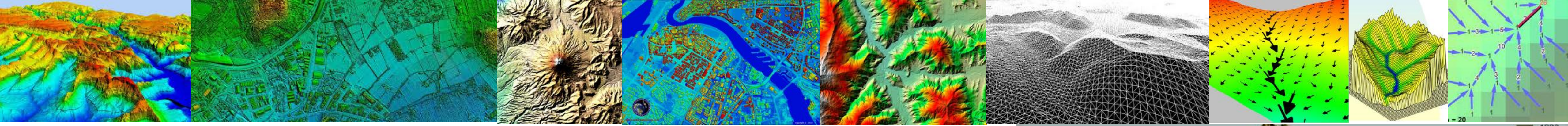
Representación

Estructuras Vectoriales vs Raster

Característica	Vectorial	Raster
Contornos	Polilíneas de altitud constante	
TIN	Red de triángulos irregulares adosados	
Matrices regulares		Malla de celda cuadrada
Quadtrees		Matrices irregulares

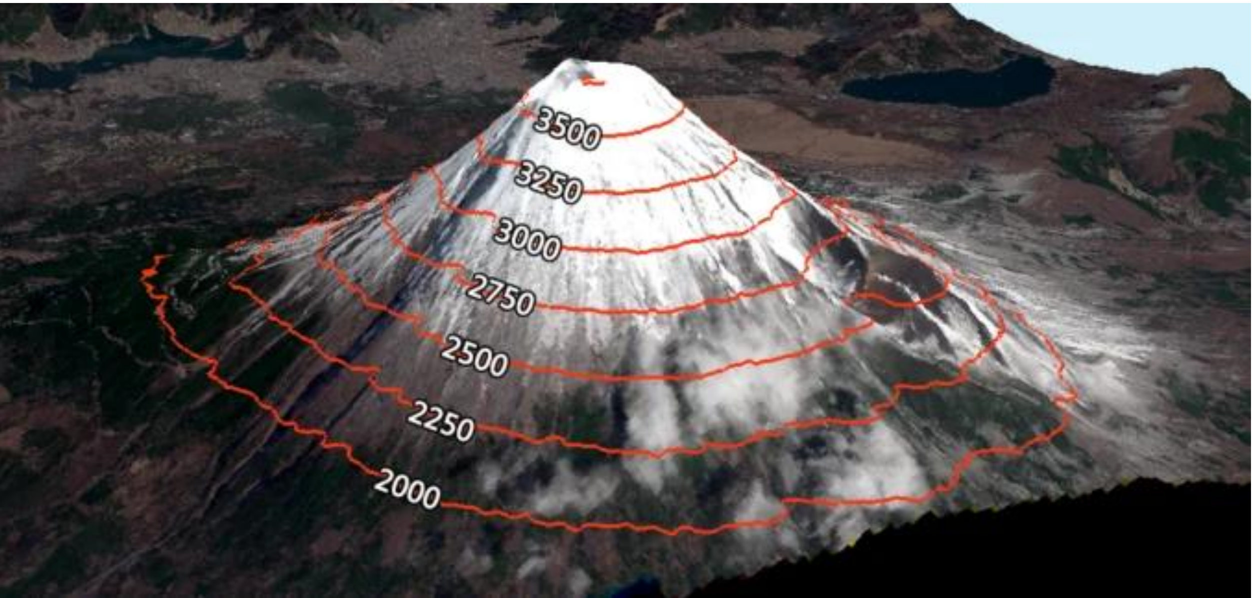


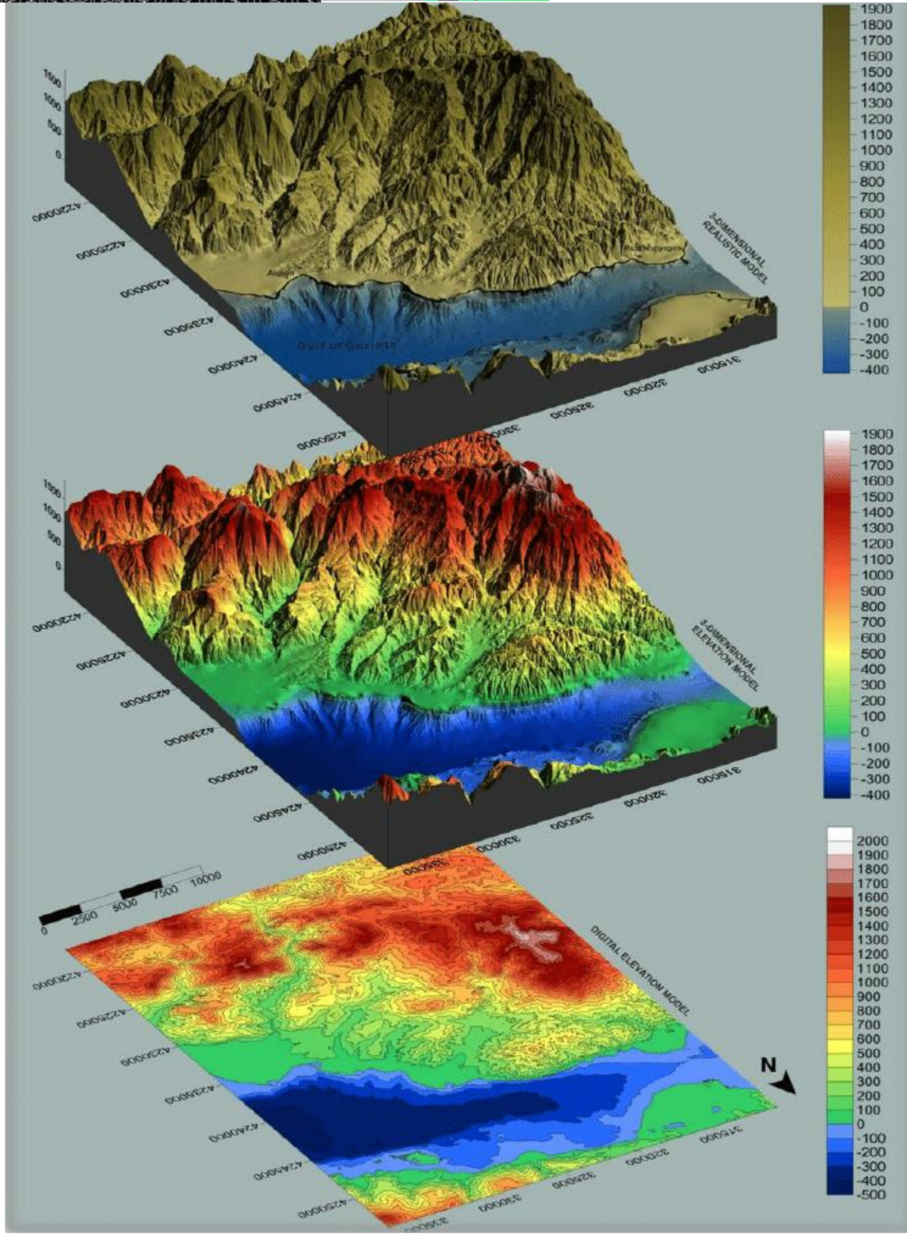


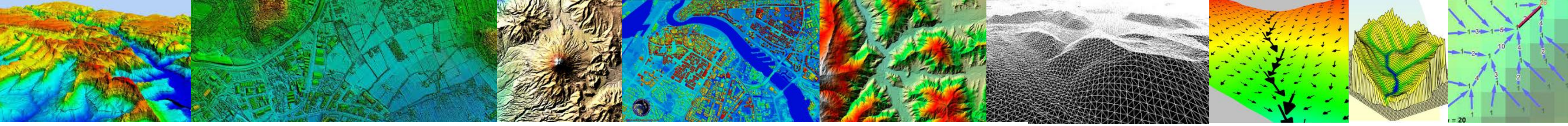


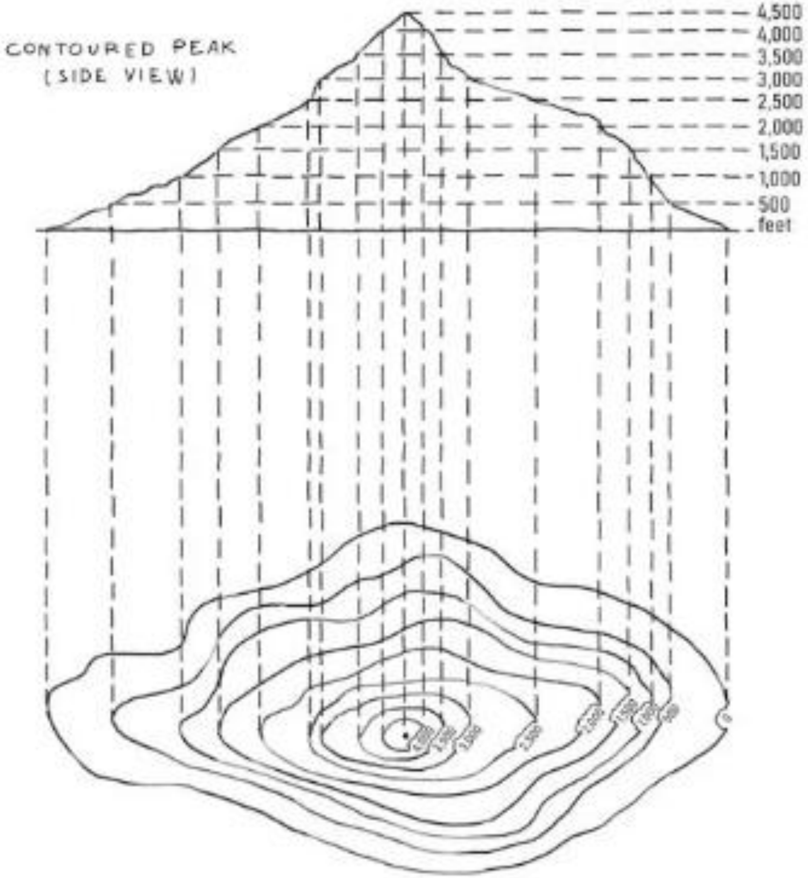
Vectorial

El modelo de contornos es el utilizado habitualmente en los mapas impresos

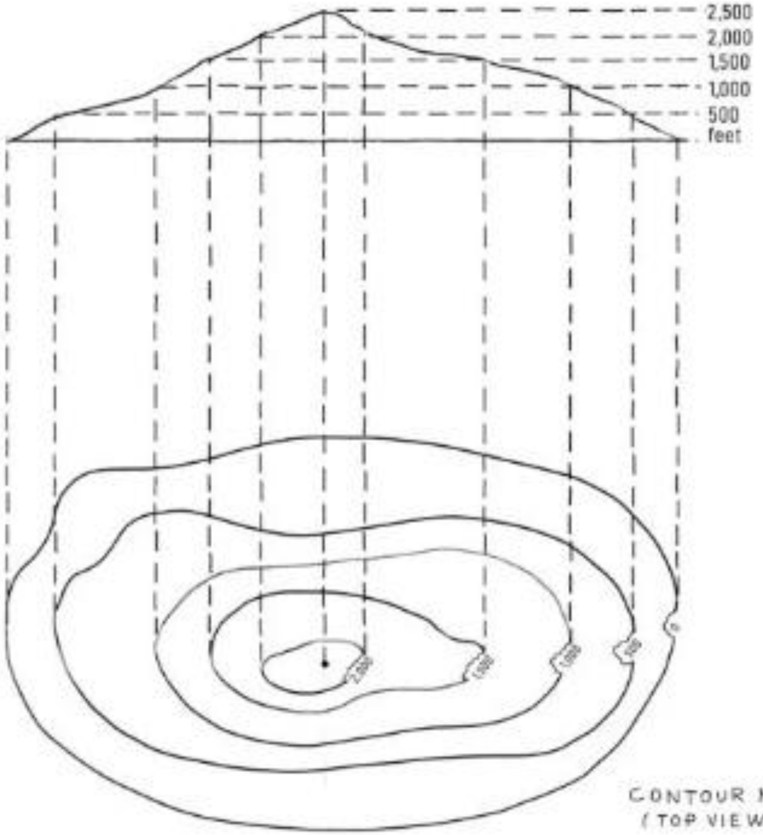




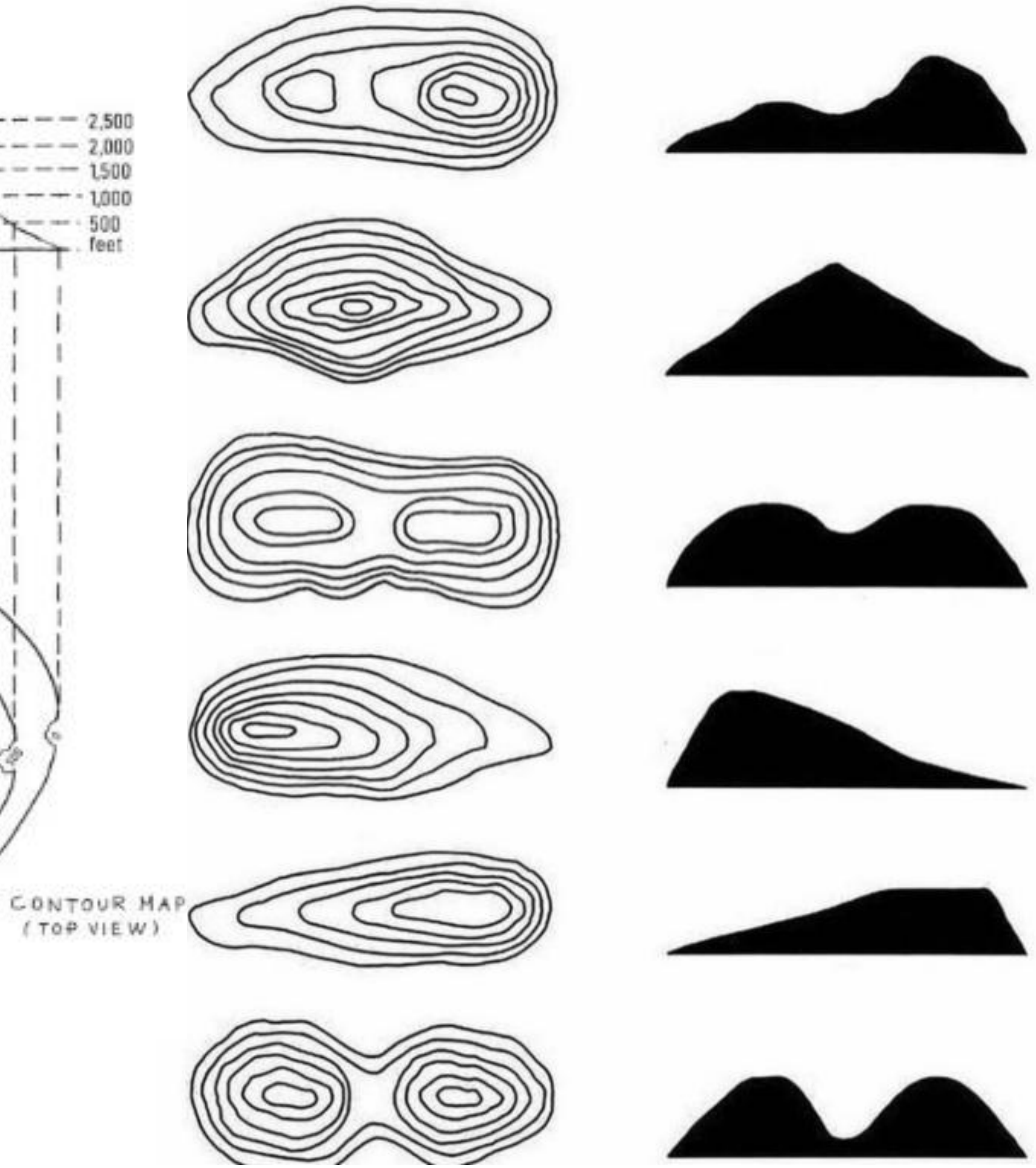


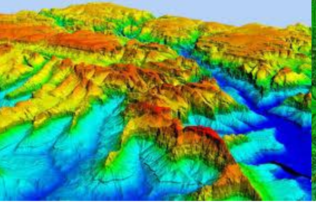


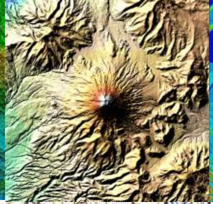
STEEP SLOPE

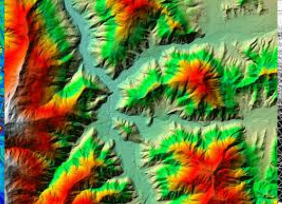
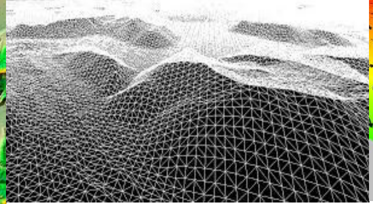
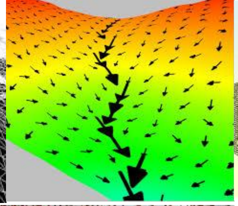
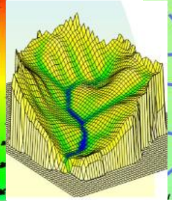
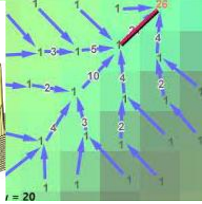


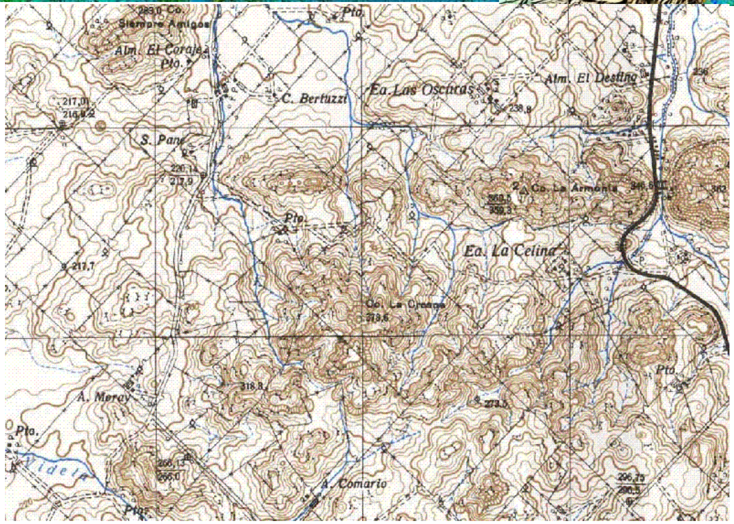
GENTLE SLOPE

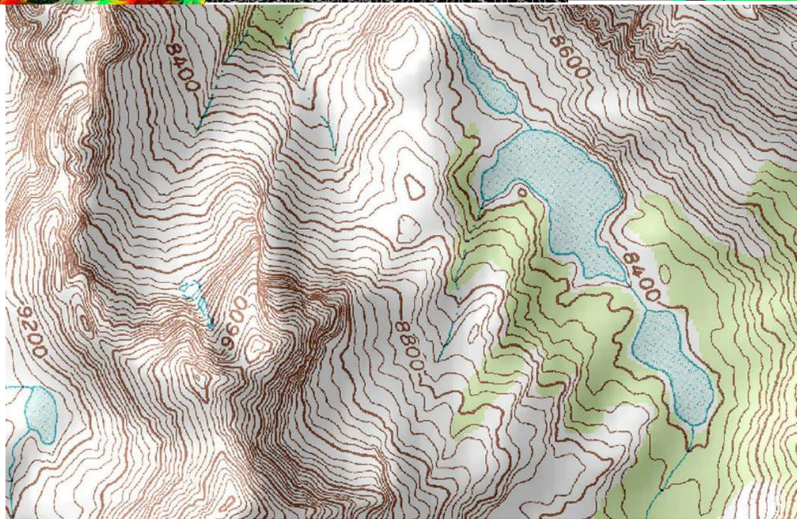


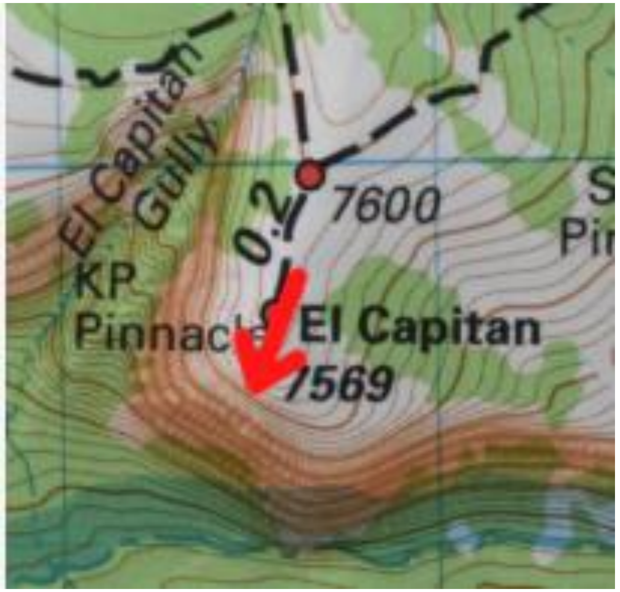




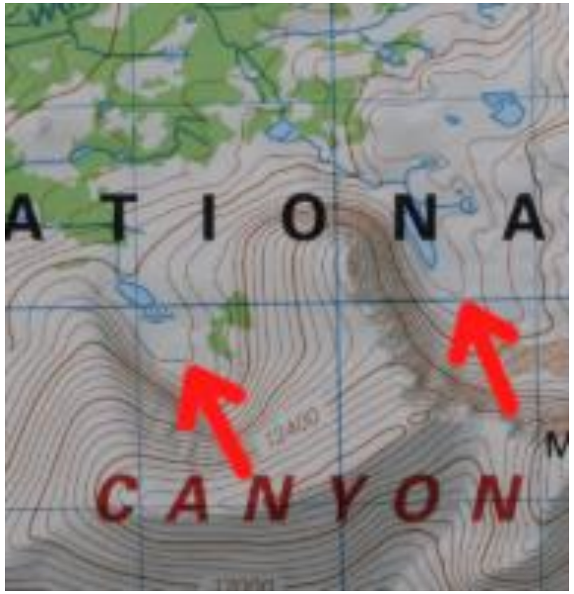












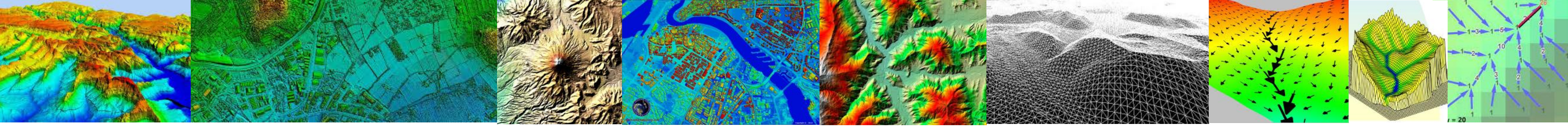
Acantilado



Red de drenaje

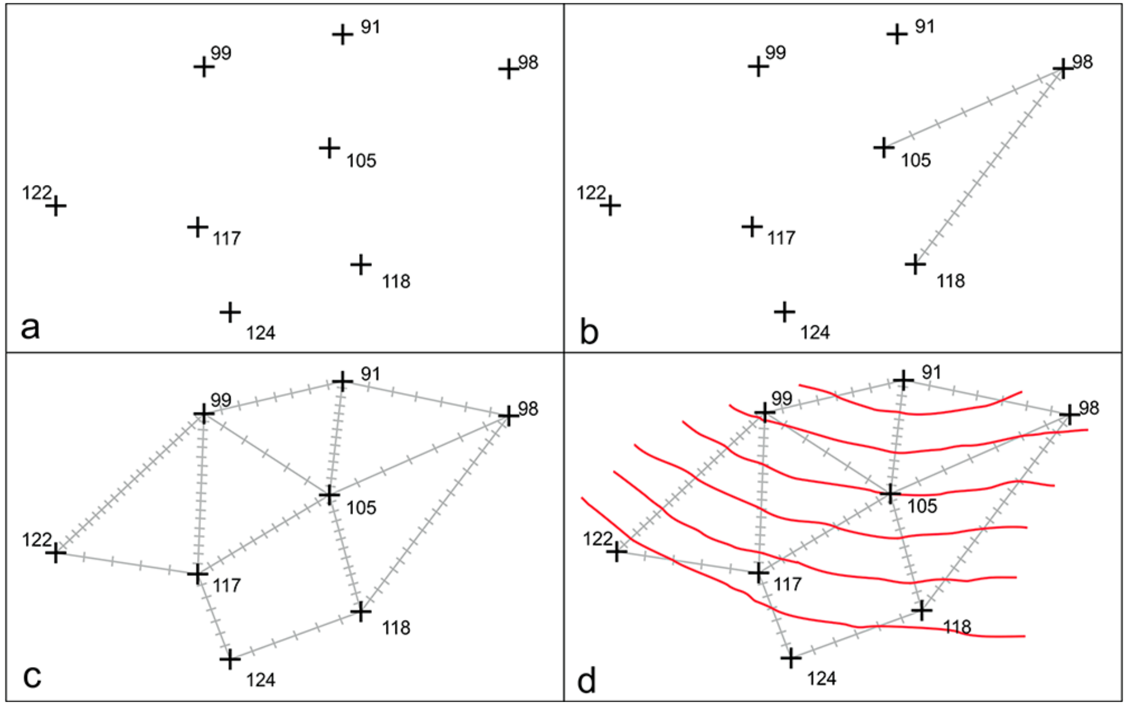
Cresta montaña

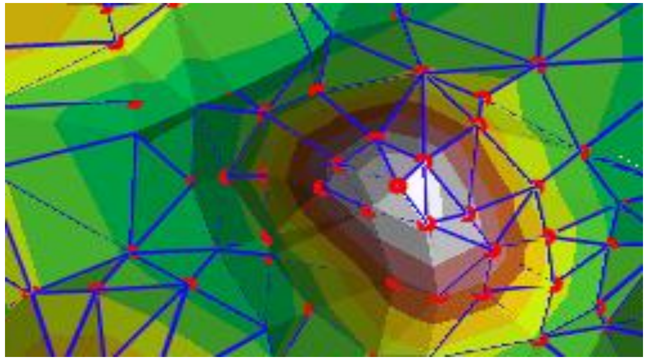
Zona Baja

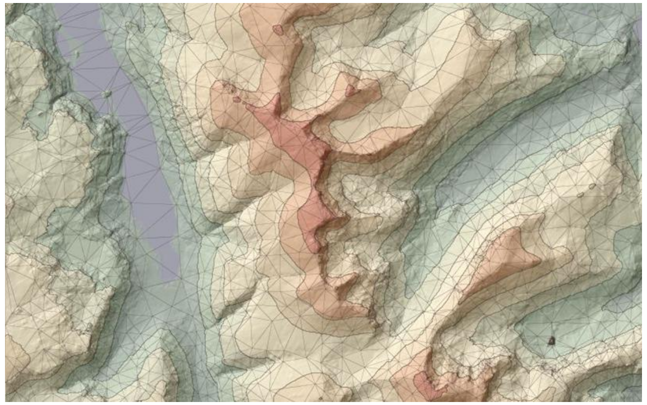


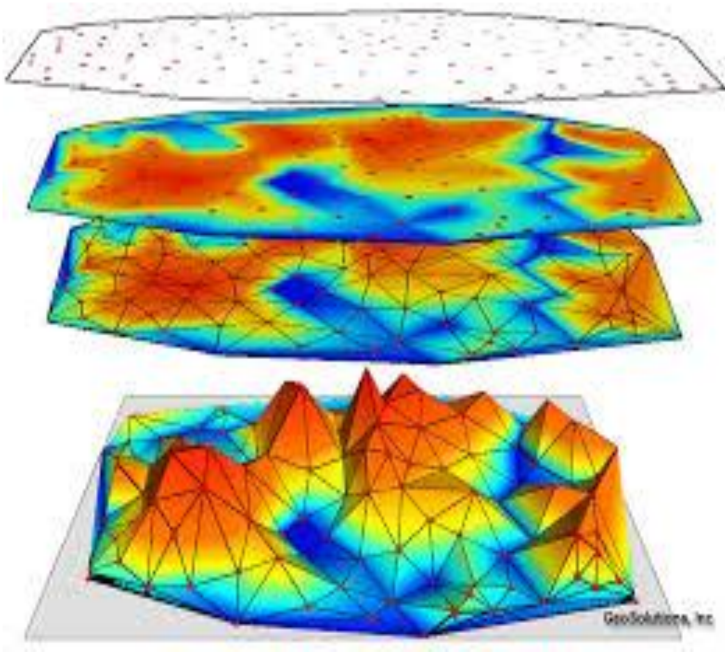
TIN

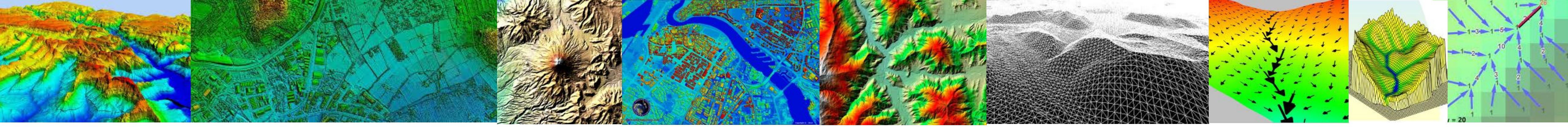
Los triángulos se construyen ajustando un plano a tres puntos cercanos no colineales, y se adosan sobre el terreno formando un mosaico que puede adaptarse a la superficie con diferente grado de detalle, en función de la complejidad del relieve





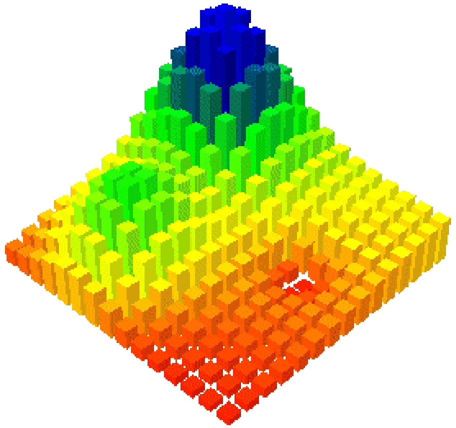


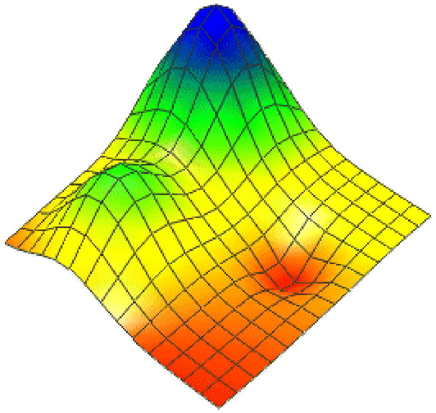




Matrices Regulares

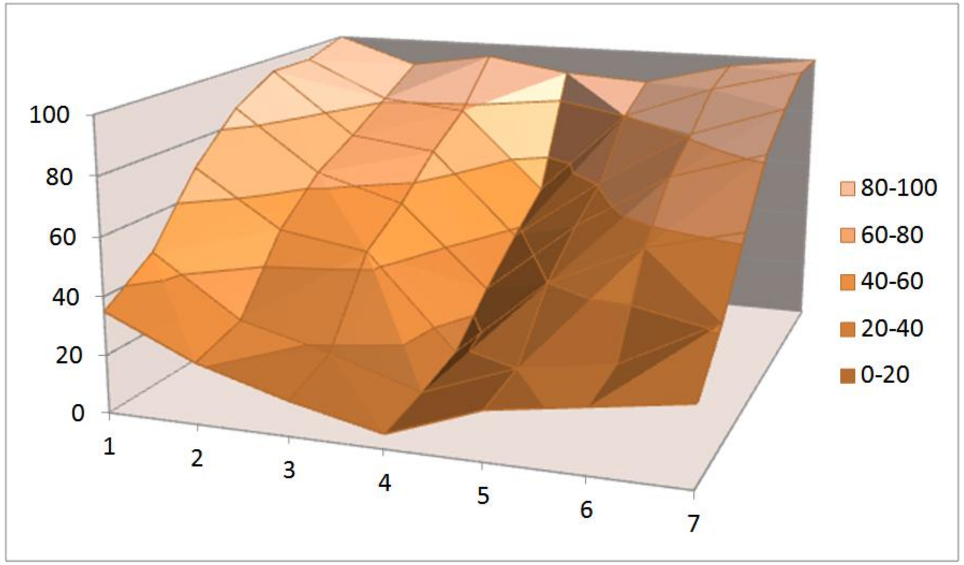
Esta estructura es el resultado de superponer una retícula sobre el terreno y extraer la altitud media de cada celda. La retícula adopta normalmente la forma de una red regular de malla cuadrada



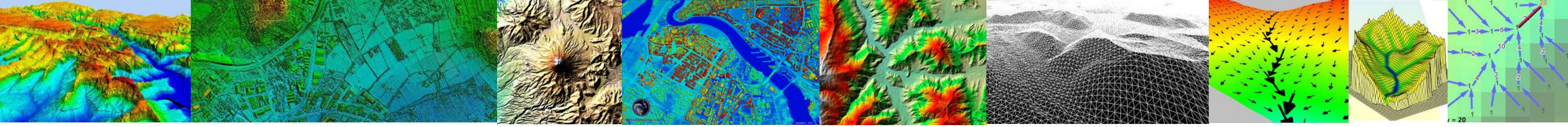


100	90	95	90	88	96	100
95	81	78	49	80	92	100
95	72	68	38	61	81	92
86	64	55	26	52	72	82
70	50	45	12	40	55	63
47	26	18	8	20	25	42
35	21	12	5	17	22	27

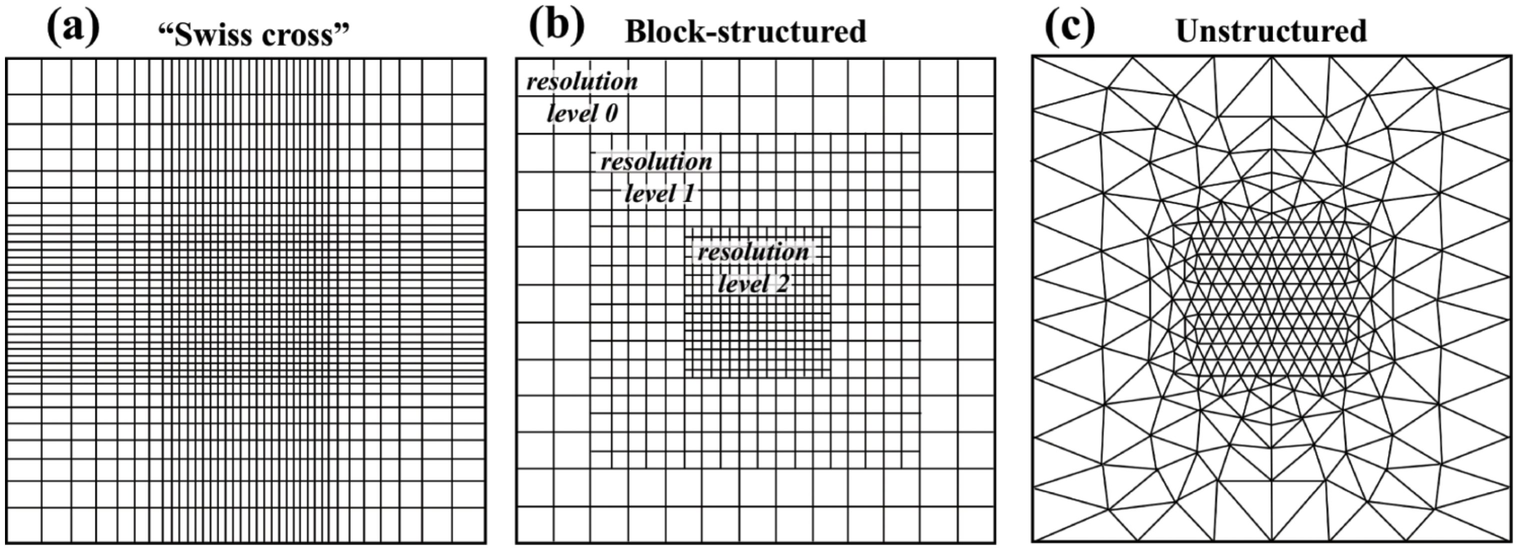
(a)

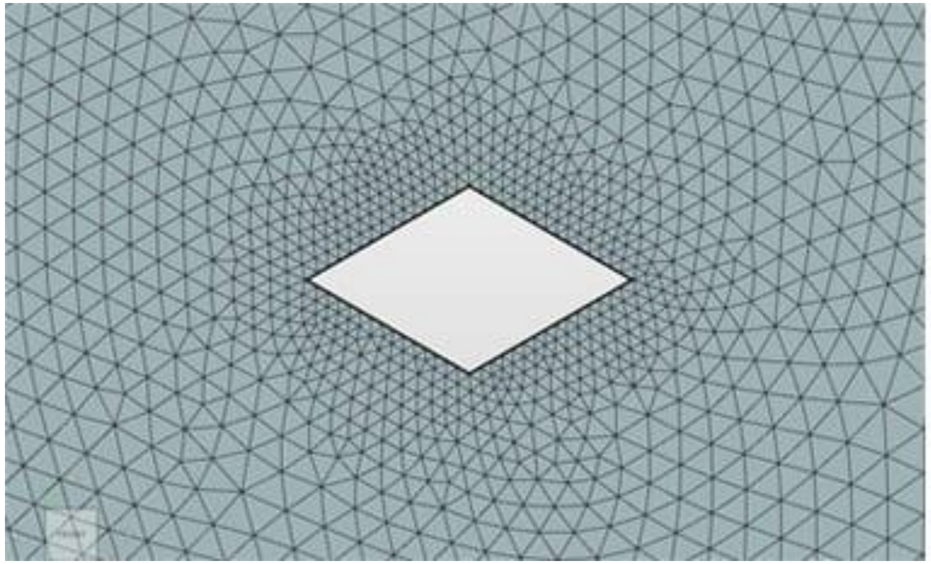


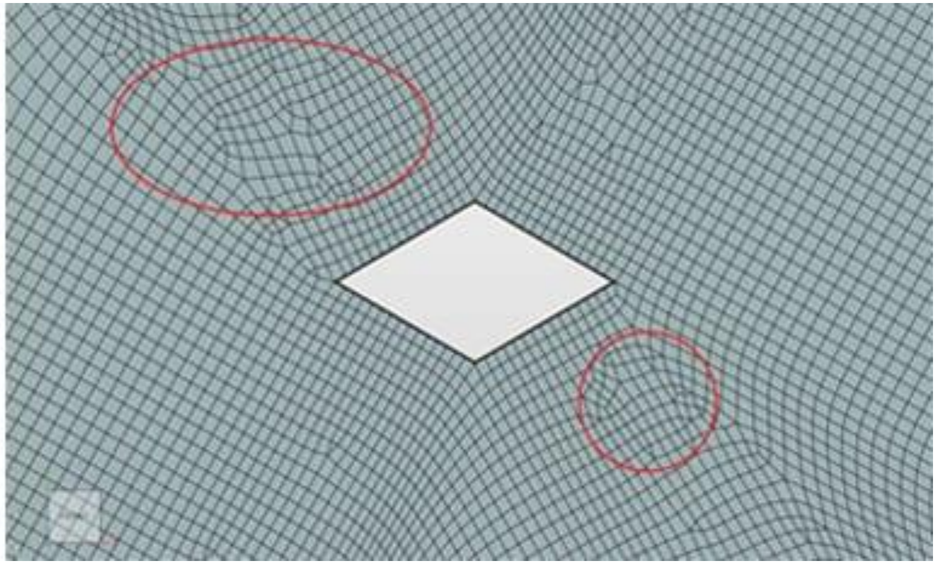
(b)

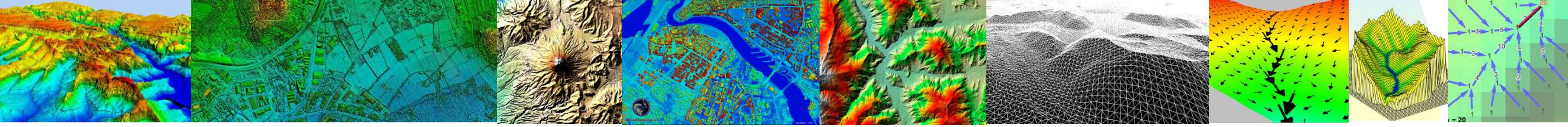


Matrices No Regulares









Pregunta de opción múltiple

En aplicaciones de ingeniería, los TIN son más eficientes que los rásters porque:

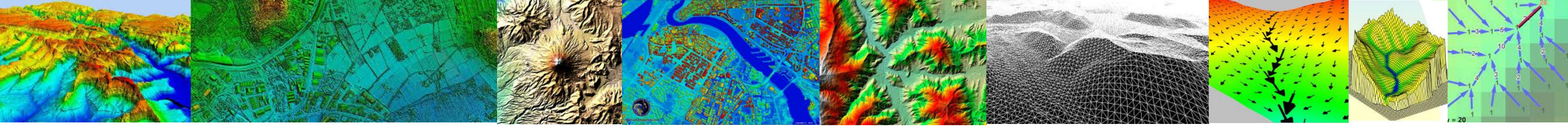
a. Los nodos que indican el eje del triángulo se pueden colocar irregularmente.

b. Las superficies se representan mediante caras que pueden sombrearse en diferentes colores para mostrar el relieve.

c. Son archivos mucho más pequeños y, por lo tanto, son más rápidos de representar que sus contrapartes ráster.

d. Todos los anteriores son verdaderos de TINs.

e. Solo hay dos respuestas correctas en las preguntas a y c.



Métodos para generarlos

1. **Métodos directos:** medida directa de la altitud sobre el terreno (fuentes primarias)

   - Altimetría: altímetros radares o laser transportados por plataformas aéreas o satélites
   - GPS: global positioning system, sistema de localización por triangulación
   - Levantamiento topográfico: estaciones topográficas con salida digital

2. **Métodos indirectos:** medida estimada a partir de documentos previos (fuentes secundarias)

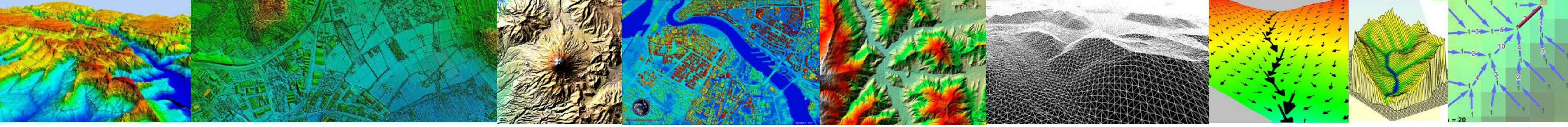
Restitución a partir de pares de imágenes

- Estereo-imágenes digitales: imágenes tomadas por satélites
- Estereo-imágenes analógicas : imágenes fotográficas convencionales
- Interferometría radar: imágenes de interferencia de sensores radar

Digitalización de mapas topográficos

Automática: mediante escáner y vectorización

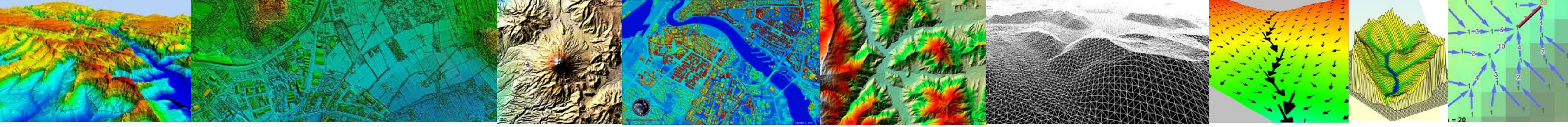
Manual: mediante tablero digitalizador



Levantamiento topográfico

GPS diferencial

Altímetros radar y láser

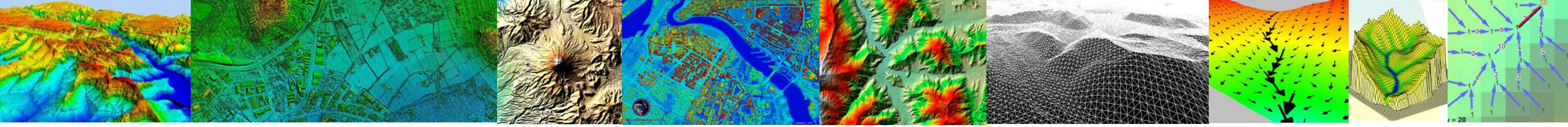


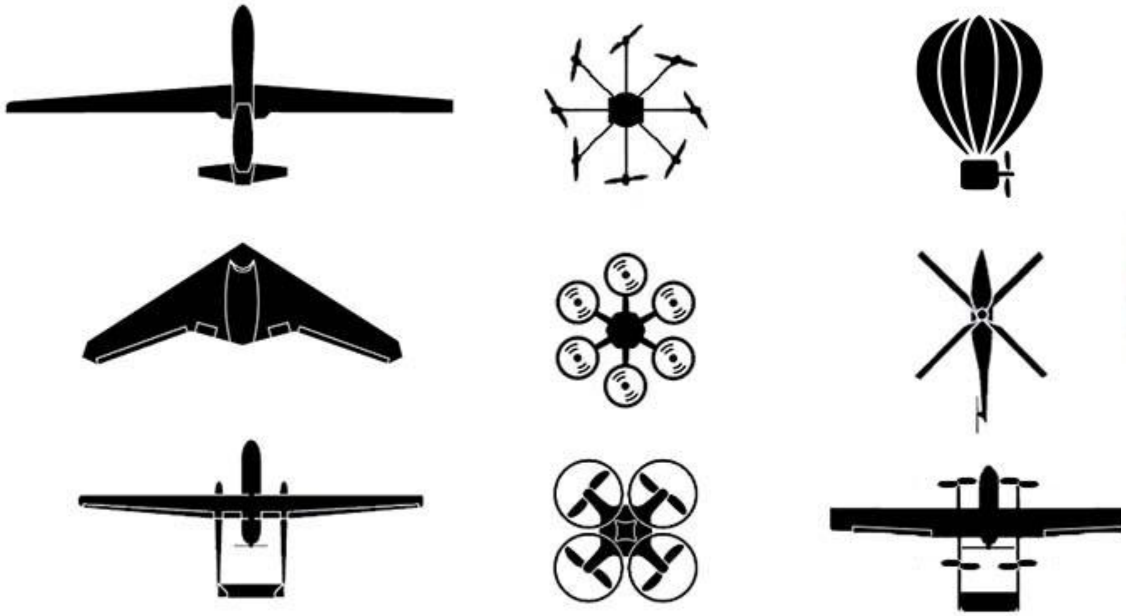
SEGÚN LA SUSTENTACIÓN

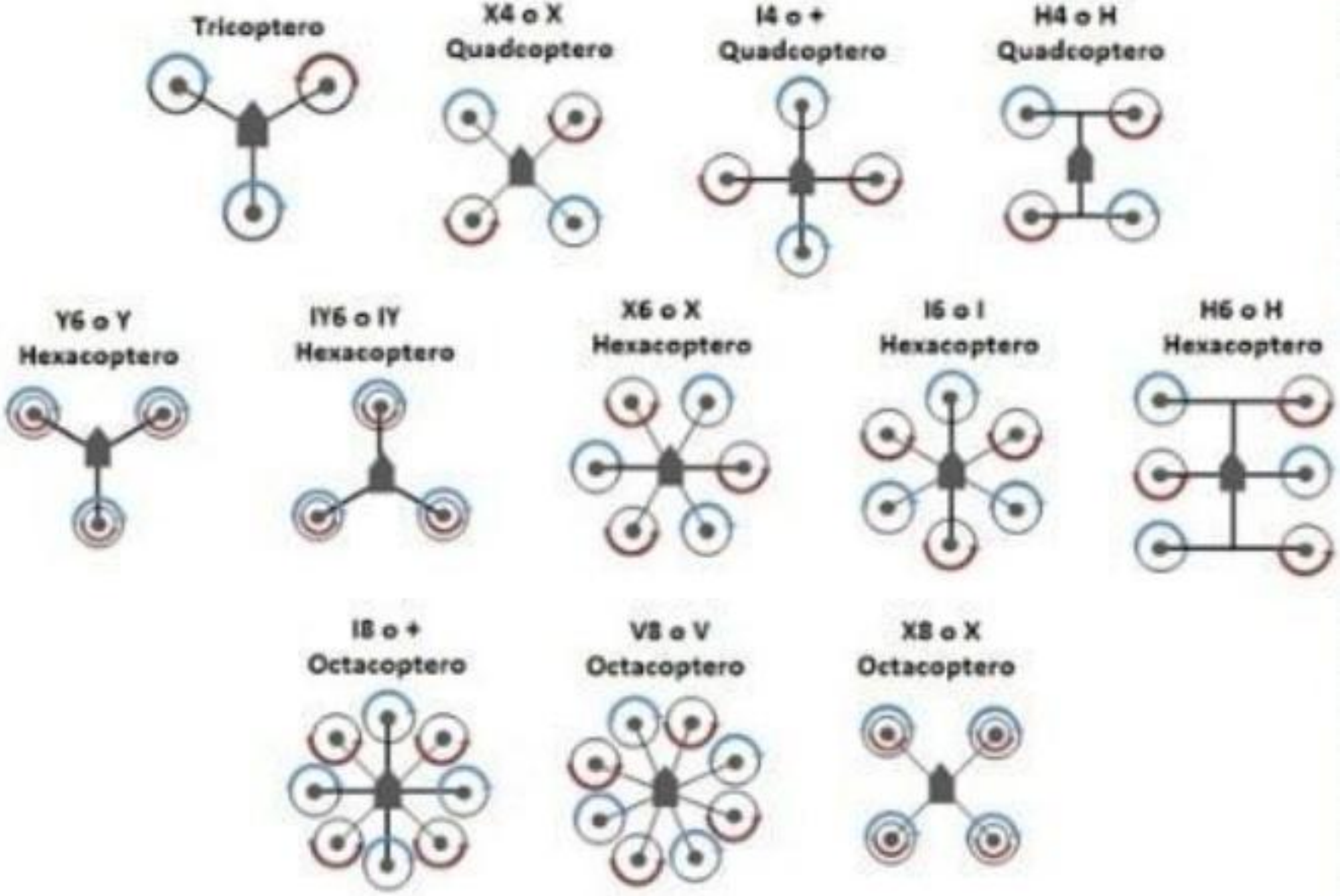
POR EL TIPO DE ALA

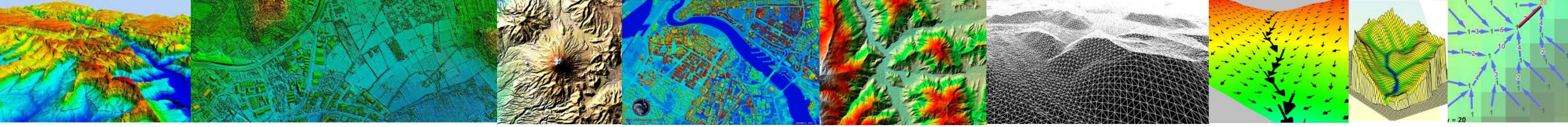
SEGÚN SU USO

SEGÚN SU APLICACIÓN






Tricoptero
X4 o X
Quadcoptero
I4 o +
Quadcoptero
H4 o H
Quadcoptero
Y6 o Y
Hexacoptero
IY6 o IY
Hexacoptero
X6 o X
Hexacoptero
I6 o I
Hexacoptero
H6 o H
Hexacoptero
I8 o +
Octacoptero
V8 o V
Octacoptero
X8 o X
Octacoptero



Sistema Lidar

Componentes del Sistema Lidar

LIDAR (Light Detection and Ranging o Laser Imaging Detection and Ranging) es un dispositivo que permite determinar la distancia desde un emisor láser a un objeto o superficie utilizando un haz láser pulsado. La distancia al objeto se determina midiendo el tiempo de retraso entre la emisión del pulso y su detección a través de la señal reflejada

Medio Aéreo

Puede ser un avión o un helicóptero. Cuando se quiere primar la productividad y el área es grande se utiliza el avión, y cuando se quiere mayor densidad de puntos se usa el helicóptero, debido a que éste puede volar más lento y bajo.

ALS

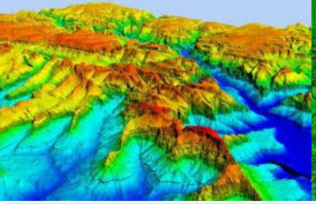
Emite pulsos de luz infrarroja que sirven para determinar la distancia entre el sensor y el terreno.

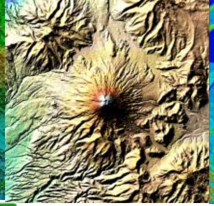
INS

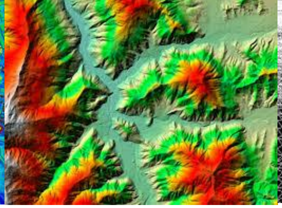
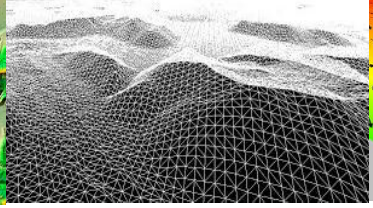
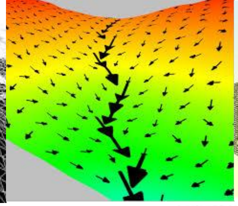
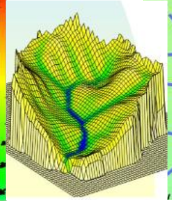
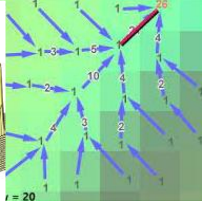
Sistema Inercial de Navegación. Nos informa de los giros y de la trayectoria del avión.

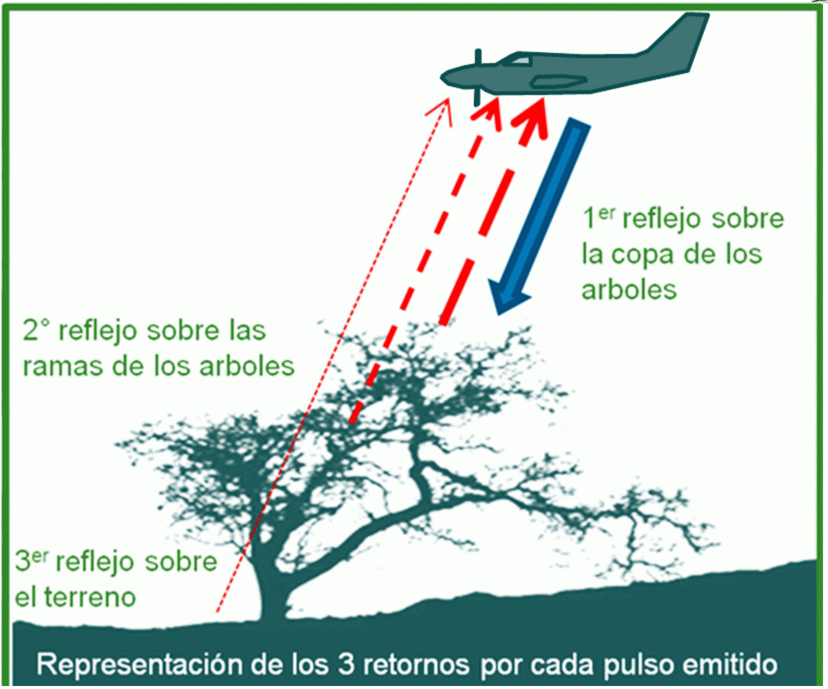
GPS Diferencial

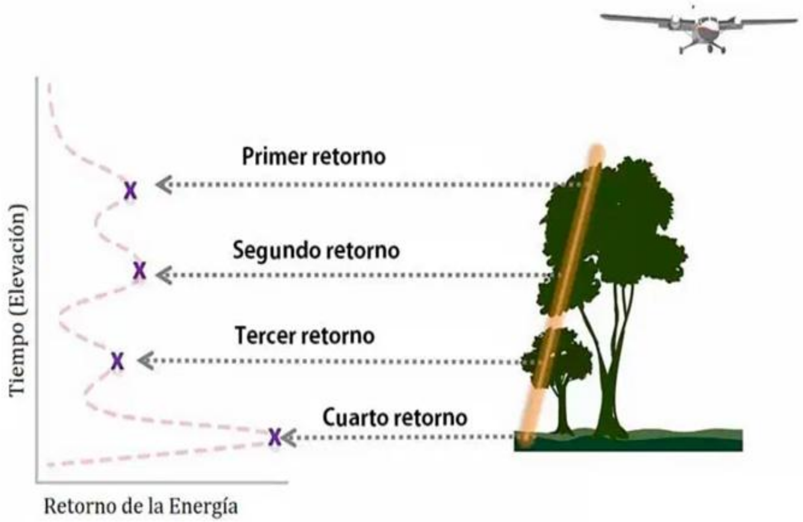
Mediante el uso de un receptor en el avión y uno o varios en estaciones de control terrestres (en puntos de coordenadas conocidas), se obtiene la posición y altura del avión.

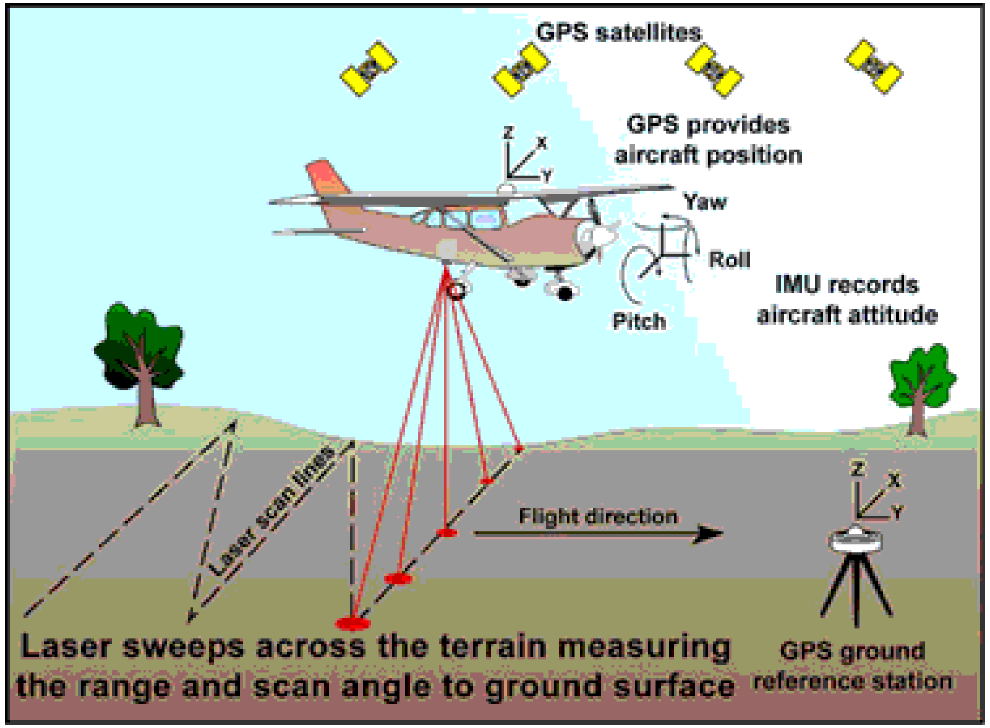


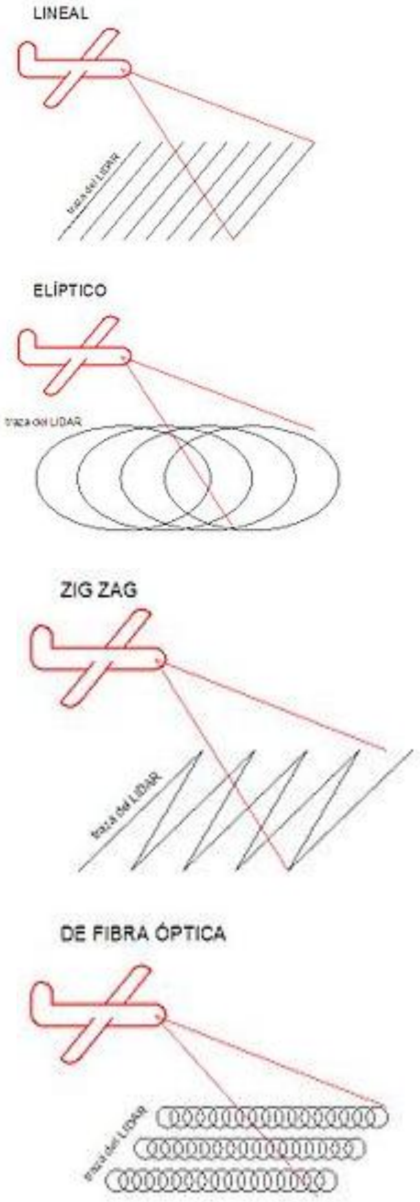


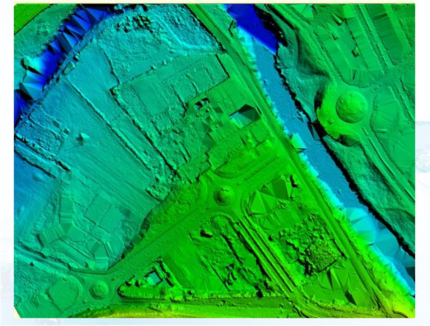






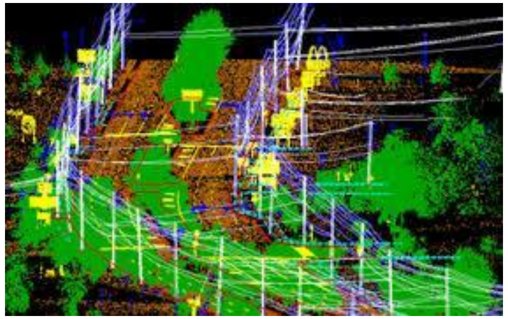
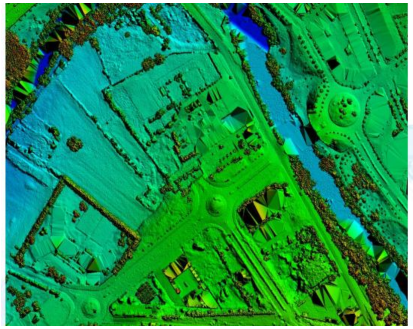
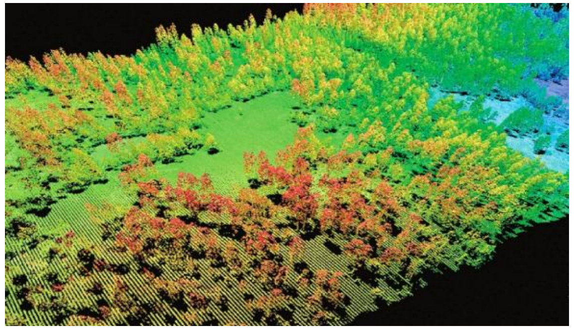
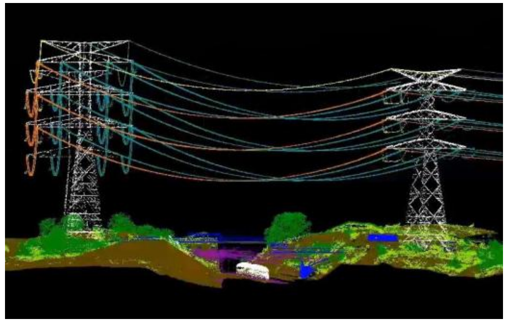
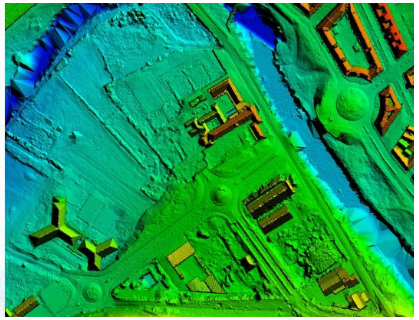
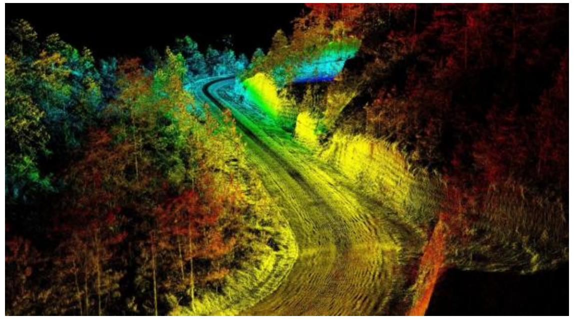
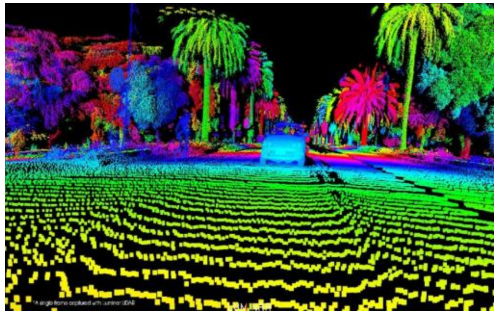
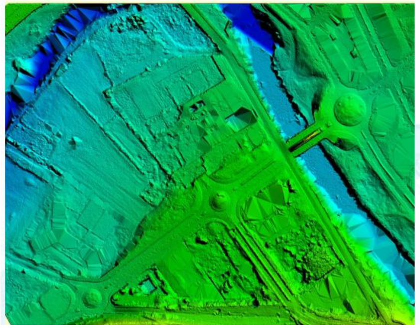
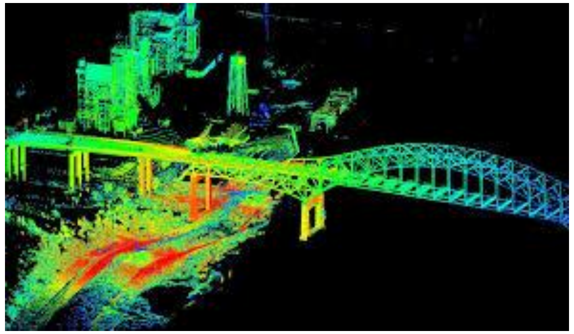
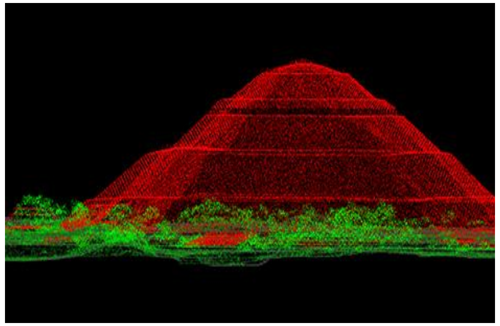

1er reflejo sobre la copa de los arboles
2° reflejo sobre las ramas de los arboles
3er reflejo sobre el terreno
Representación de los 3 retornos por cada pulso emitido

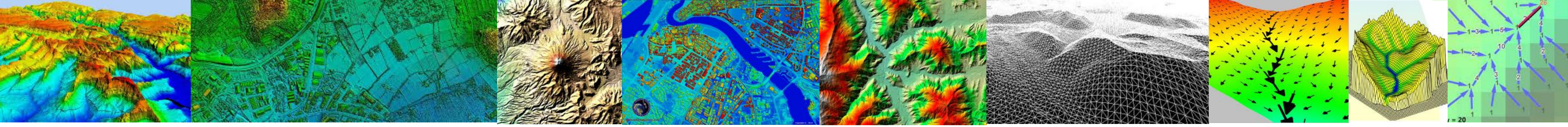

Primer retorno
Segundo retorno
Tercer retorno
Cuarto retorno
Tiempo (Elevación)
Retorno de la Energía


GPS satellites
GPS provides aircraft position
Yaw
Roll
Pitch
IMU records aircraft attitude
Laser scan lines
Flight direction
Laser sweeps across the terrain measuring the range and scan angle to ground surface
GPS ground reference station


LINEAL
ELÍPTICO
ZIG ZAG
DE FIBRA ÓPTICA

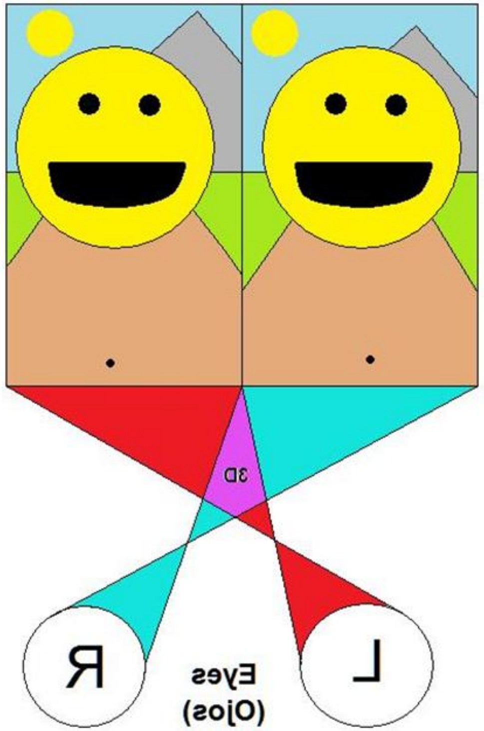


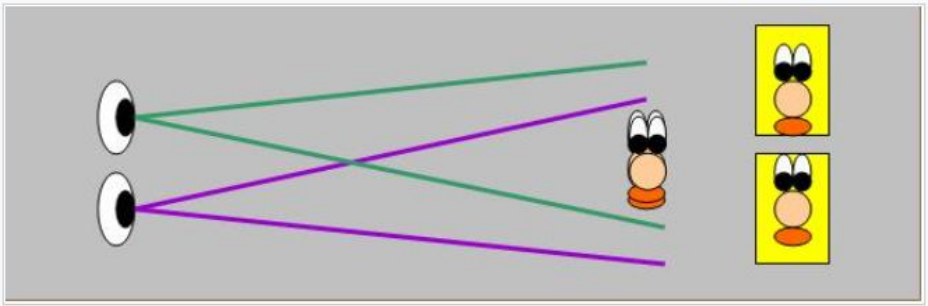













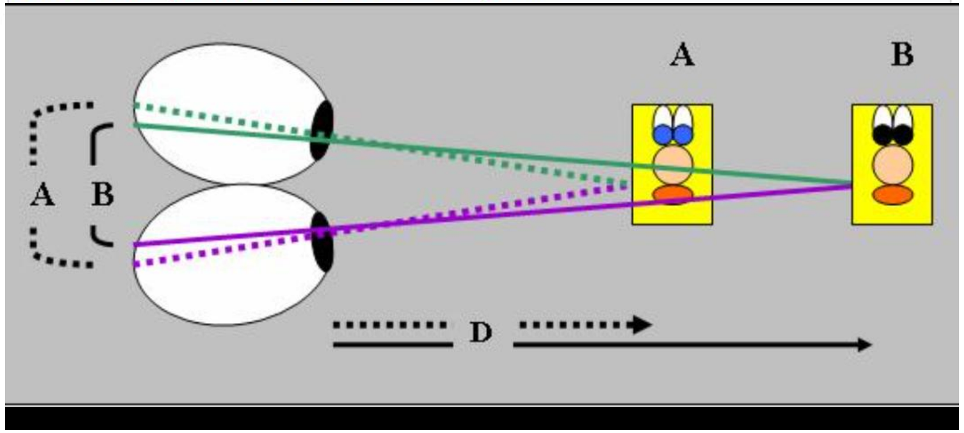
Estereoscopia

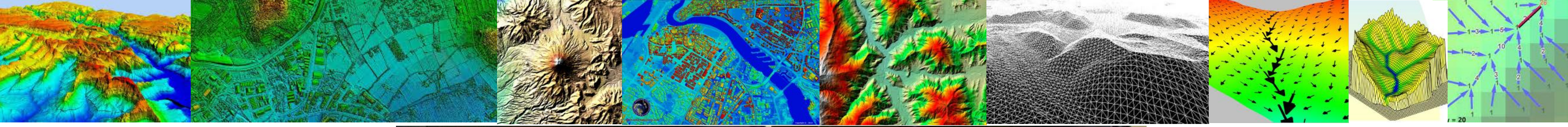
Visión estereoscópica es aquella que integra dos imágenes que, por medio del cerebro, el ser humano es capaz de integrar en una sola, y crear una imagen tridimensional. La visión estereoscópica es una habilidad físico-psicológica del ser humano, permite ver imágenes tridimensionales con una visión binocular. En la retina de cada ojo se forma una imagen de un objeto, las dos imágenes son ligeramente distintas por la diferencia posición de los ojos, formando un efecto de relieve.





El ojo percibe los objetos en diferentes ángulos, creando la ilusión de profundidad de los objetos.

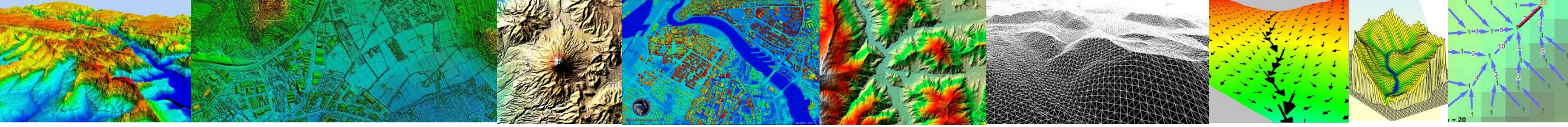





INDIANI
INDIANI


INDIANI


INDIANI

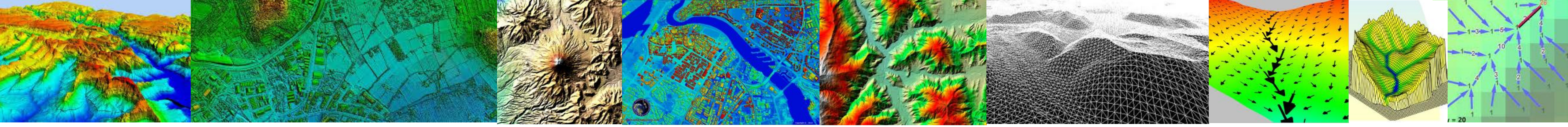


Fotogrametría

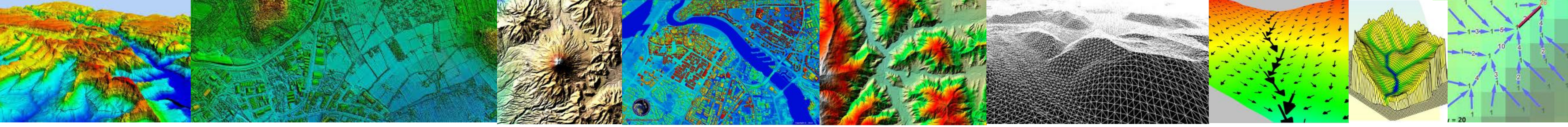
No es preciso acceder físicamente a la zona de estudio, pues utilizan documentos preexistentes, y la generación de datos se hace de forma relativamente rápida, cuestión básica cuando el volumen de información es muy elevado.

Estereo-imágenes analogicas

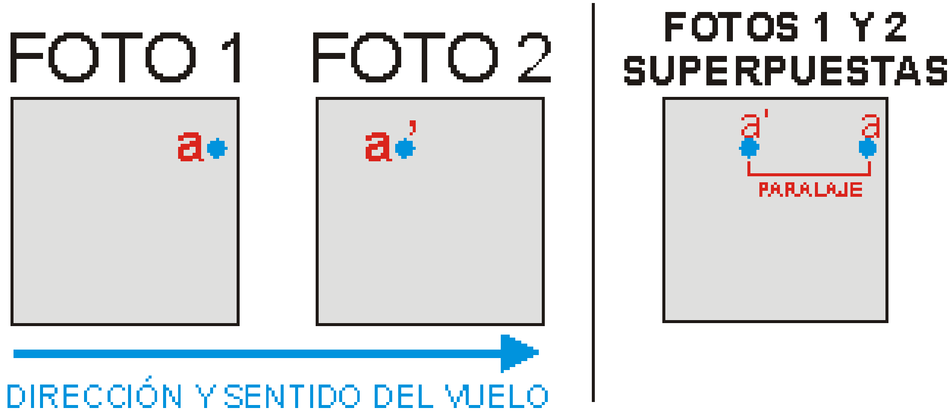
En la restitución fotogramétrica son fotos aéreas tomadas desde orientaciones diferentes, que forman los denominados pares estereoscópicos. Examinando puntos homólogos en los pares estereoscópicos es posible deducir de su paralaje las cotas de referencia necesarias para reconstruir la topografía.

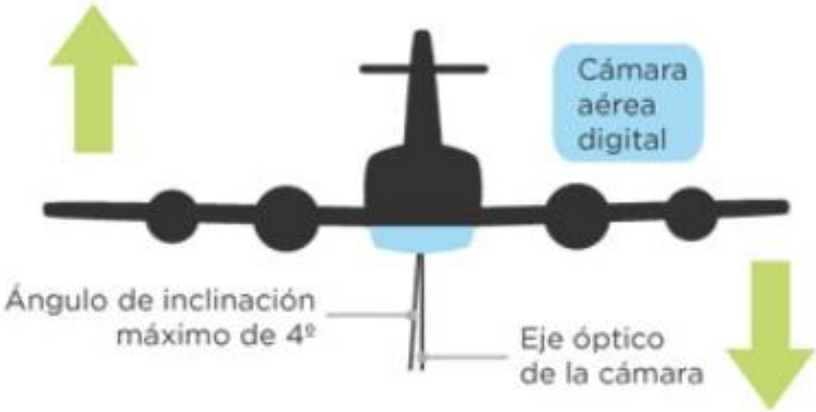


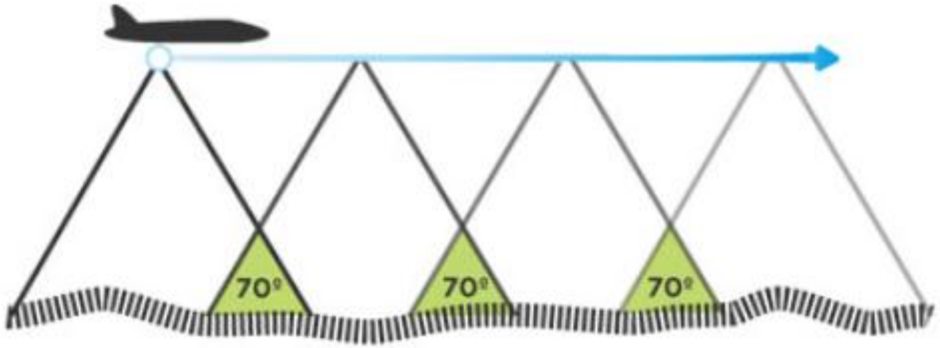
Estereoscopio es un aparato que presenta una doble imagen para ser unidas a nivel perceptual en nuestro cerebro y así se ve una sola imagen estereoscópica, que permite percibir las tres dimensiones

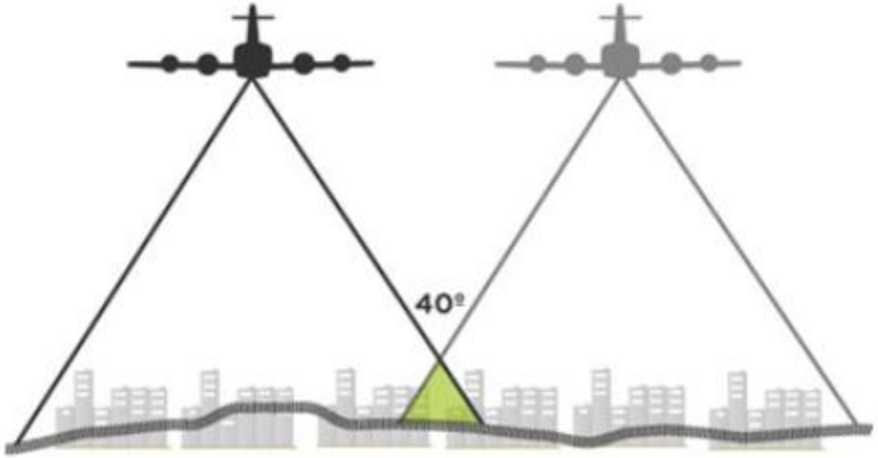


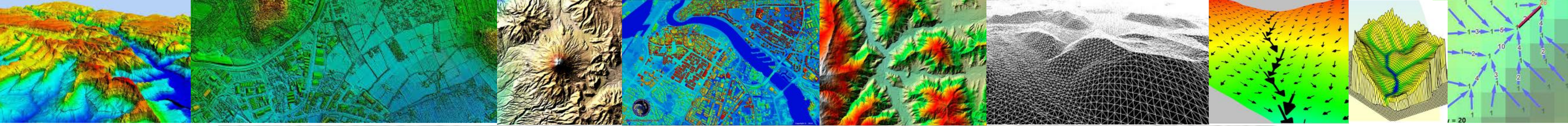
Existen sistemas automatizados que realizan la restitución, los resultados en un formato digital son compatibles con sistemas de información geográfica. El acceso al terreno es necesario para establecer un conjunto de puntos de apoyo que permitan fijar valores de altitud en una escala absoluta.

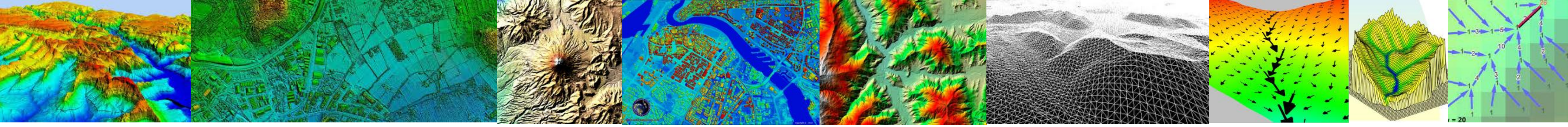




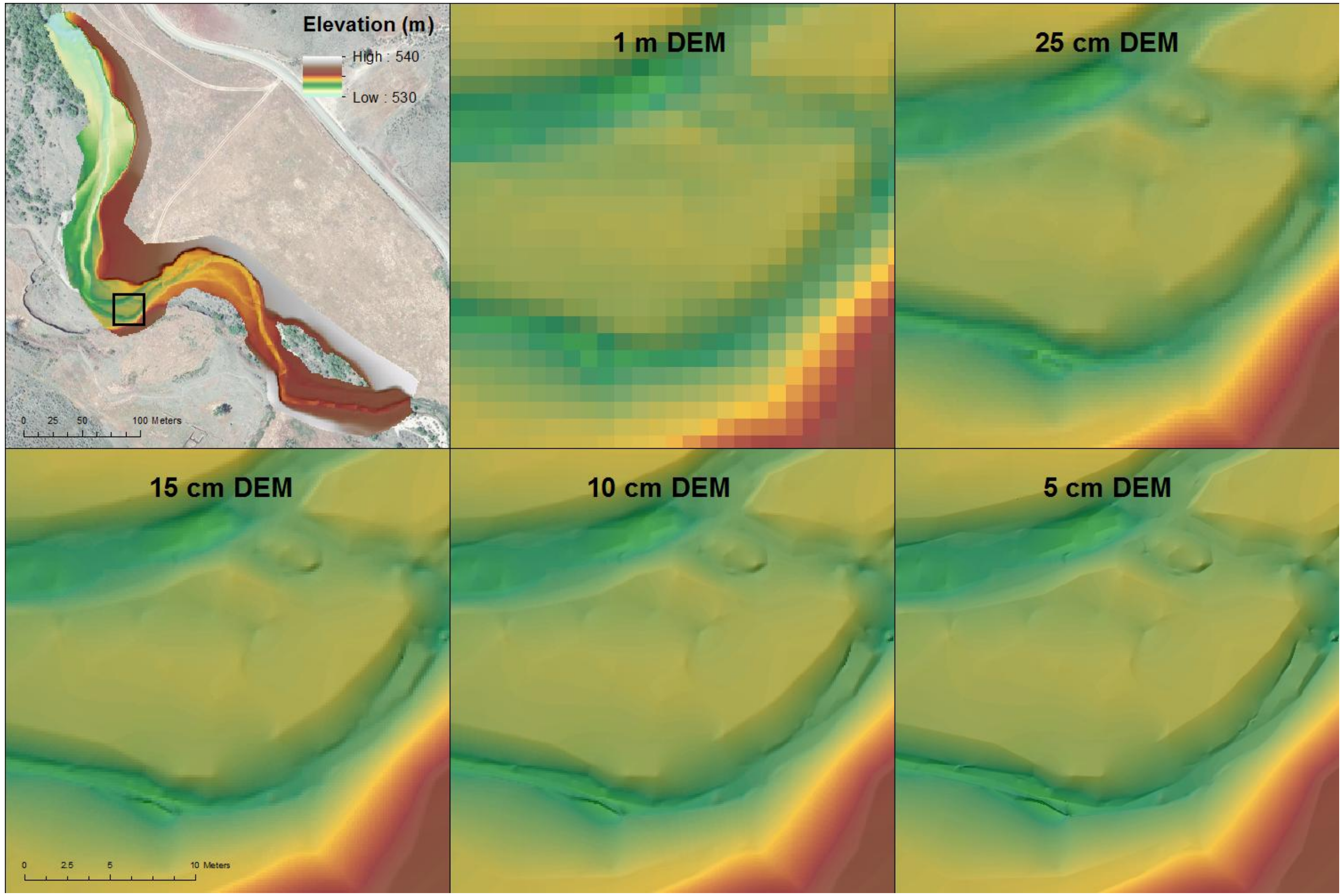


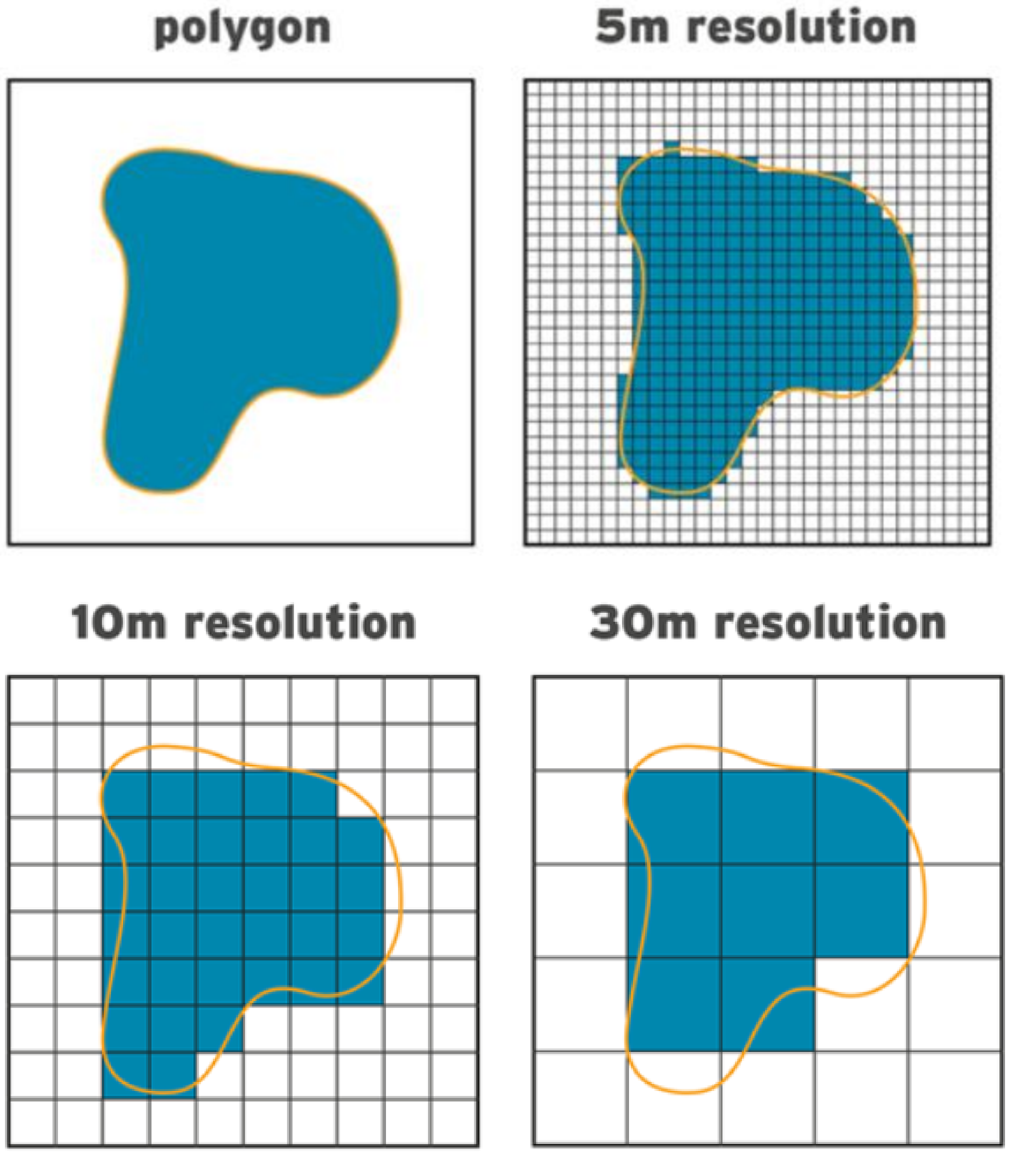


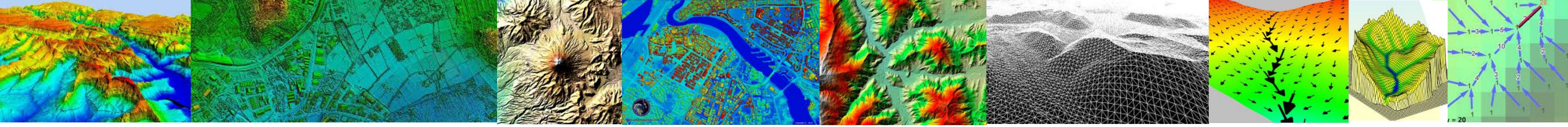




MDE o DEM

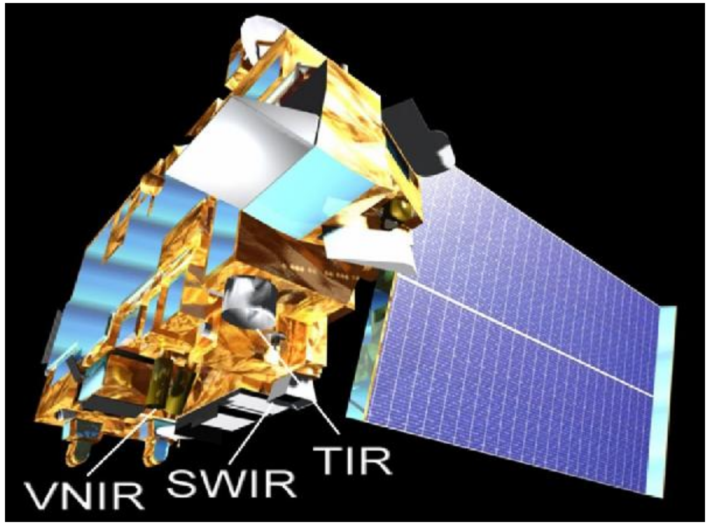

Elevation (m)
High : 540
Low : 530
0 25 50 100 Meters
1 m DEM
25 cm DEM
15 cm DEM
10 cm DEM
5 cm DEM
0 2.5 5 10 Meters


polygon
5m resolution
10m resolution
30m resolution

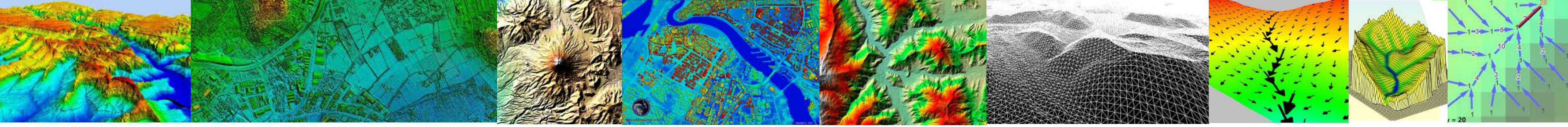


Radiómetro espacial avanzado de emisiones térmicas y reflexión (ASTER)

Creado por el Ministerio de Economía, Comercio e Industria de Japón (METI), usa una banda espectral infrarroja generando imágenes estéreo, con una relación de base-altura de 0,6 (Tachikawa et al., 2011) que cubre la superficie terrestre mundial desde 83 ° N a 83 ° S (Abrams et al., 2015). Este MDT tiene una referencia espacial de 1 segundo de arco (~ 30 m) cobertura mundial. El modelo de elevación digital (GDEM) versión 2, fue lanzado en octubre de 2011 (tres años después de que su predecesor, la versión 1), además de la adición de 200.000 nuevas imágenes para el proceso estereoscópico, varias anomalías fueron corregidas, y la precisión global se mejoró. Este MDT tiene una precisión vertical ± 17 metros

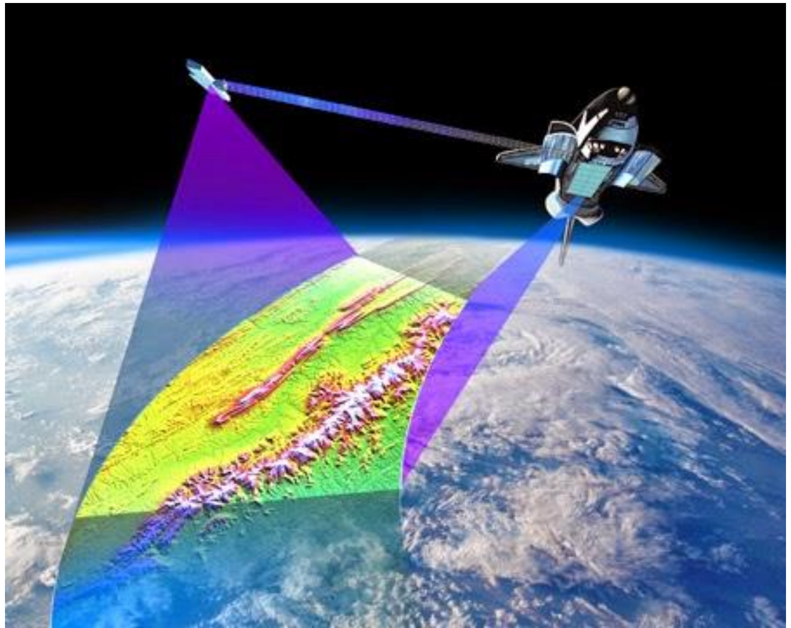


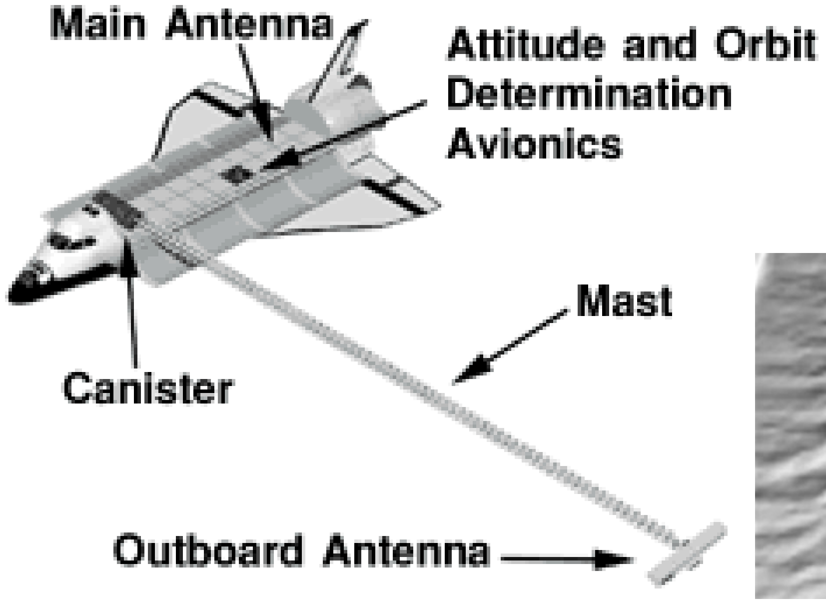
VNIR: Radiómetro visible
SWIR: Radiómetro infrarrojo
TIR: Radiómetro infrarrojo termal

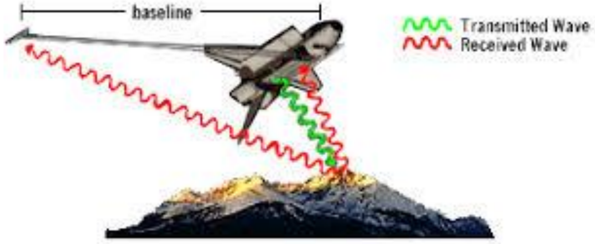


Misión Topográfica Radar Shuttle (SRTM)

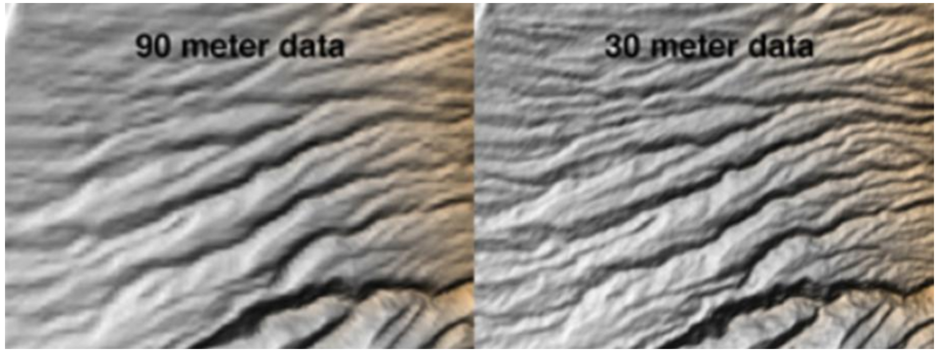
Es un proyecto cooperativo entre la administración espacial y aeronáutica nacional de Estados Unidos (NASA) y la agencia de imágenes y cartografía nacional (NIMA) del departamento de defensa de Estados Unidos. La misión fue diseñada para utilizar un interferómetro de radar, para producir un modelo de elevación digital de la superficie de la tierra entre aproximadamente 60° N y 56° S, esto es alrededor del 80 por ciento de la masa terrestre del planeta (Rabus et al., 2003). Esta misión se llevó a cabo durante 11 días del mes de febrero del 2000. Este MDT tiene una referencia espacial de 1 segundo de arco (~ 30 m), y tiene una precisión vertical ± 16 metros (Gesch, 2006; Kellndorfer et al., 2004; Sun et al., 2003).

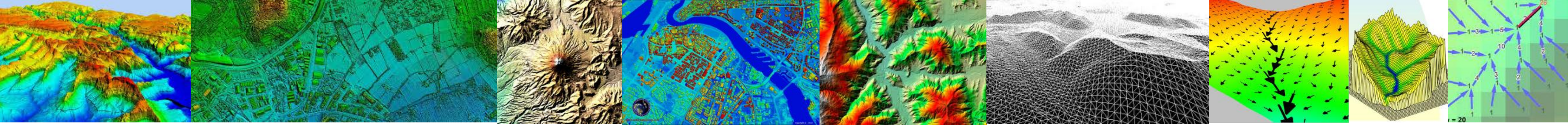


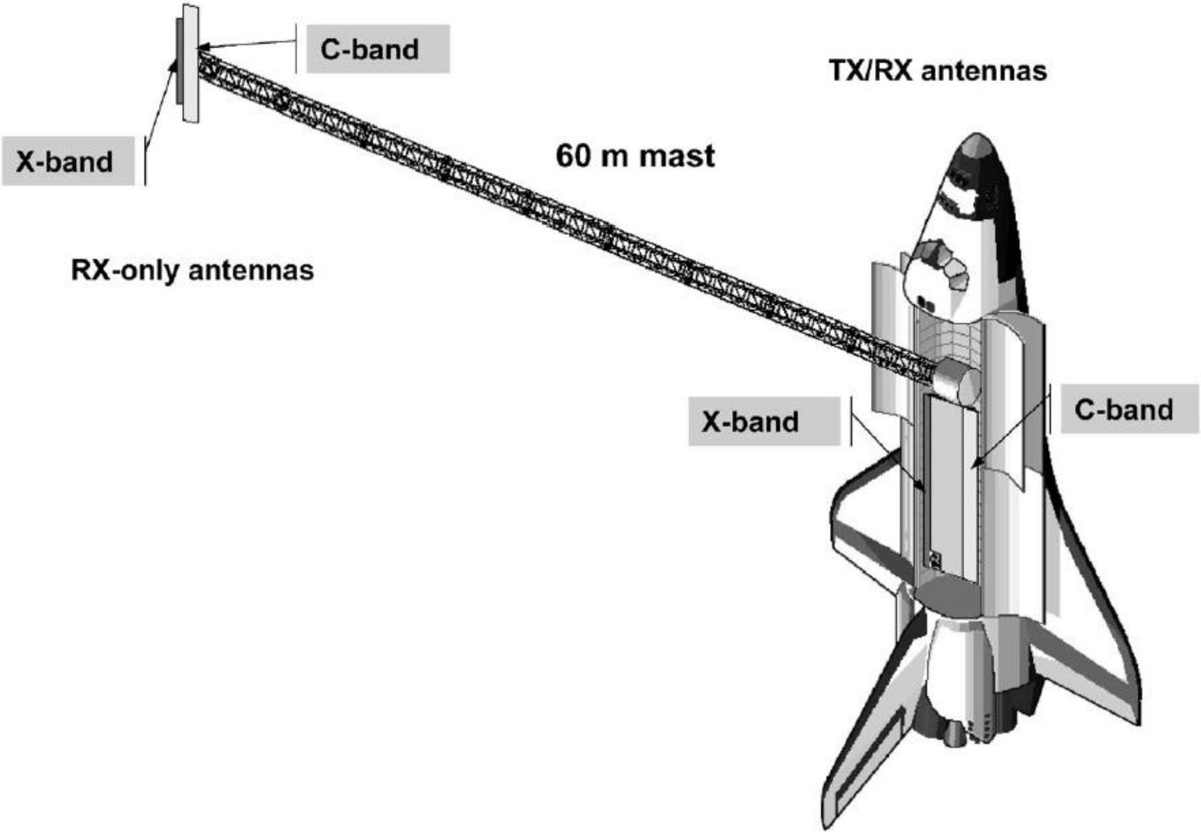


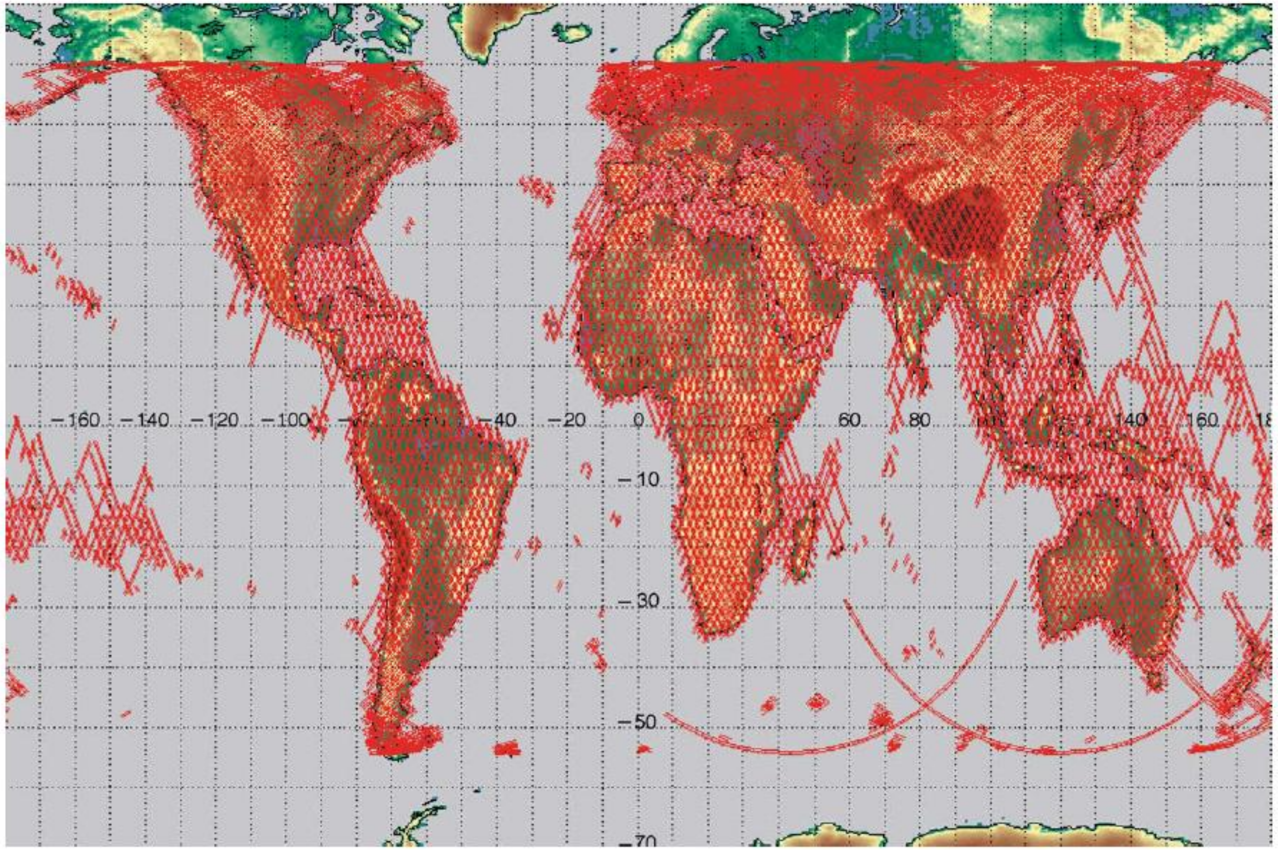


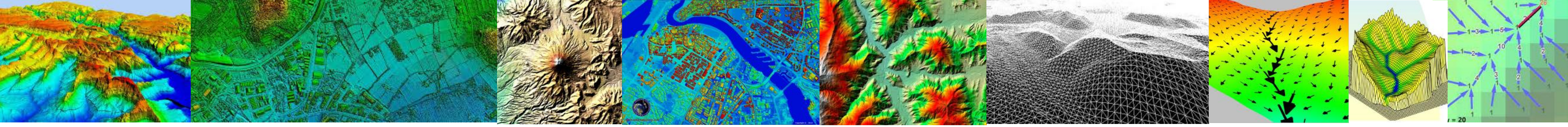
Radar signals being transmitted and recieved in the SRTM mission (image not to scale).





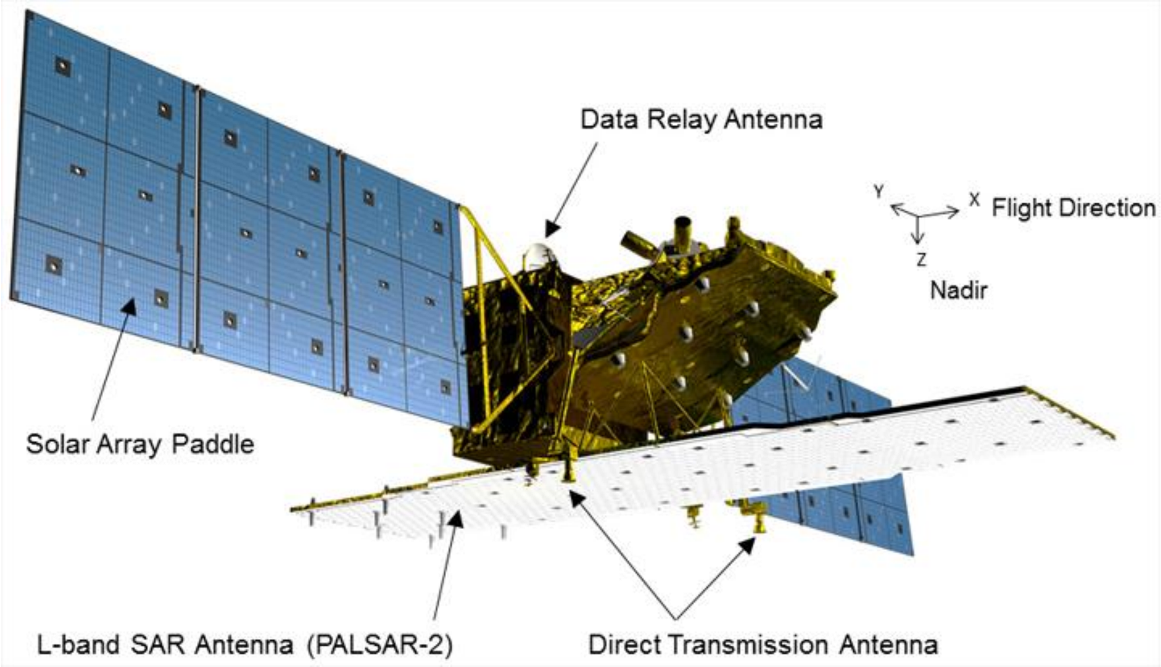

C-band
TX/RX antennas
X-band
60 m mast
RX-only antennas
X-band
C-band

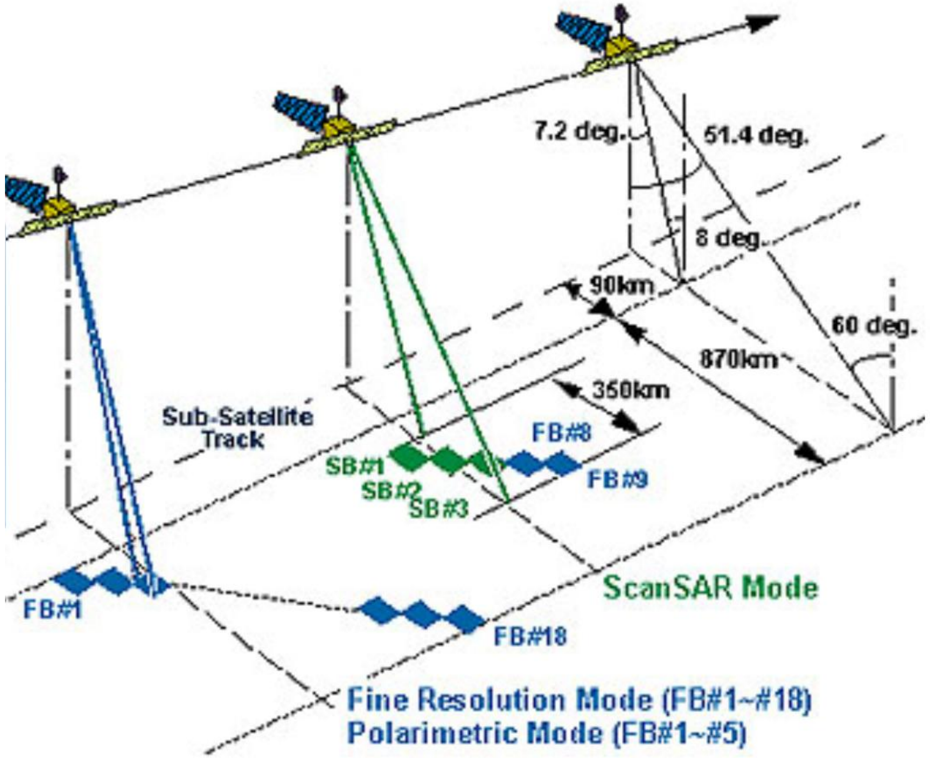

−160
−140
−120
−100
−40
−20
0
60
80
140
160
−10
−30
−50
−70

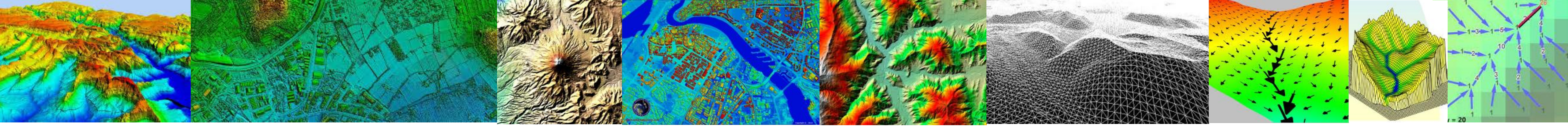


ALOS PALSAR

El MDE del satélite de observación avanzado de la tierra (ALOS) de la Agencia de Exploración Espacial Japonesa (JAXA) utiliza el sensor de apertura sintética de banda L de tipo escalonado (PALSAR) que opera con un ancho de banda y una frecuencia de 14 a 28 MHz. Este MDE tiene una resolución espacial de 12,5 m y cubre un área de 156,25 m^2 por pixel (Chu y Lindenschmidt, 2017; Rosenqvist et al., 2007; Kimura y Ito, 2000)



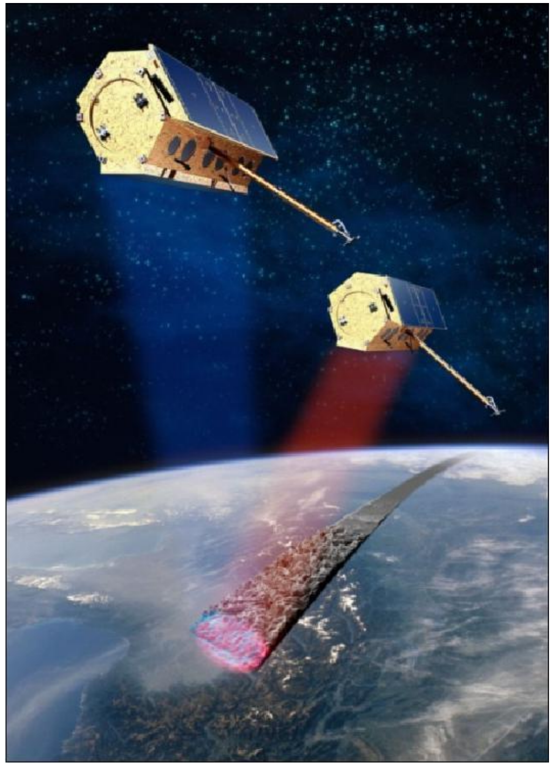


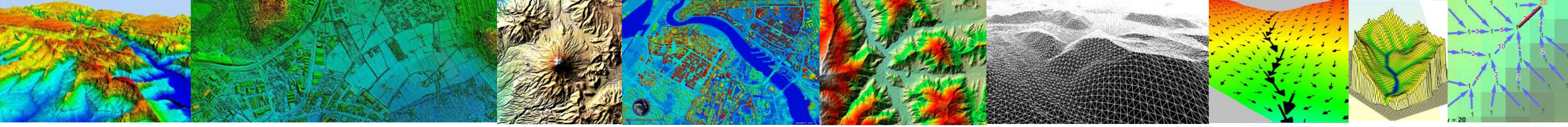


TanDEM-X

Interferómetro de radar de apertura sintética (SAR), lanzado gracias a la colaboración público-privada entre el Centro Aeroespacial Alemán (DLR) y EADS Astrium.

La fiabilidad de la adquisición de los datos del sistema TanDEM-X es inmensa, porque operan de manera independiente a la cobertura nubosa, y a las condiciones de iluminación. La homogeneidad de la adquisición, garantiza un Modelo Digital sin líneas de rotura en fronteras ya sean regionales o nacionales.





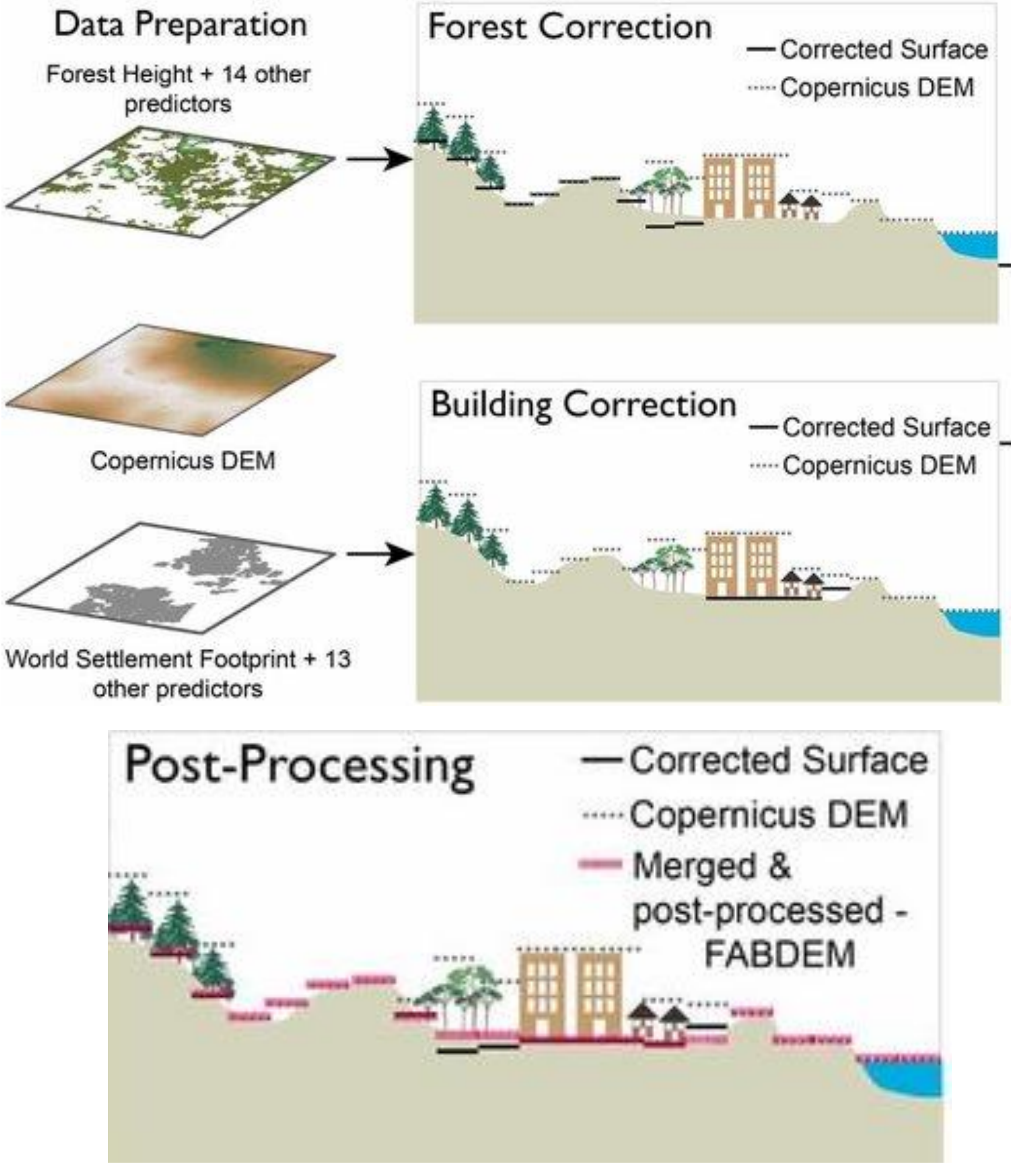
FABDEM

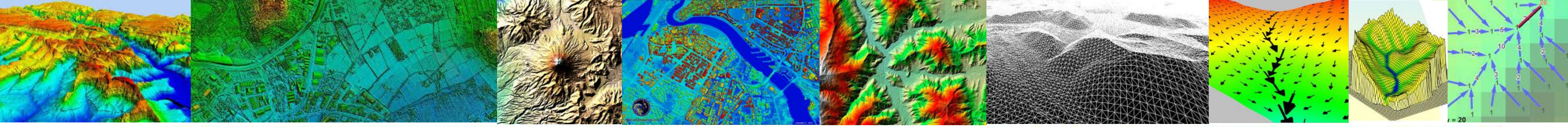
Forest And Buildings removed Copernicus DEM) es un modelo digital del terreno (DTM) global a 1 arc-second (~30 m) que elimina sistemáticamente la señal de copas de árboles y edificaciones presente en el Copernicus GLO-30 DEM mediante técnicas de *machine learning*. El resultado es una superficie "bare-earth" más coherente y precisa para aplicaciones hidrológicas, geomorfológicas y de gestión del riesgo.

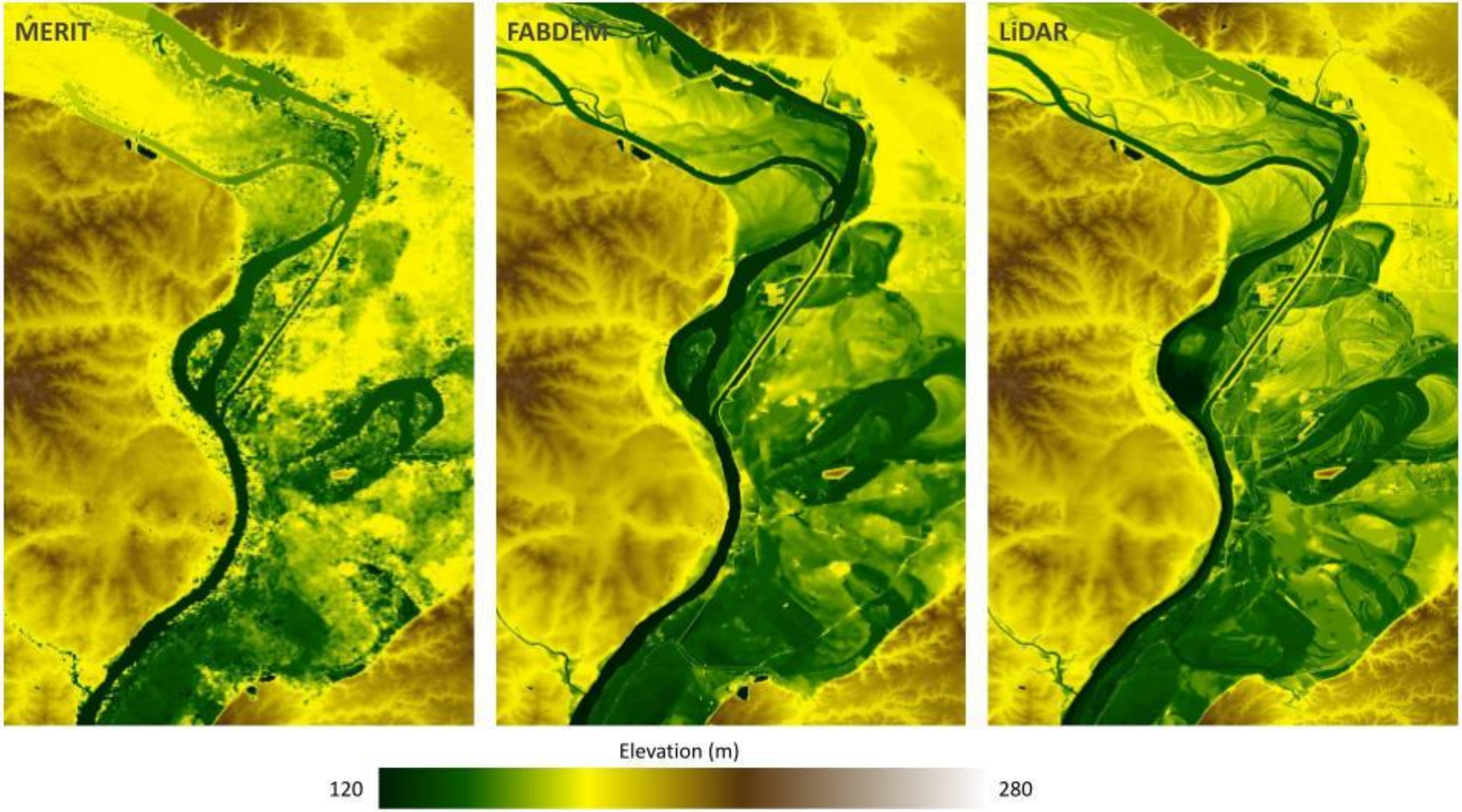
1. Origen y método de generación

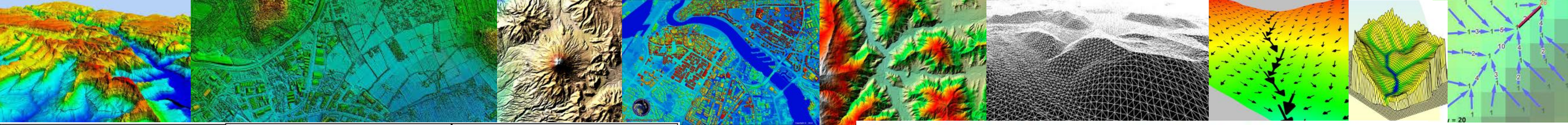
•Parte del DSM Copernicus GLO-30 (derivado de TanDEM-X).

•Un algoritmo de *Random Forest* entrenado con ~78 millones de puntos de referencia LiDAR y SRTM bare-earth identifica y corrige alturas anómalas asociadas a vegetación y construcciones.






MERIT
FABDEM
LiDAR
Elevation (m)
120
280



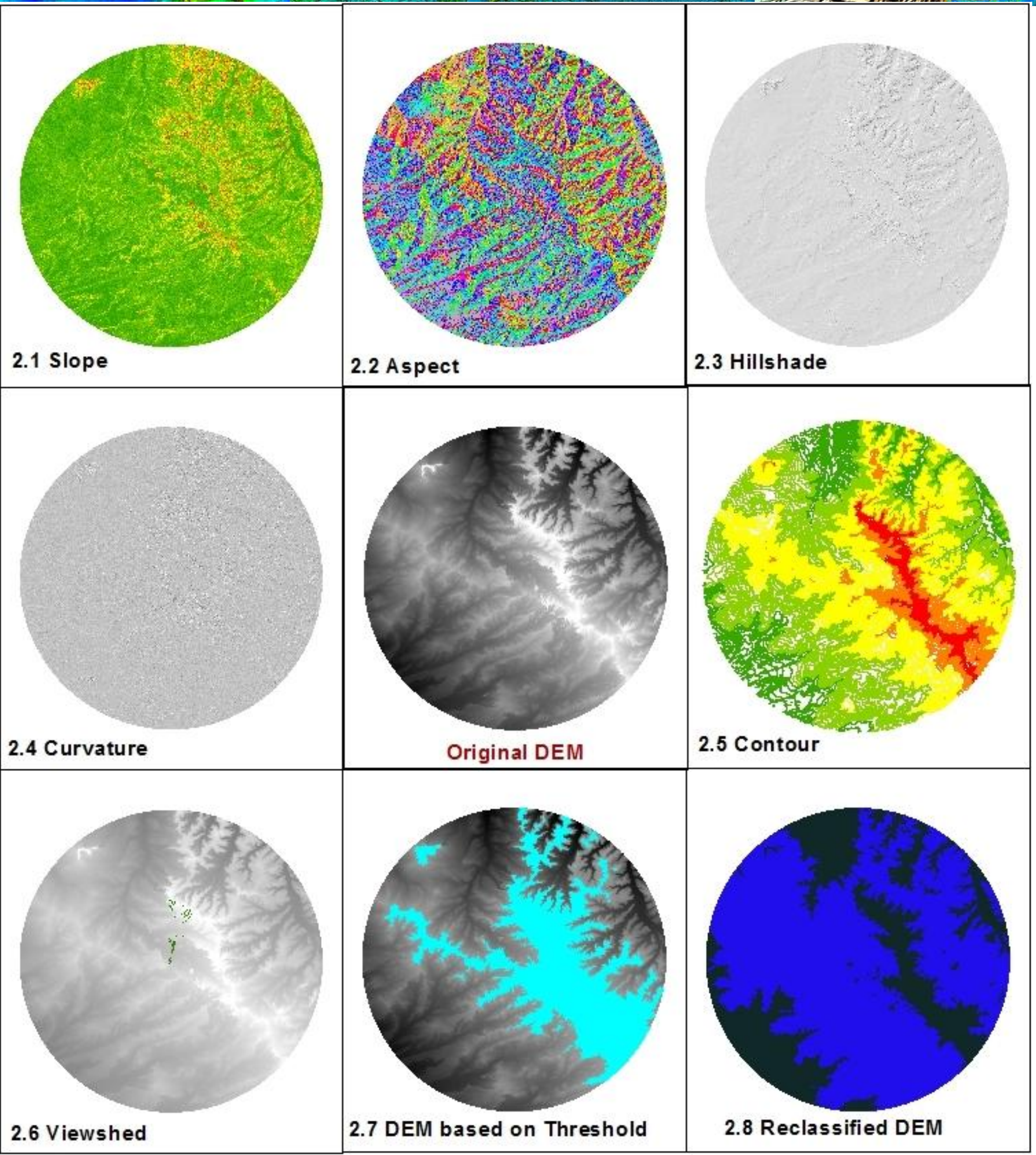
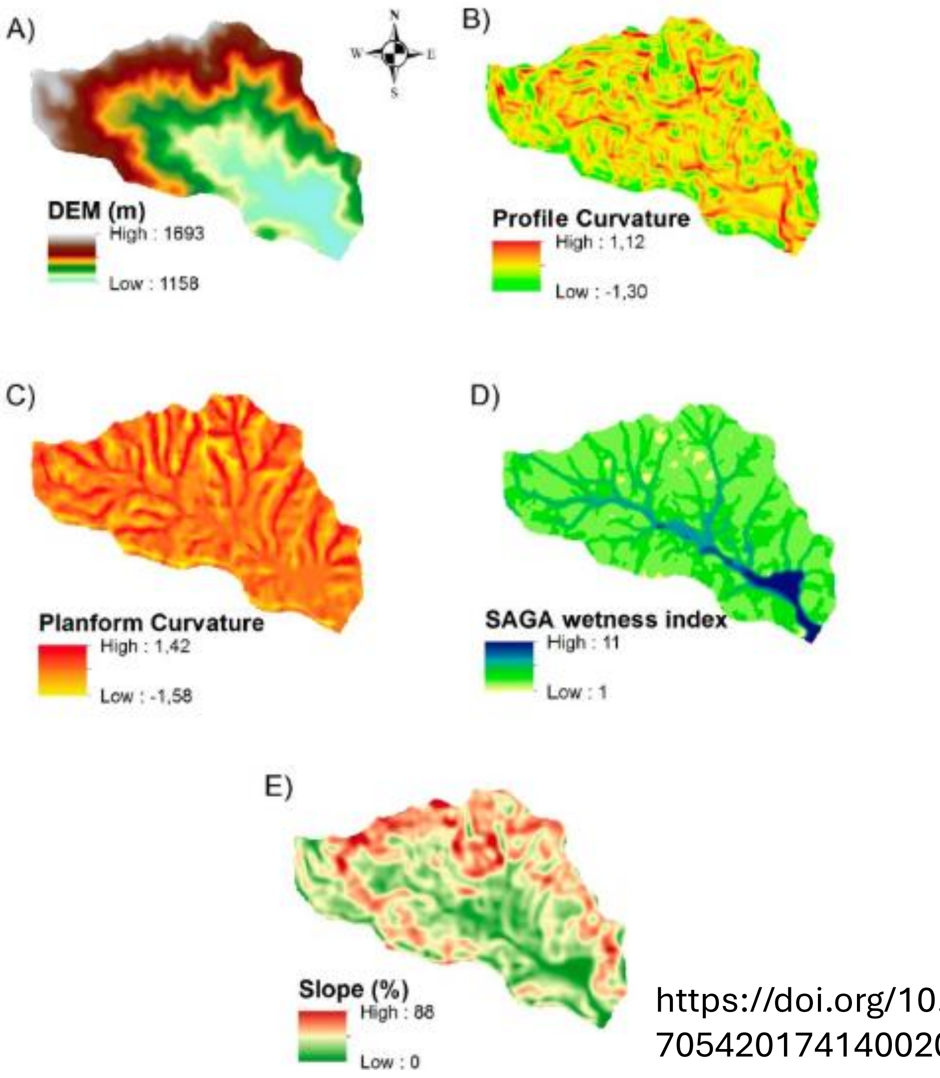
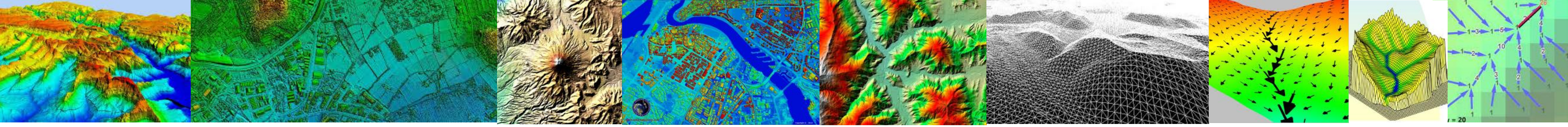


Figure 2: Digital elevation model (DEM) (A) and terrain attributes (TA): profile curvature (B), planform curvature (C), SAGA wetness index (D) and slope (E) for LCW.

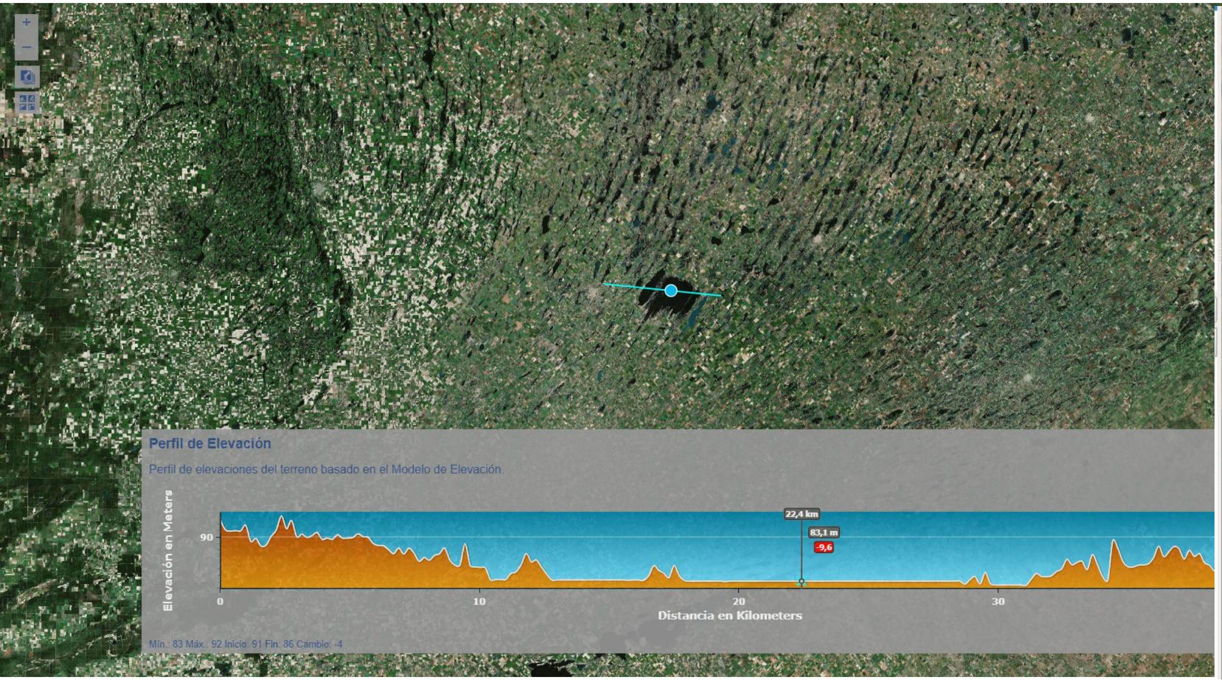


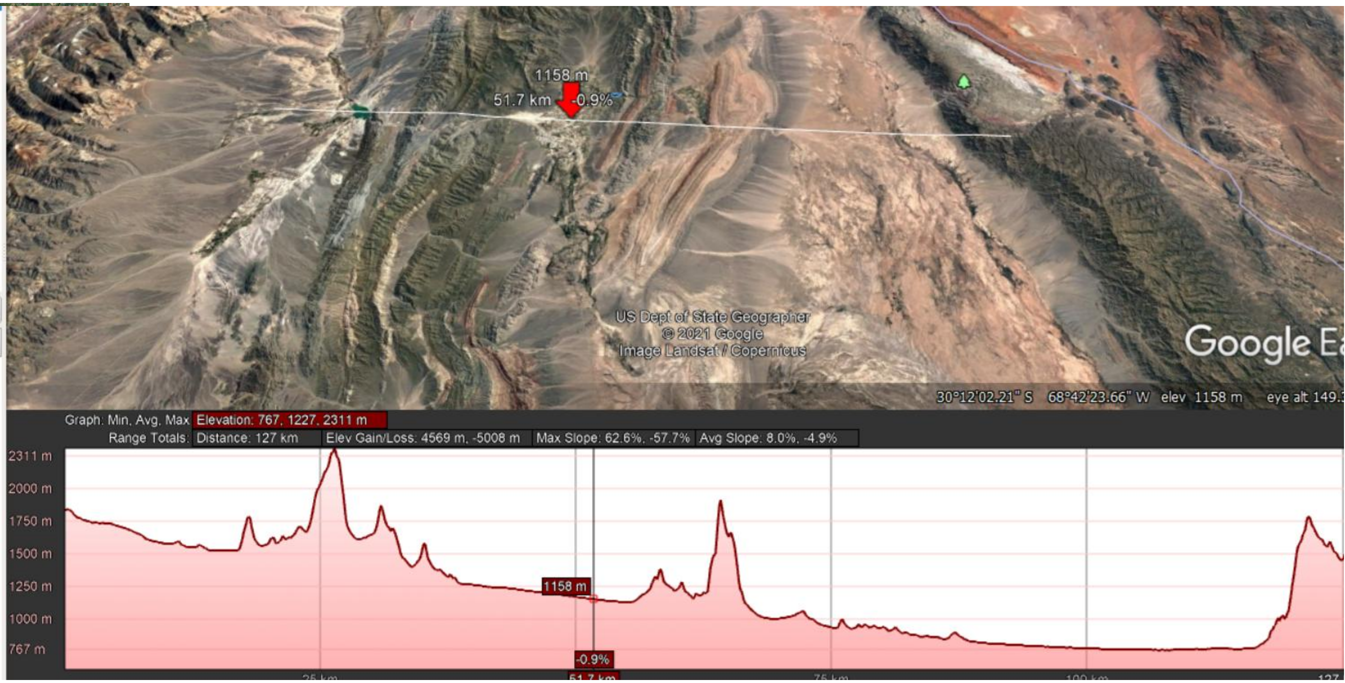
https://doi.org/10.1590/1413-70542017414002017

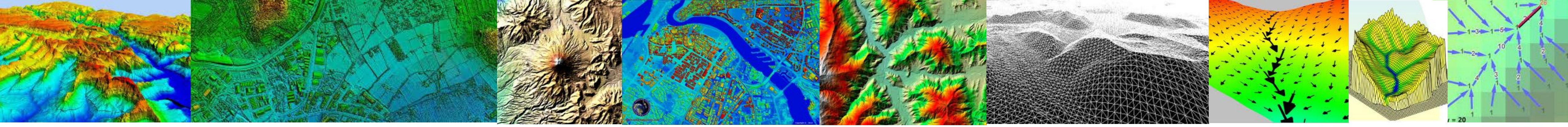


https://ide.ign.gob.ar/portal/apps/Profile/index.html?appid=89b630dced514e8a922d0f9957ce925a

https://www.google.es/intl/es/earth/versions/

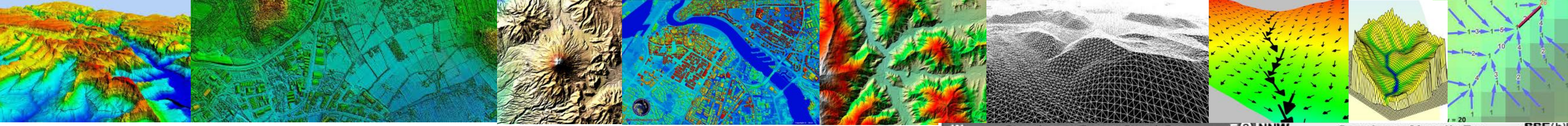






Aplicaciones de los MDEs

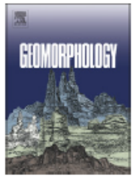
- Generación de curvas de nivel
- Generación de mapas de pendiente
- Generación de mapas en relieve
- Planificación de vuelos en tres dimensiones
- Rectificación geométrica de fotografías aéreas o de imágenes satelitales
- Reducción de las medidas de gravedad, también denominada corrección de terreno o topográfica
- Apoyo a obras de ingeniería civil
- Trazado de perfiles topográficos del terreno
- Determinación de volúmenes y/o movimientos de suelo
- Análisis de riesgos ambientales
- Modelación hidrodinámica
- Modelación hidrológica
- Modelación hidrogeológica
- Análisis geomorfológicos
- Análisis geológicos



Contents lists available at ScienceDirect

Geomorphology

journal homepage: www.elsevier.com/locate/geomorph



The use of new elevation data (SRTM/ASTER) for the detection and morphometric quantification of Pleistocene megadunes (draa) in the eastern Sahara and the southern Namib

Olaf Bubenzer [a,*], Andreas Bolten [b]

[a] Department of Geography, University of Heidelberg, Im Neuenheimer Feld 348, 69120 Heidelberg, Germany
[b] Department of Geography, University of Cologne, Albertus-Magnus-Platz, 50923 Cologne, Germany

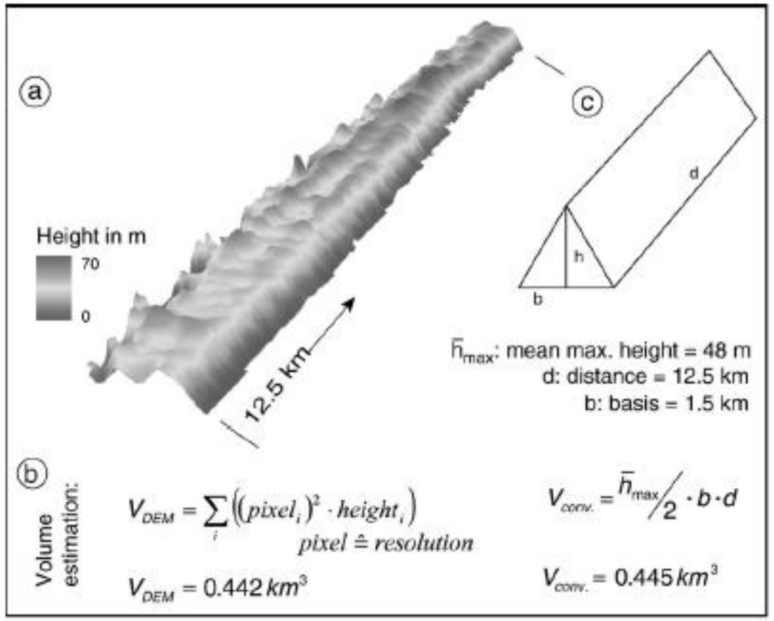
ARTICLE INFO

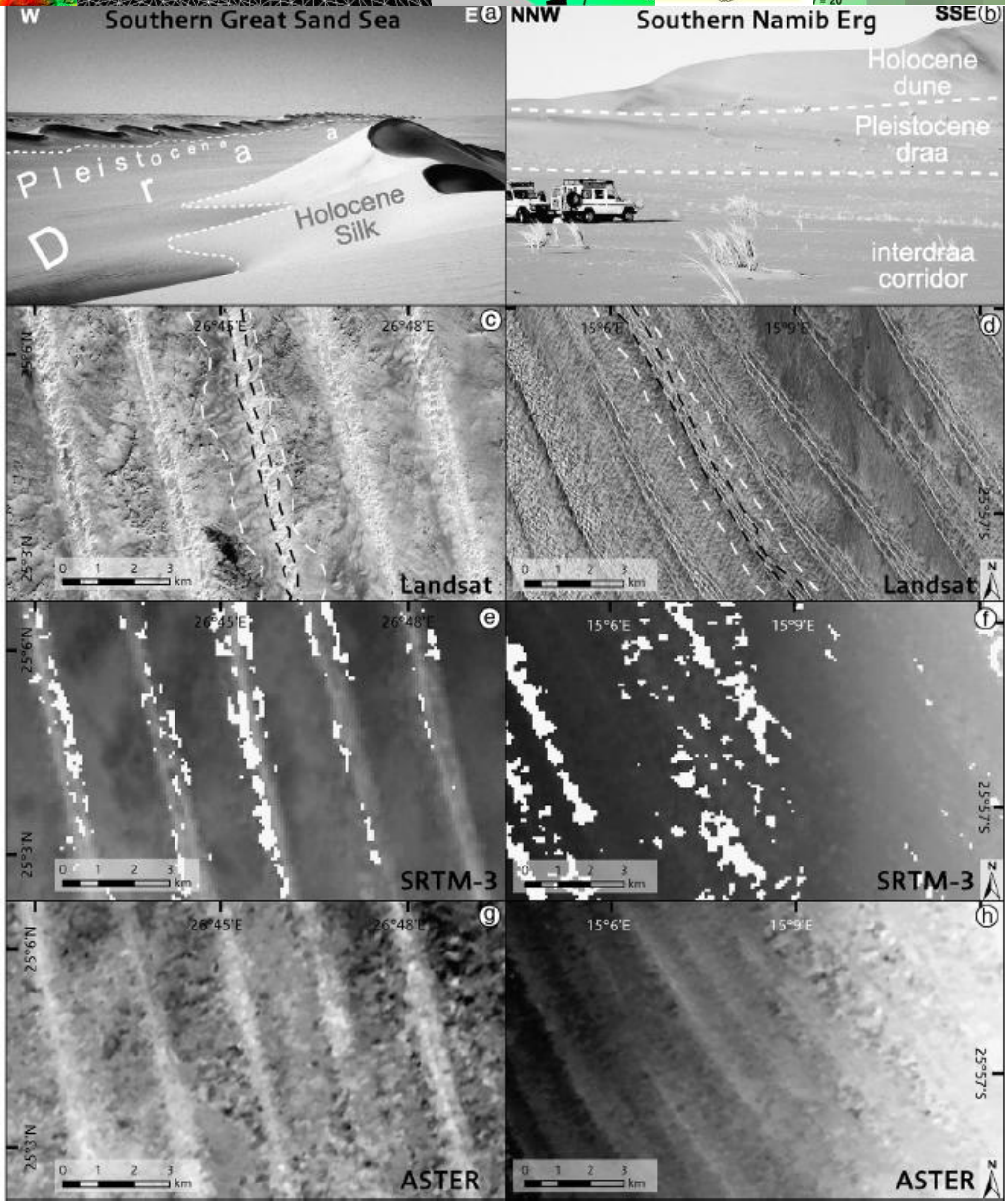
Article history:
Accepted 15 May 2007
Available online 16 May 2008

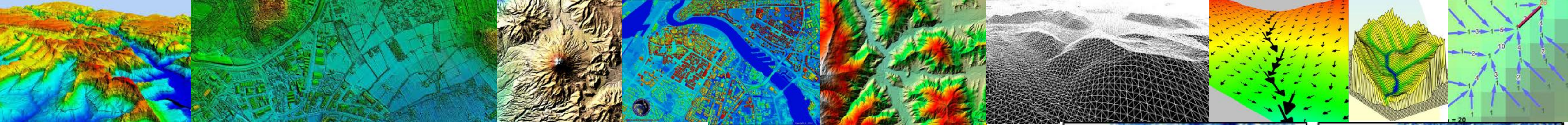
Keywords:
Aeolian
Digital elevation model
Sand seas
Complex dunes
Desert geomorphology
Remote sensing
SRTM ASTER

ABSTRACT

New freely available digital elevation dat
the Advanced Spaceborne Thermal Emis
use of area-wide high-resolution elevat
regions that are devoid of detailed
sedimentological, geochronological, and
ACACIA are presented, showing that the
complex dune pattern of Pleistocene meg
suggesting draa, formed by a strong uni
vortices. The draa differ conspicuously in
dunes, which arise from the recent bim
southern Namib Erg (Namibia) yielded si
these results, the second part of the pap
availability for the determination of draa
their different resolution (SRTM-3: 90 m
ASTER: stereo data), both models have
calculation of geomorphometric parame



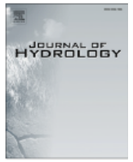




Contents lists available at ScienceDirect

Journal of Hydrology

journal homepage: www.elsevier.com/locate/jhydrol



Satellite-derived Digital Elevation Model (DEM) selection, preparation and correction for hydrodynamic modelling in large, low-gradient and data-sparse catchments

Abdollah A. Jarihani [a,*], John N. Callow [b], Tim R. McVicar [c], Thomas G. Van Niel [d], Joshua R. Larsen [a]

[a] School of Geography, Planning and Environmental Management, University of Queensland, Brisbane, QLD 4072, Australia
[b] Environmental Dynamics and Ecohydrology, School of Earth and Environment, University of Western Australia, 35 Stirling Highway, Crawley, Perth, WA 6009, Australia
[c] CSIRO Land and Water, GPO Box 1666, Canberra, ACT 2601, Australia
[d] CSIRO Land and Water, Private Bag No. 5, Wembley, WA 6913, Australia

ARTICLE INFO

Article history:
Received 6 December 2014
Received in revised form 25 February 2015
Accepted 26 February 2015
Available online 7 March 2015
This manuscript was handled by Geoff Syme, Editor-in-Chief

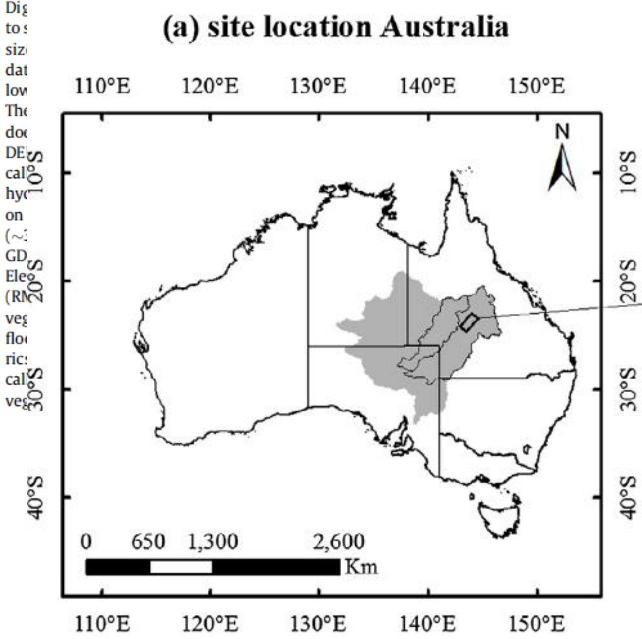
Keywords:
Digital elevation models
Hydrodynamic modelling
DEM preparation
Spatial resolution
Low-gradient river systems

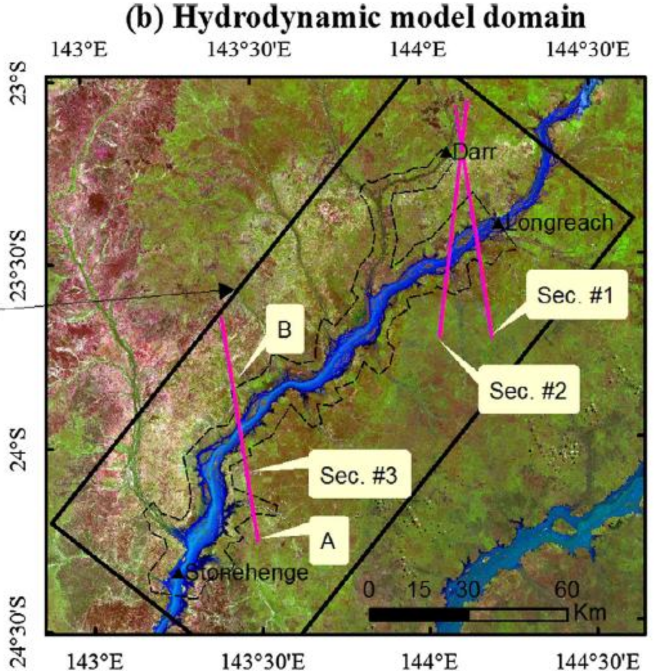
SUMMARY

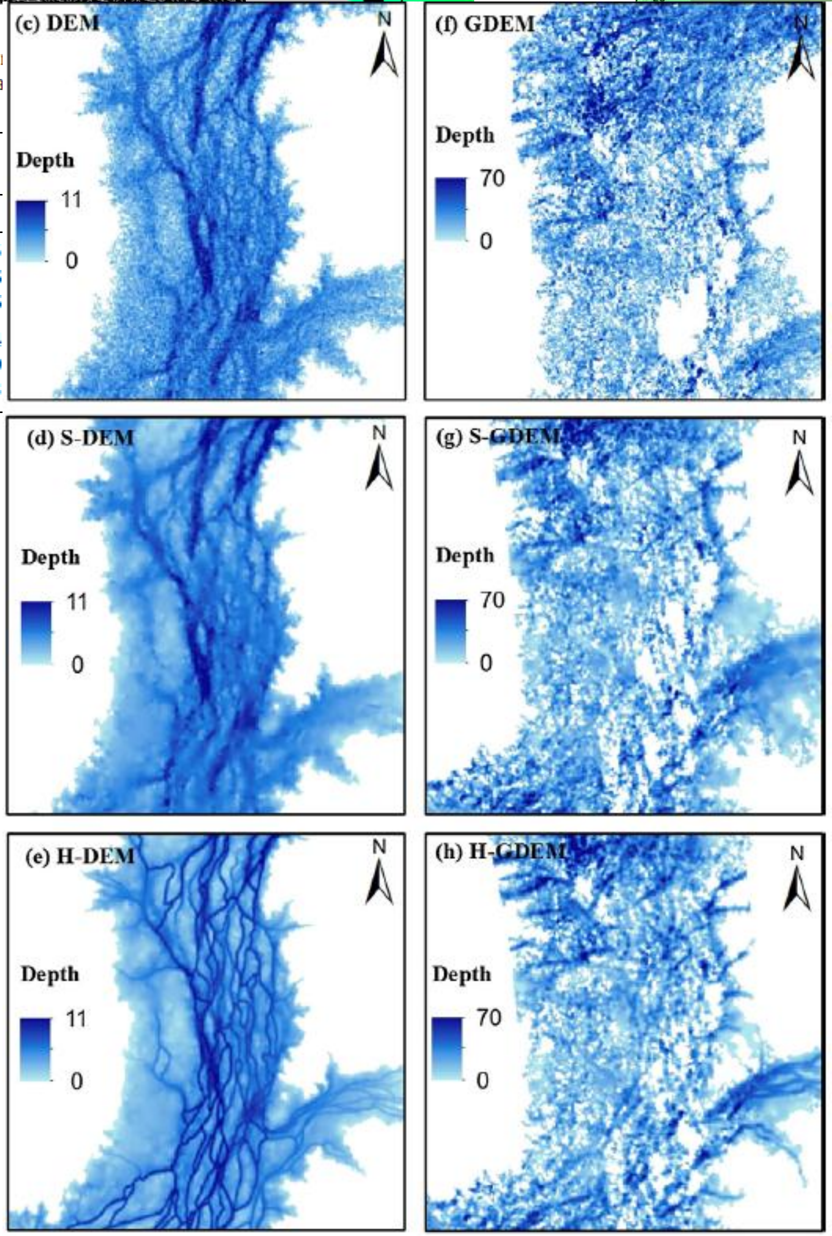
Dig
to s
siz
dat
low
The
doc
DE
cal
hy
on
(~
GD
Ele
(RN
veg
flo
ric
cal
veg

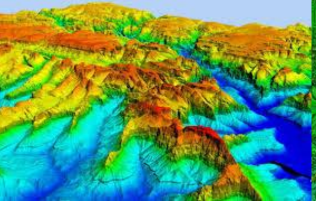
Table 4
Elevation difference between ICESat and three cross sections afte
The locations of the three sections are shown on Fig. 1b. Mean bia
are in metres.

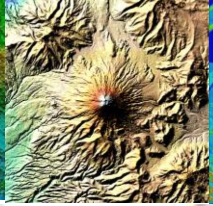
DEM	Section #	Original			
		Bias	SD	RMSD	R^2
SRTM DEM	#1	+2.97	1.39	3.28	0.995
	#2	+2.69	1.53	3.09	0.996
	#3	+2.49	1.26	2.79	0.996
ASTER GDEM	#1	−7.94	3.46	8.66	0.964
	#2	−7.70	3.13	8.31	0.980
	#3	−3.60	3.69	5.15	0.963

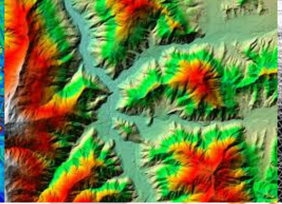
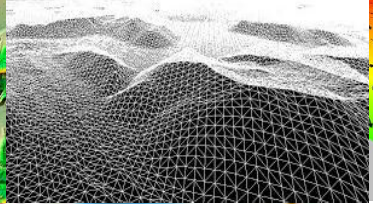
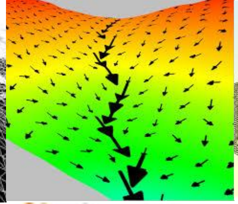
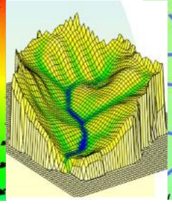
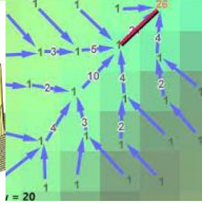






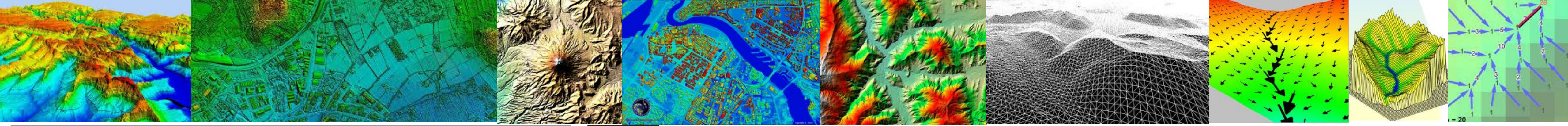




30m

5m

Contents lists available at SciVerse ScienceDirect

Computers & Geosciences

journal homepage: www.elsevier.com/locate/cageo

Comparison of drainage-constrained methods for DEM generalization

Yumin Chen[a,*], John P. Wilson[b], Quansheng Zhu[a], Qiming Zhou[c]

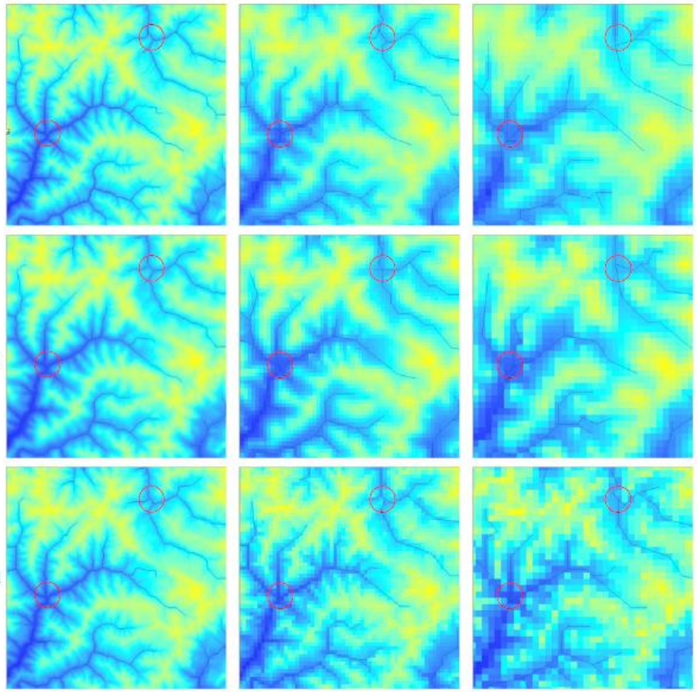
[a] School of Resource and Environment Science, Wuhan University, 129 Luoyu Road, Wuhan, 430079, China
[b] Spatial Sciences Institute, University of Southern California, Los Angeles, CA 90089-0374, USA
[c] Department of Geography, Hong Kong Baptist University, Kowloon Tong, Kowloon, Hong Kong

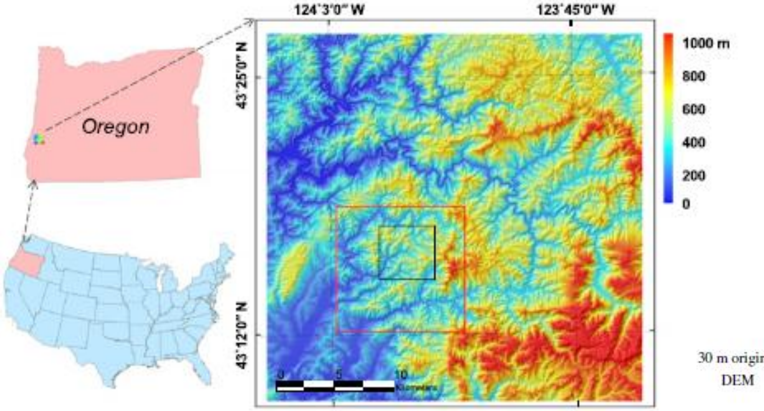
ARTICLE INFO

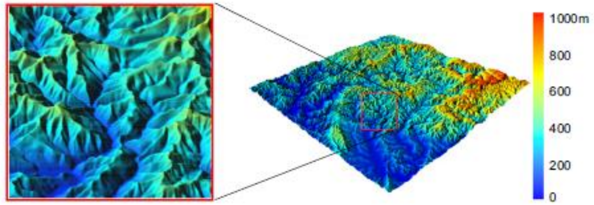
Article history:
Received 13 February 2012
Received in revised form
1 May 2012
Accepted 3 May 2012
Available online 14 May 2012

Keywords:
Digital terrain modeling
Digital terrain analysis
DEM generalization
TINs
Drainage-constrained methods

ABSTRACT







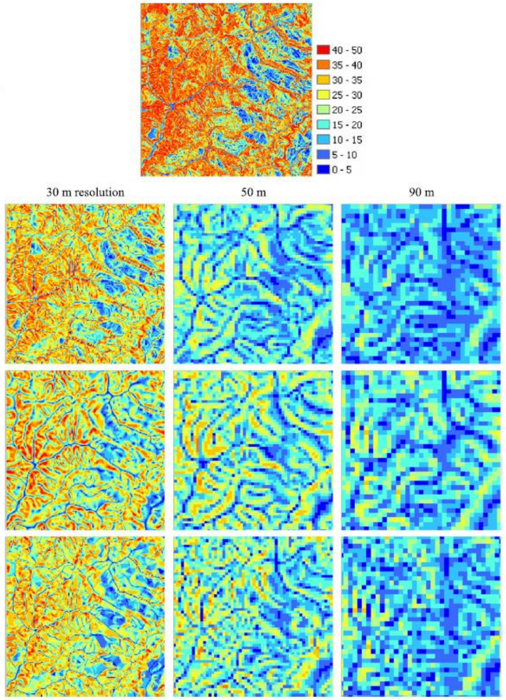


Fig. 6. The slope of DEMs generated from original 3 m DEM by three methods.

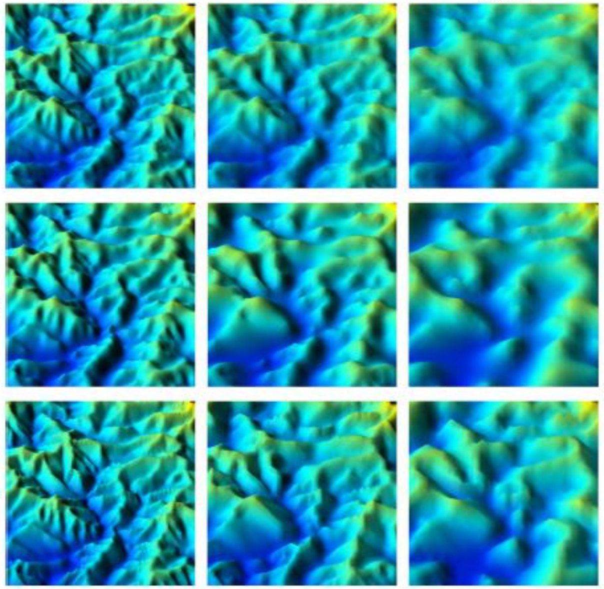
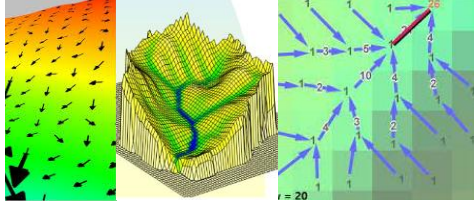


Fig. 5. 3D surfaces of DEMs generated from original 30 m DEM by three methods.



Contents lists available at ScienceDirect

Applied Geography

journal homepage: www.elsevier.com/locate/apgeog

Impacts of DEM resolution, source, and resampling technique on SWAT-simulated streamflow

Mou Leong Tan [a,*], Darren L. Ficklin [b], Barnali Dixon [c], Ab Latif Ibrahim [a], Zulkifli Yusop [d], Vincent Chaplot [e]

[a] Geoscience and Digital Earth Centre, Research Institute for Sustainable Environment, Universiti Teknologi Malaysia, 81310 Skudai, Johor Bahru, Malaysia
[b] Department of Geography, Indiana University, 701. E. Kirkwood Ave., Bloomington, IN 47405, USA
[c] Department of Environmental Science, Policy and Geography Davis 209, University of South Florida St. Petersburg 140 Seventh Ave South, St. Petersburg, Fl 33701, USA
[d] Centre for Environmental Sustainability and Water Security, Research Institute for Sustainable Environment, Universiti Teknologi Malaysia, 81310 Skudai, Johor Bahru, Malaysia
[e] IRD-LOCEAN, Université Pierre et Marie Curie, boîte 100, 4, Place Jussieu, 75252 Paris Cedex 05, France

A R T I C L E I N F O

Article history:
Received 12 May 2015
Received in revised form
24 July 2015
Accepted 24 July 2015
Available online 3 August 2015

Keywords:
DEM
SWAT model
ASTER
SRTM
Hydrology
Sensitivity analysis

A B S T R A C T

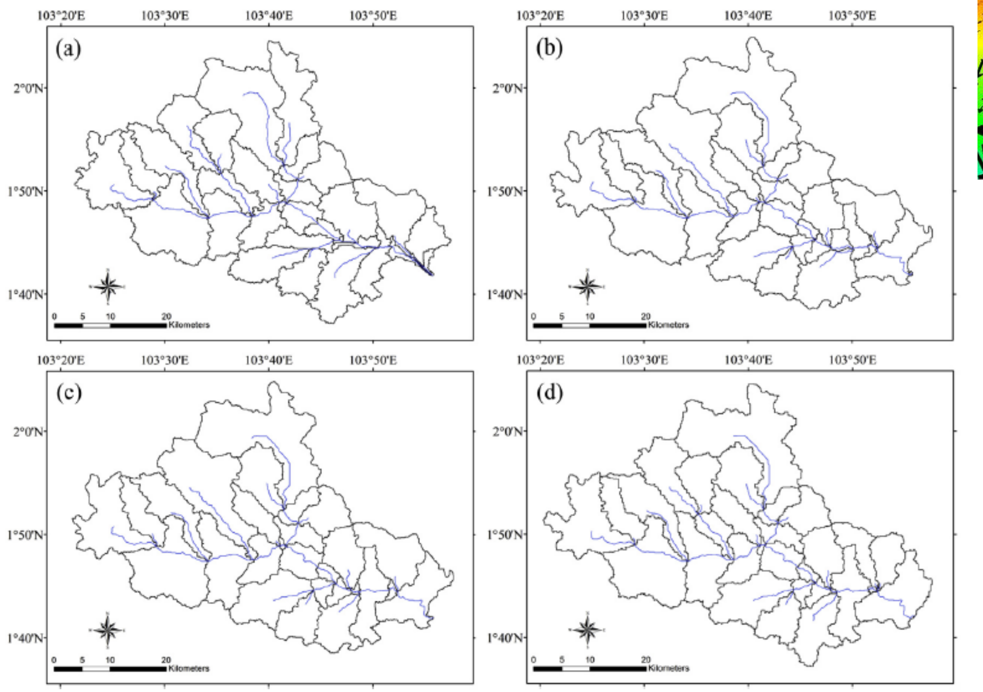


Fig. 5. Sub-basins derived from the four DEMs: (a) ASTER GDEM2 (30 m), (b) SRTM v4.1 (90 m), (c) EarthEnv-DEM90 (90 m) and (d) GMTED2010 (90 m

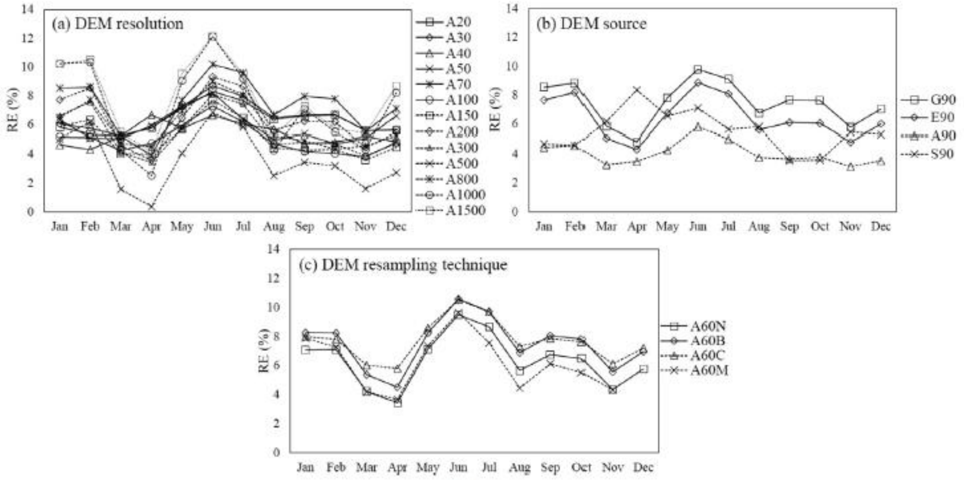


Fig. 7. The relative error of average monthly streamflow (1980–2012) between the baseline (C20) and (a) DEM resolution, (b) DEM source and (c) resampling technique.

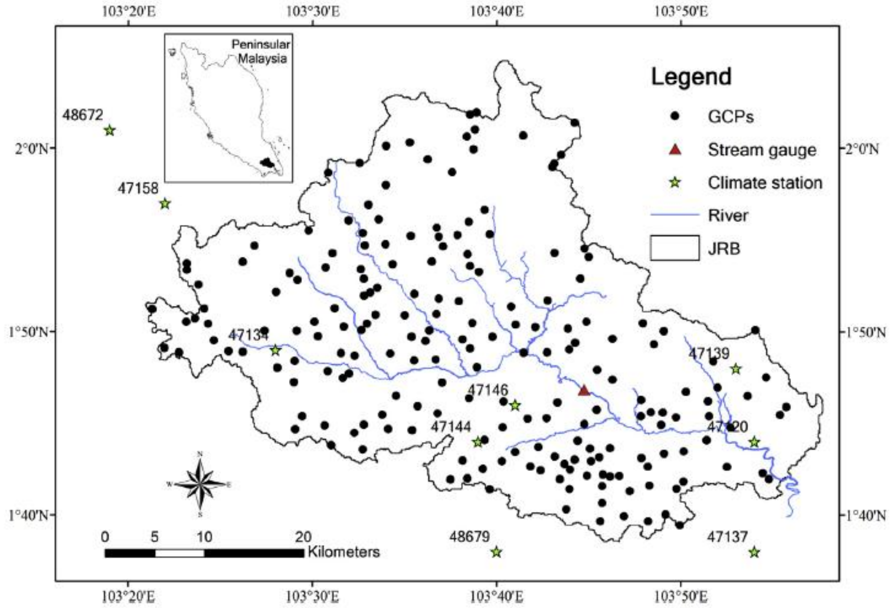
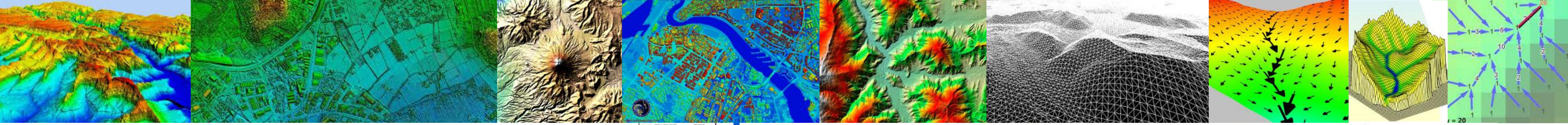


Fig. 1. Location of the Johor River Basin. (GCPs: ground control points).



Contents lists available at SciVerse ScienceDirect

International Journal of Applied Earth Observation and Geoinformation

journal homepage: www.elsevier.com/locate/jag

On the suitability of the SRTM DEM and ASTER GDEM for the compilation of topographic parameters in glacier inventories

Holger Frey*, Frank Paul[1]

Department of Geography, University of Zurich, Winterthurerstrasse 190, CH-8057 Zurich, Switzerland

ARTICLE INFO

Article history:
Received 9 July 2010
Accepted 29 September 2011

Keywords:
SRTM
ASTER GDEM
Glacier inventory
Topographic parameter
GLIMS

ABSTRACT

Topographic parameters in glacier inventories are of vital importance for several subsequent applications. In combination with digital elevation models (DEMs) with near global coverage, such parameters can in principle be derived for all digital glacier outlines. In this study we investigate the suitability of the SRTM DEM and the ASTER GDEM for the compilation of seven glacier-specific topographic parameters for a sample of 1786 Swiss glaciers. Both DEMs were applied in two different resolutions, and the obtained parameter values were compared to values derived from the Swiss national DEM (DHM25), which served as the reference. The analysis revealed that large differences of the values can occur on individual glaciers, but they average out for larger glacier samples. Parameters that depend on a single DEM value (e.g. minimum or maximum elevation) show a larger variability than parameters that are averaged over the entire glacier area (e.g. mean elevation or mean slope). Besides artifacts, also changes due to different acquisition dates and techniques (radar, optical) have an influence on the inventory parameters. Although the SRTM DEM yielded slightly better results than the ASTER GDEM, both DEMs are suitable for the compilation of topographic parameters in glacier inventories.

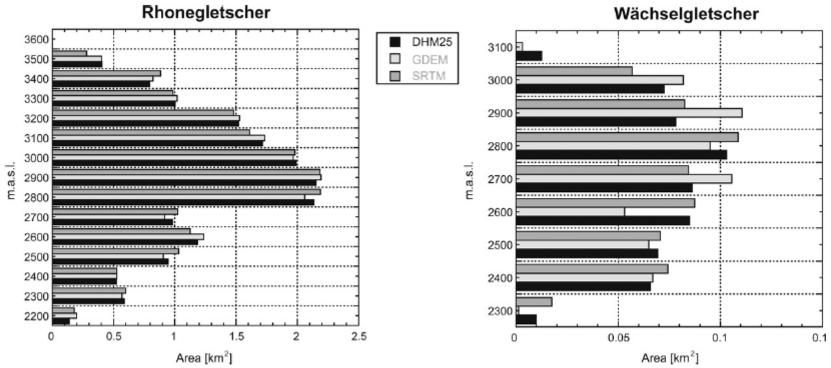


Fig. 8. Hypsographies of Rhonegletscher and Wächselgletscher derived from the three different DEMs.

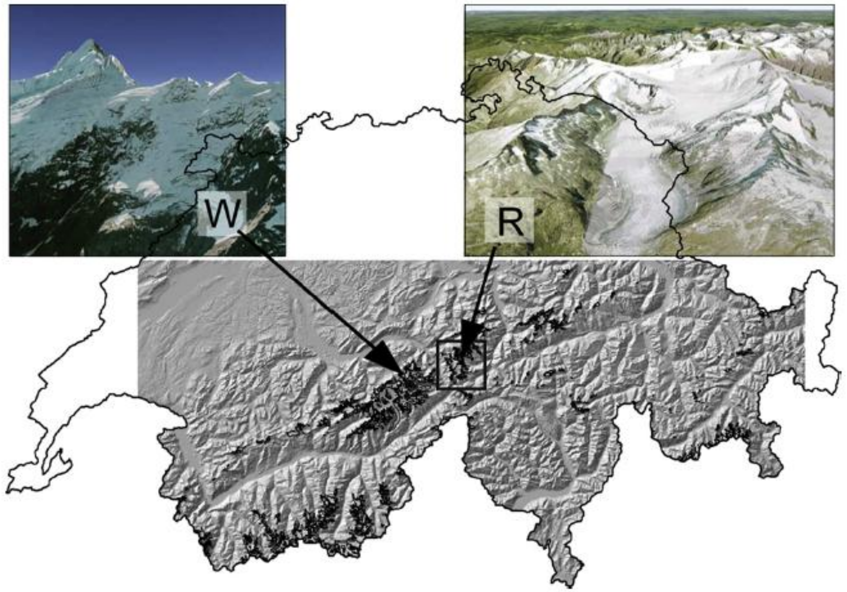


Fig. 1. Overview of the study region. Outlines of Switzerland are shown, the hillshade-view (from SRTM) indicates the extent of the DEMs. Small inset pictures are bird's eye-views of Rhonegletscher (R) and Wächselgletscher (W) from GoogleEarth™, which were used for comparisons of hypsographies (cf. Figs. 7 and 8). The rectangle around Rhonegletscher indicates the extent of Fig. 3.

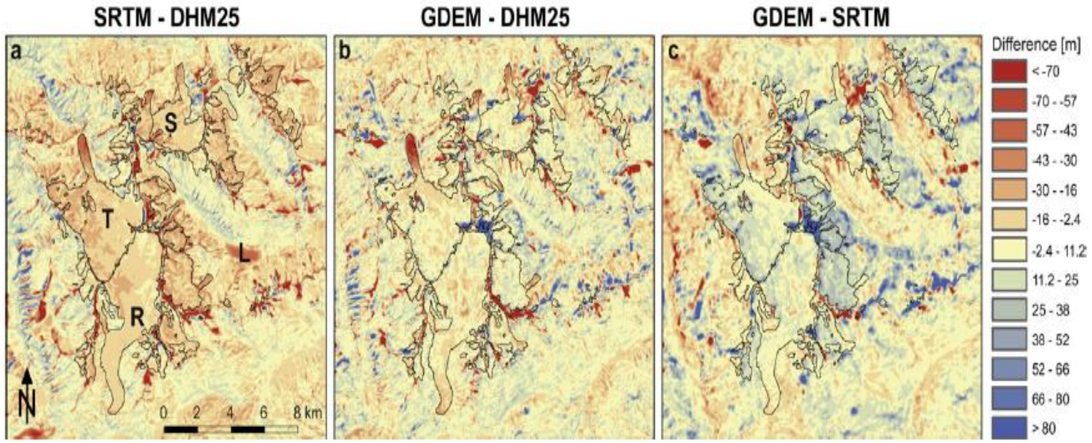
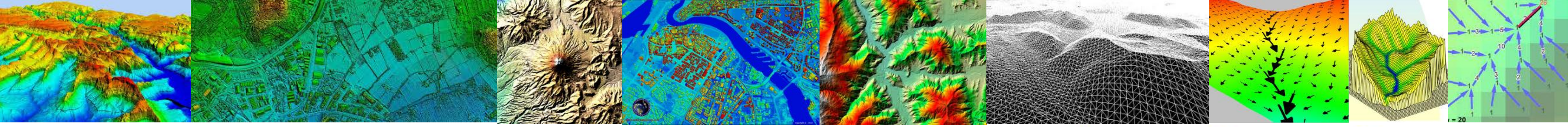


Fig. 3. Differences of the three DEMs in the region of Rhonegletscher (R), Triftgletscher (T), and Steingletscher (S). (L) indicates the location of the Göscheneralp storage Lake. The color table corresponds to half standard deviations of SRTM–DHM25.



ELSEVIER

Available online at www.sciencedirect.com
ScienceDirect

Advances in Water Resources 30 (2007) 1831–1843

Advances in Water Resources

www.elsevier.com/locate/advwatres

Evaluation of on-line DEMs for flood inundation modeling

Brett F. Sanders *,1

Department of Civil and Environmental Engineering, University of California, Irvine, CA, United States

Received 19 October 2006; received in revised form 6 February 2007; accepted 8 February 2007
Available online 21 February 2007

Abstract

Recent and highly accurate … n time
and budget constraints so the … dy test
problems. DEMs are used to … lished
flood maps and observed floo… f hor-
izontal resolution, vertical acc… sed on
airborne interferometric synth… ctures
and vegetation and therefore f… nission
(SRTM) DEMs suffer from ra… s may
include non-physical pools. D… SRTM
but using NED, flood predicti… . This
study highlights utility in SR…

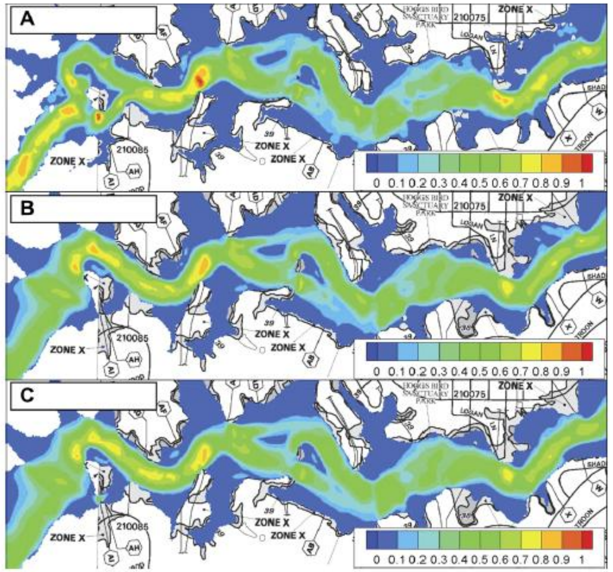


Fig. 7. BreZo prediction of 500 year flood extent at western end of reach using (A) 1/9 s NED, (B) 1/3 s NED, and (C) 1 s NED. Contours show velocity magnitude (m/s) to highlight differences in flood predictions.

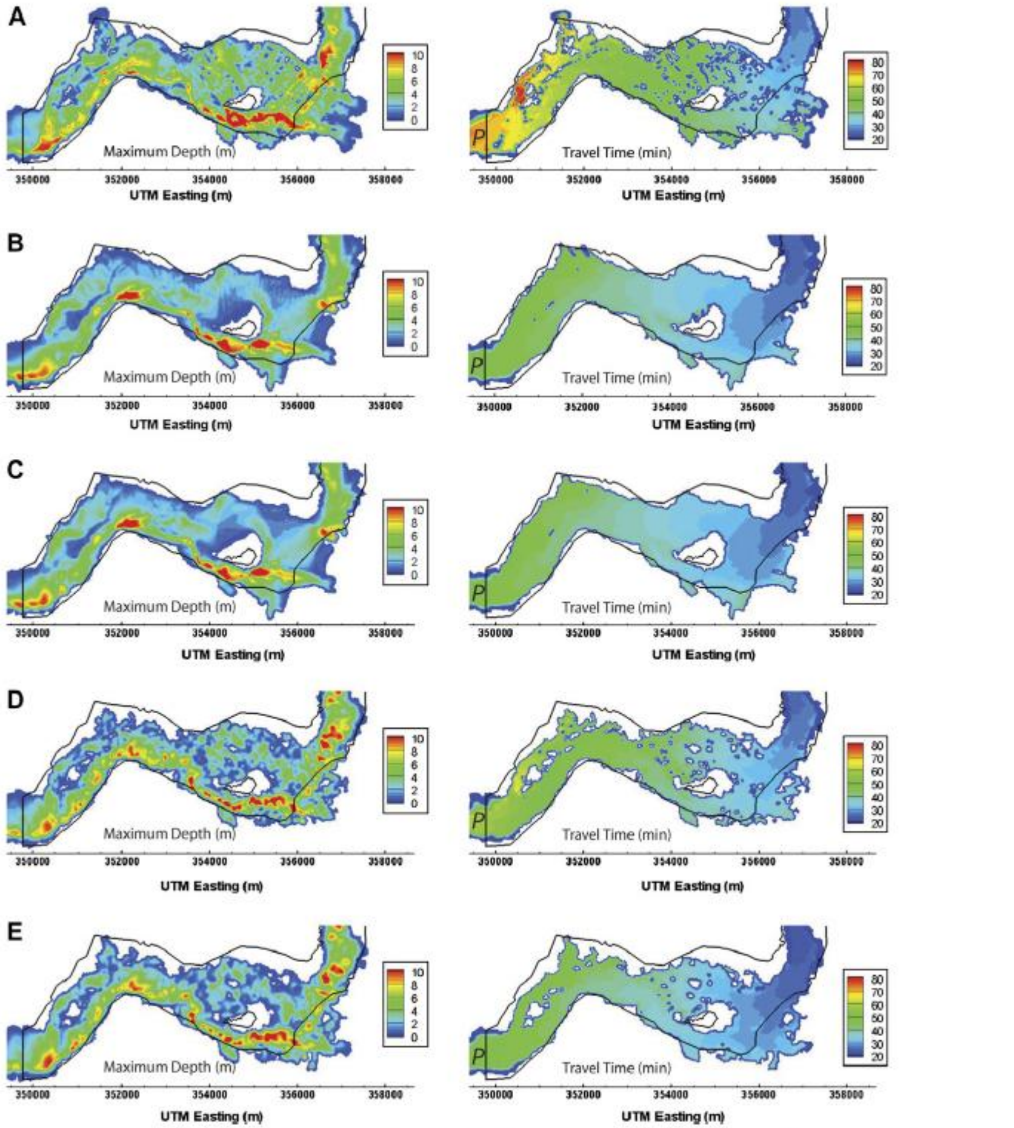
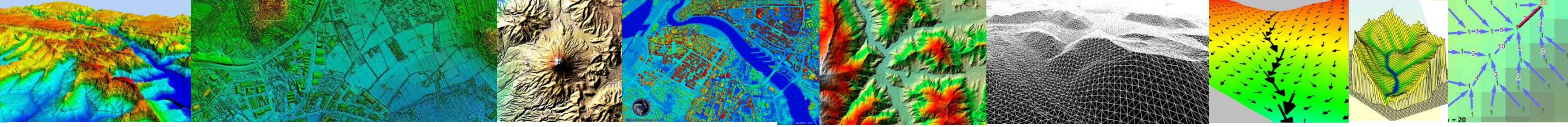


Fig. 3. Contours of maximum flood depth (left) and flood travel times (right) predicted by BreZo for St. Francis dam-break test case. The flood arrived from the north and then progressed west. Solid curves denote observed flood extent [31]. (A) 1/9 s IfSAR; (B) 1/3 s NED; (C) 1 s NED; (D) 1 s SRTM and (E) 3 s SRTM.



Available online at www.sciencedirect.com

ScienceDirect

Journal of Volcanology and Geothermal Research 170 (2008) 99–110

Journal of volcanology and geothermal research

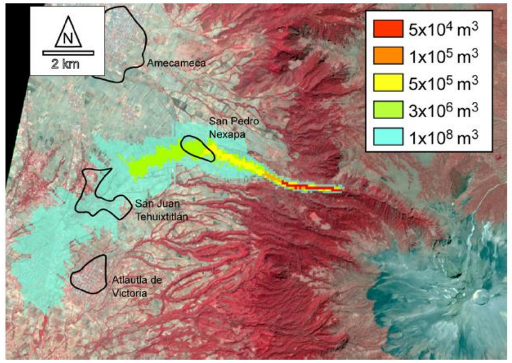
www.elsevier.com/locate/jvolgeores

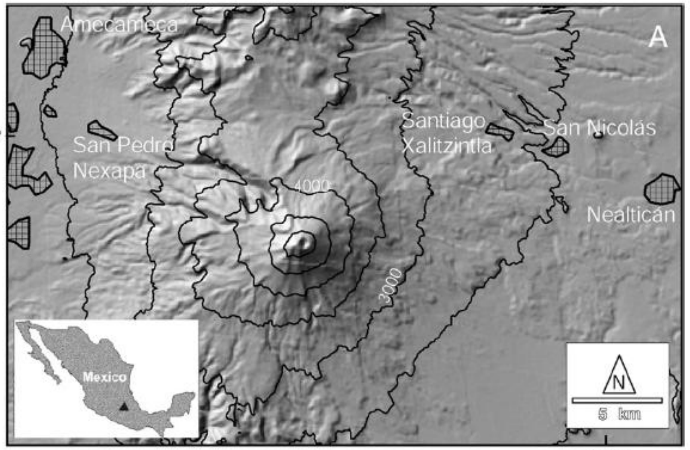
Evaluation of ASTER and SRTM DEM data for lahar modeling: A case study on lahars from Popocatépetl Volcano, Mexico

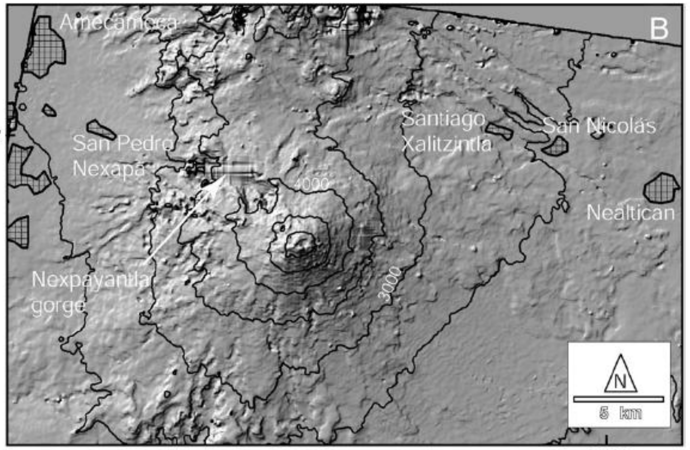
C. Huggel [a,*], D. Schneider [a], P. Julio Miranda [b], H. Delgado Granados [b], A. Kääb [c]

[a] Glaciology and Geomorphodynamics Group, Department of Geography, University of Zurich, Winterthurerstr. 190, 8057 Zurich, Switzerland
[b] Instituto de Geofsica, Universidad Nacional Autónoma de México, C.U., Coyoacán, 04510, México D.F., Mexico
[c] Department of Geoscience, University of Oslo, Norway

Received 6 September 2007
Available online 19 September 2007

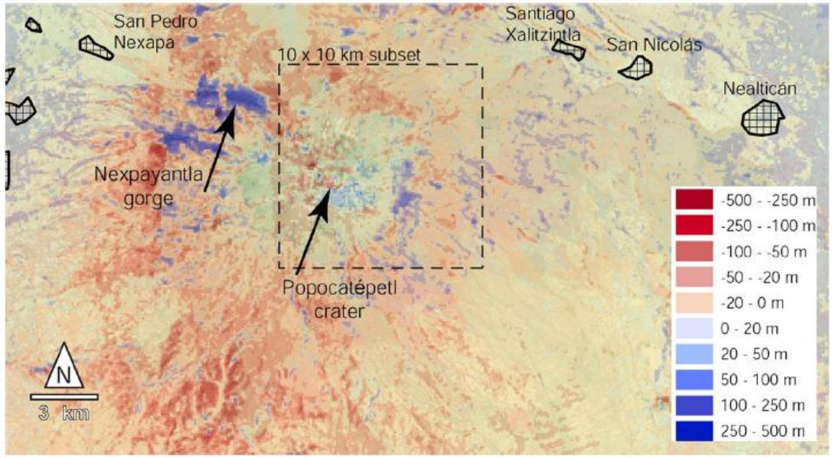


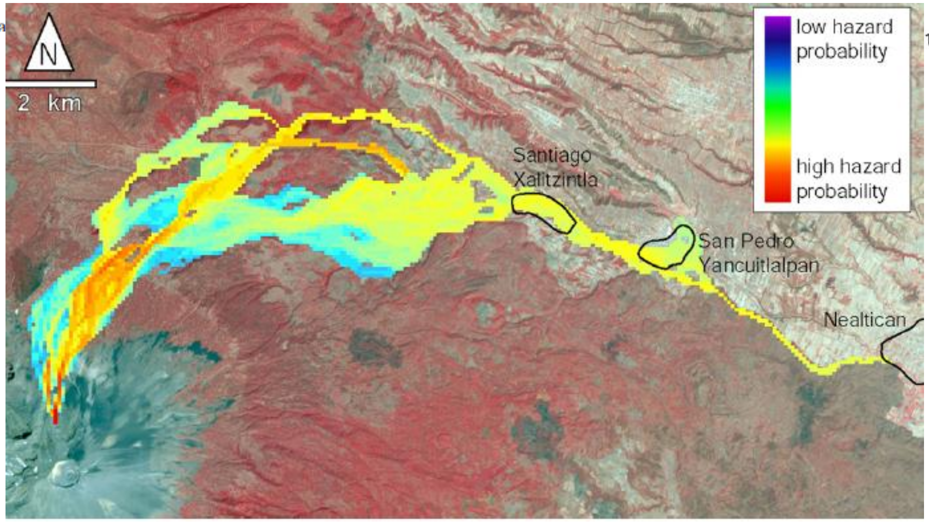


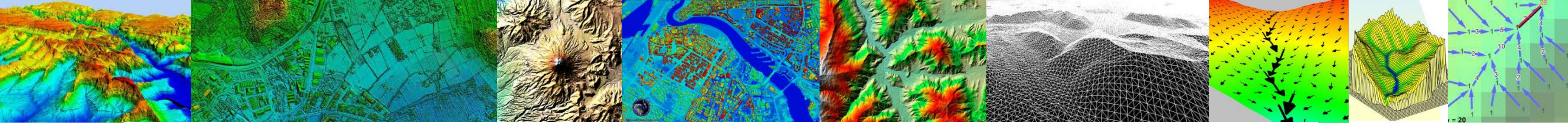


Abstract

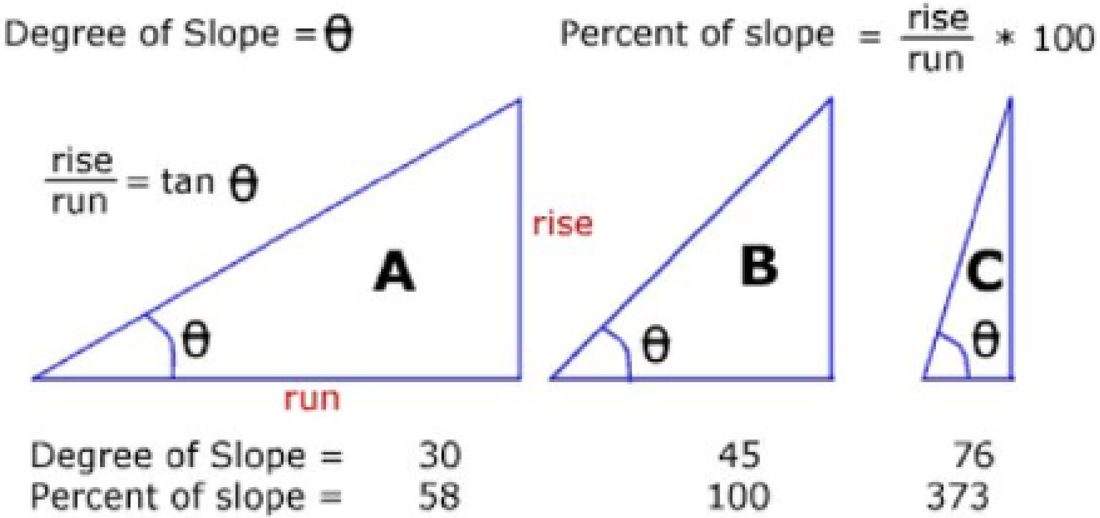
Lahars are among the most serious and far-reaching volcanic hazards. In regions with potential intera







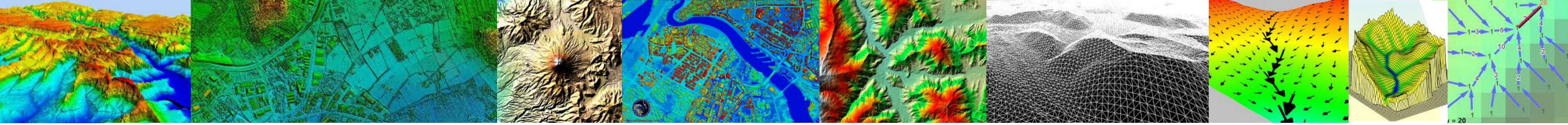
Slope



a	b	c
d	e	f
g	h	i

- Can be computed in different ways: like the aspect, it is usually calculated using a 3 x 3 matrix around each pixel
- Horn's algorithm:

$$\text{slope} = \frac{\sqrt{\left((a+2d+g)-(c+2f+i)\right)^2 + \left((a+2b+c)-(g+2h+i)\right)^2}}{8 \times \text{pixelsize}}$$



- The rate of change in the x direction for cell '*e*' is :

  [dz/dx] = ((c + 2f + i) - (a + 2d + g)) / 8

- The rate of change in the y direction for cell '*e*' is :

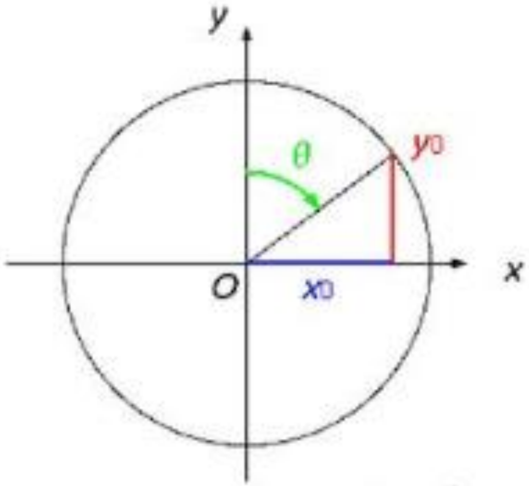
  [dz/dy] = ((g + 2h + i) - (a + 2b + c)) / 8

- Taking the rate of change in both the x and y direction for cell '*e*', aspect is calculated using:

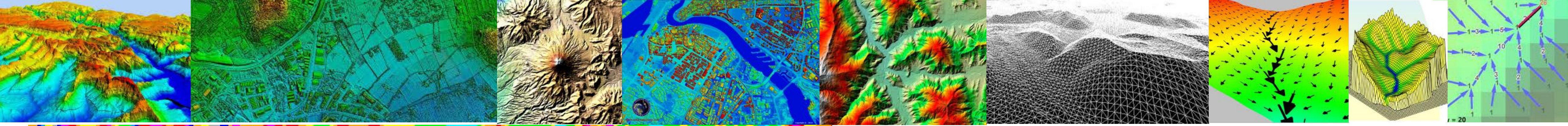
  aspect = 57.29578 * atan2 ([dz/dy], -[dz/dx])

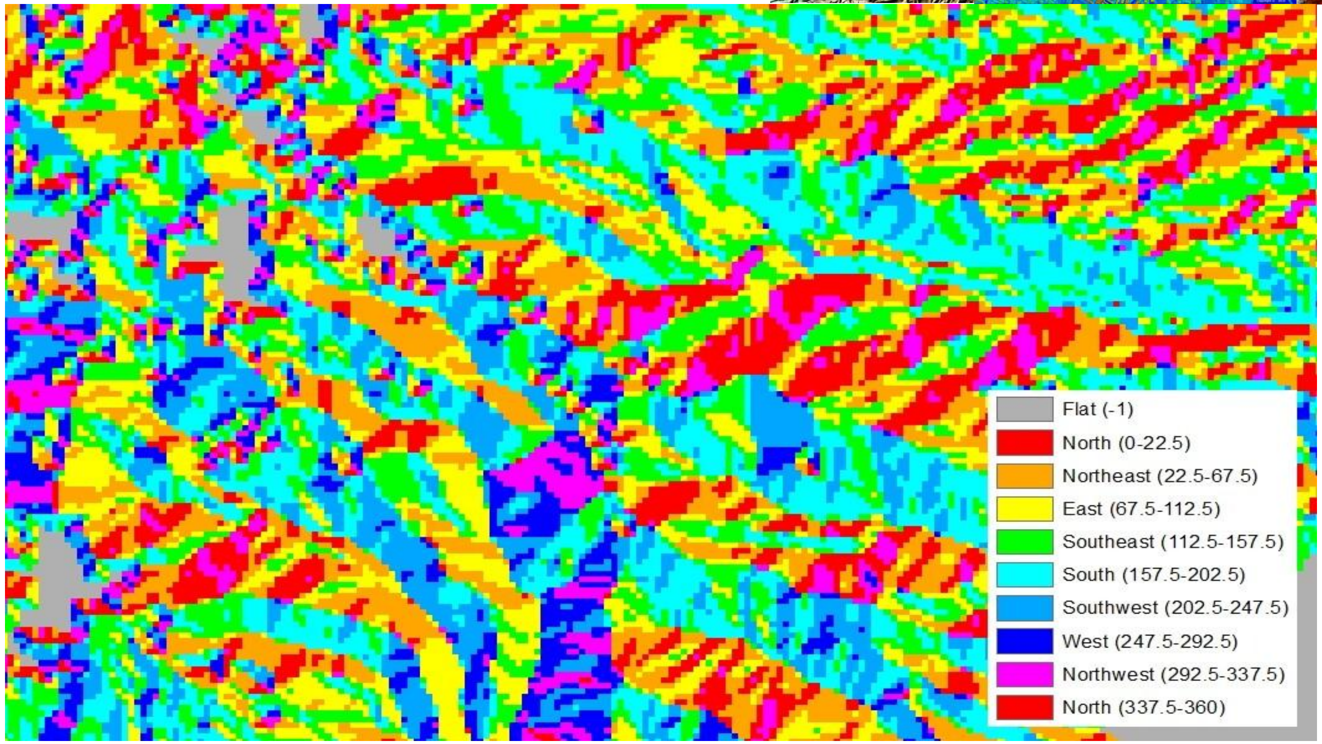
- The aspect value is converted to compass direction values (0–360 degrees, 0 and 360 indicate North)
- The aspect image is generaly classified into 8 classes, i.e. N, NE, E, SE, S, SW, W, NW
- NoData value is assigned to flat areas

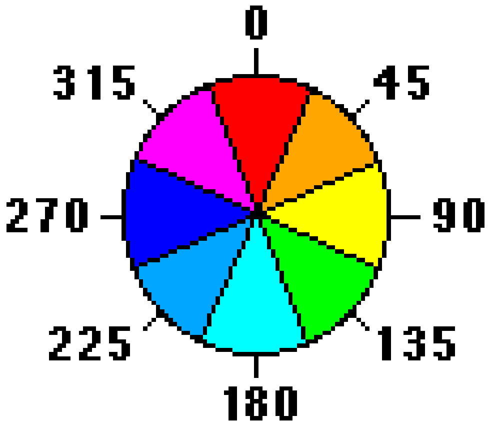
a	b	c
d	e	f
g	h	i

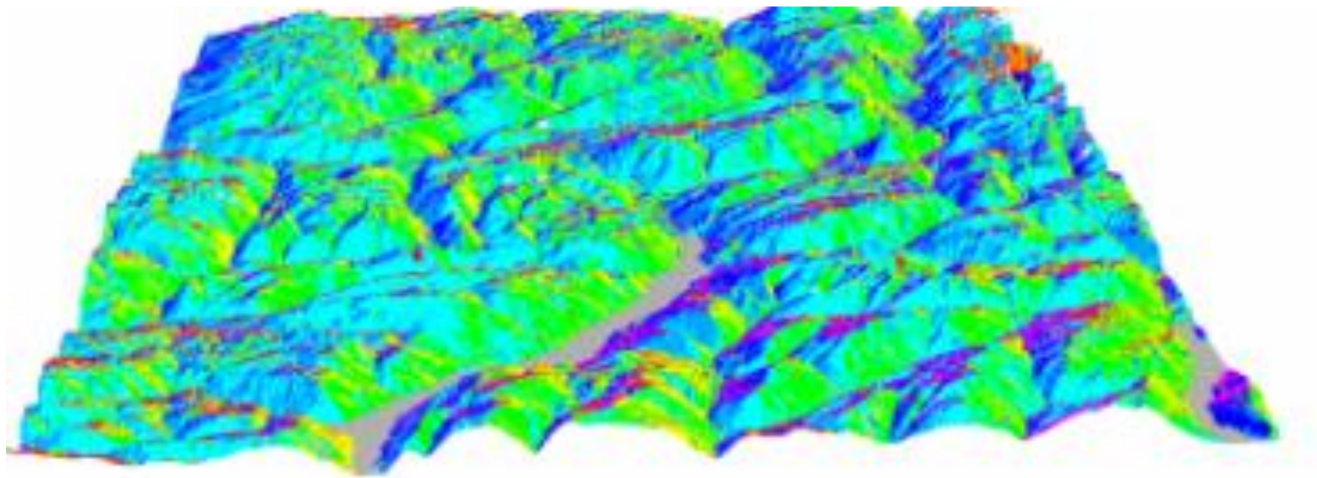


atan2



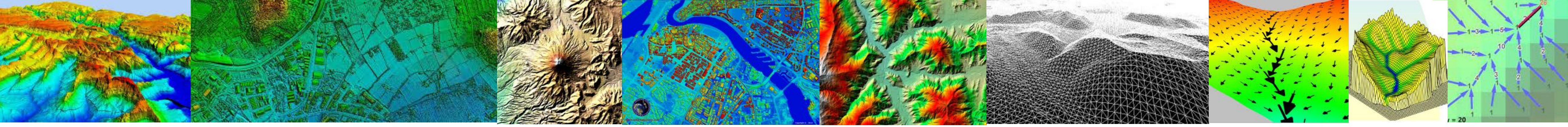


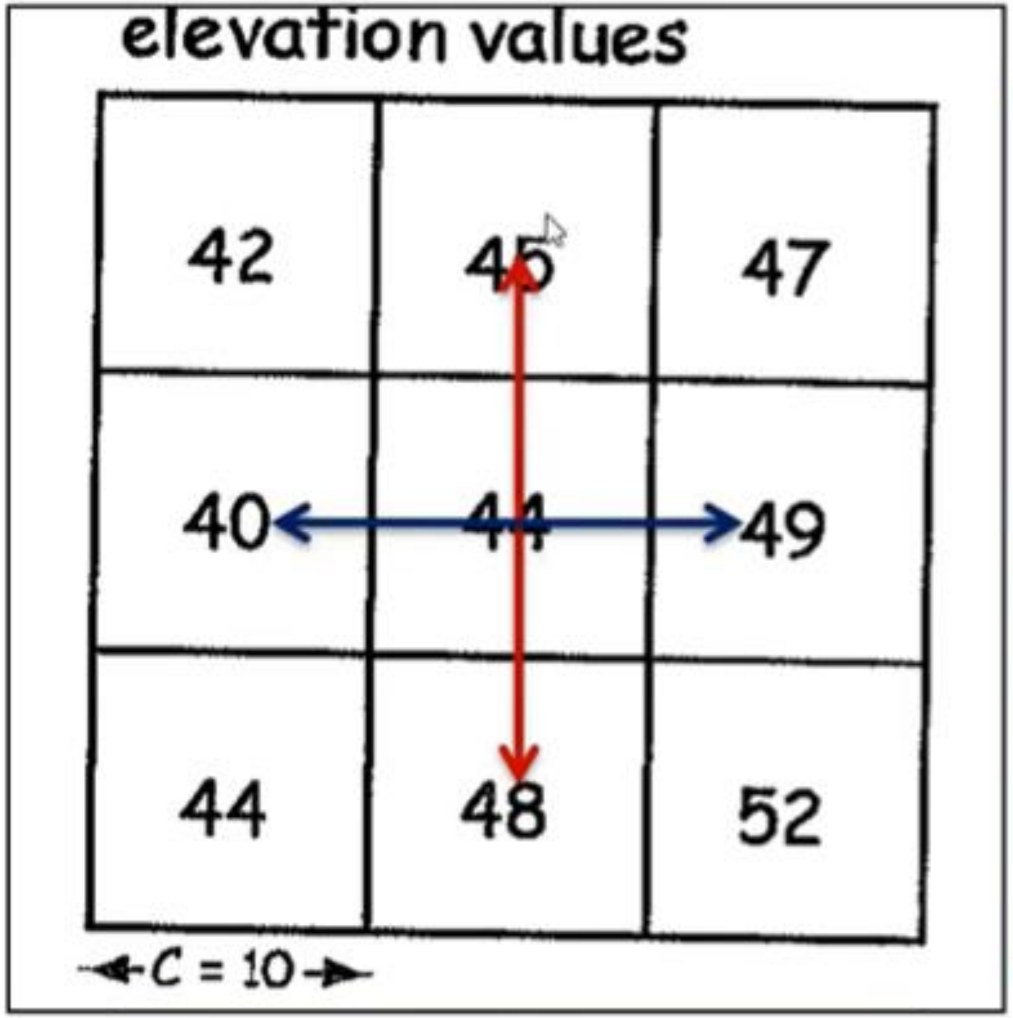




Aspect directions

- Flat (-1)
- North (0-22.5)
- Northeast (22.5-67.5)
- East (67.5-112.5)
- Southeast (112.5-157.5)
- South (157.5-202.5)
- Southwest (202.5-247.5)
- West (247.5-292.5)
- Northwest (292.5-337.5)





dZ/dy = (45 - 48)/20 = -0.15

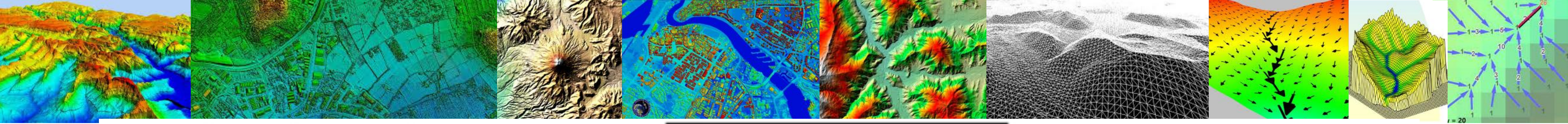
dZ/dx = (49 - 40)/20 = 0.45

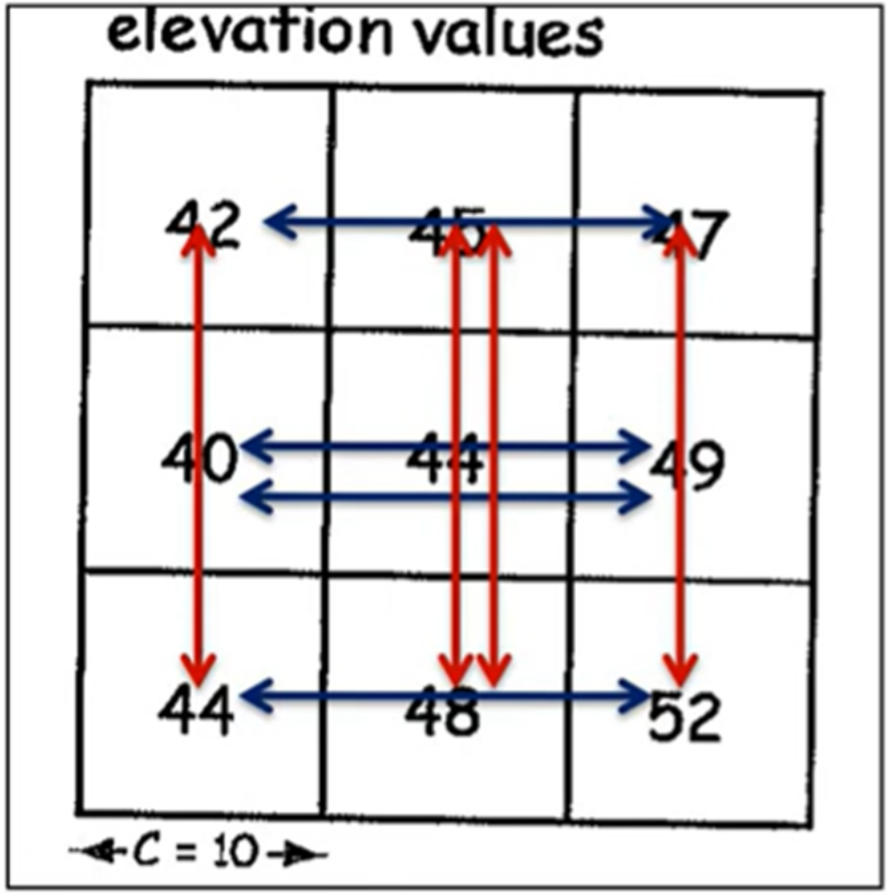
Cell size 10 m

$$\text{3D Slope} = \sqrt{\left(\frac{dZ}{dx}\right)^2 + \left(\frac{dZ}{dy}\right)^2}$$

$$= \left[(0.45)^2 + (-0.15)^2\right]^{0.5}$$

$$= 0.474$$



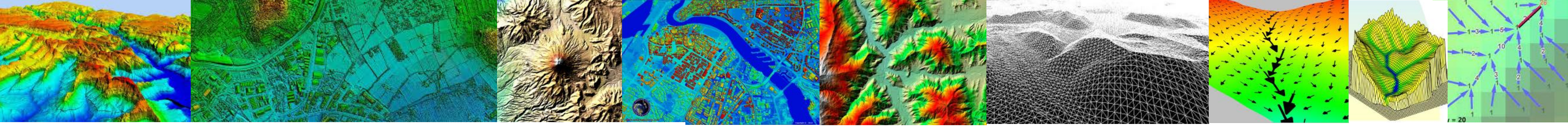


dZ/dy =
[(47 - 52) +
2 (45 - 48) +
(42 - 44)]/80
= -0.16

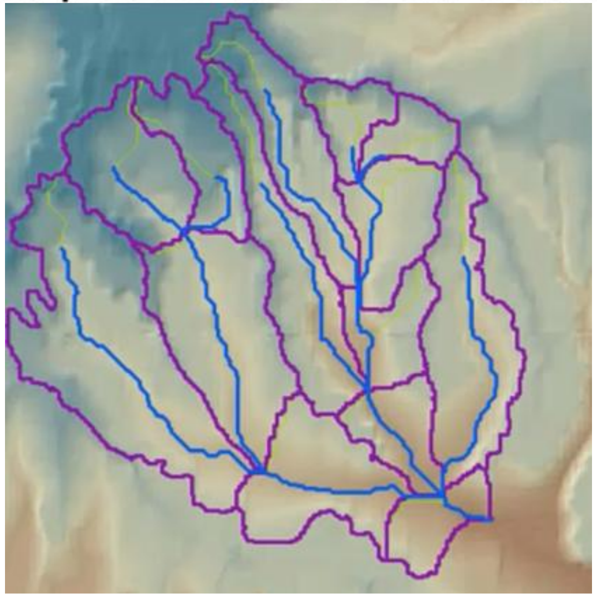
dZ/dx =
[(47 - 42) +
2 (49 - 40) +
(52 - 44)]/80
= 0.39

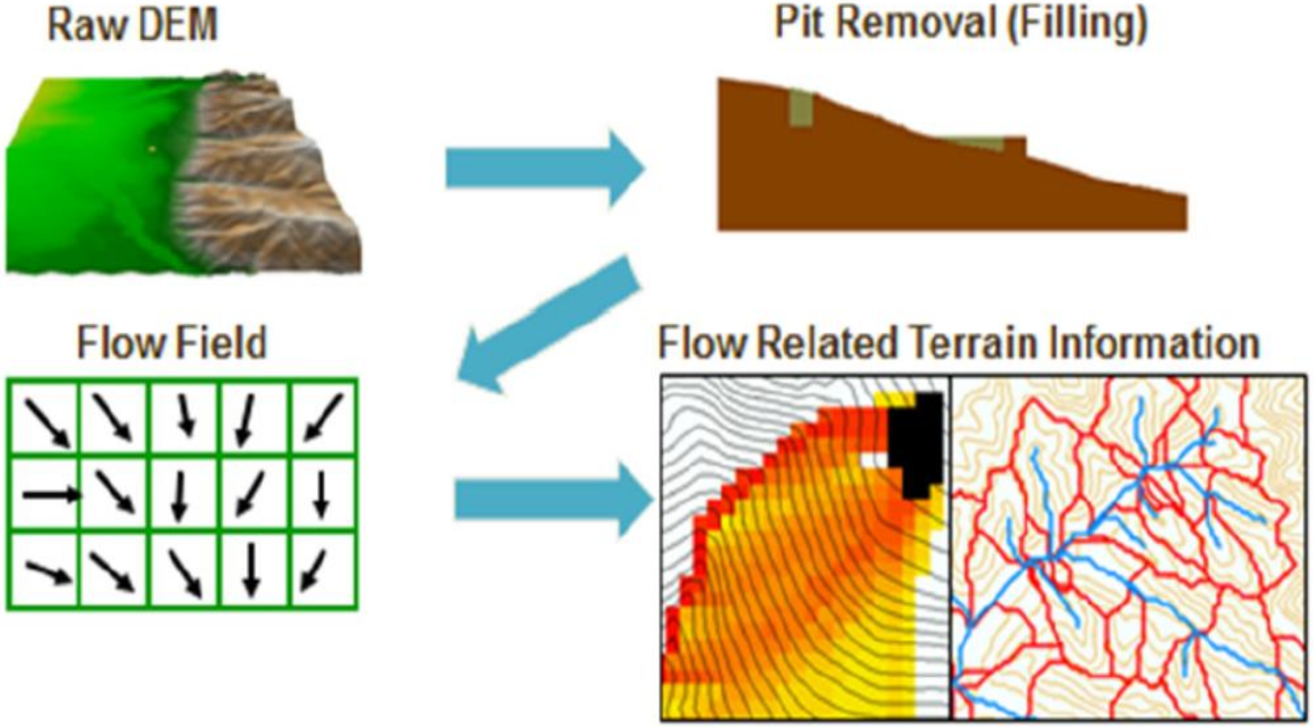
$$\textbf{3D Slope} = \sqrt{\left(\frac{dZ}{dx}\right)^2 + \left(\frac{dZ}{dy}\right)^2}$$

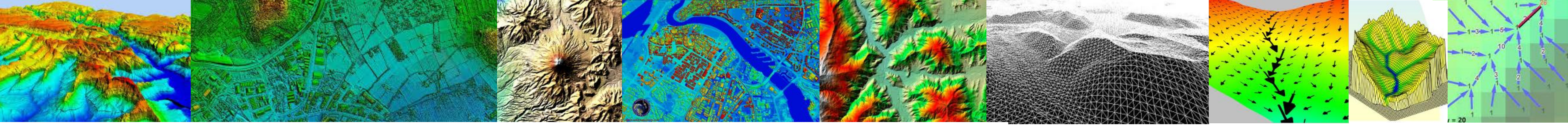
$$= \left[(0.39)^2 + (-0.16)^2\right]^{0.5}$$



1. Prepare elevation data (DEM)
   - Acquire and project DEM if needed
   - Fill pits and depressions
   - Burn in streams if desired
2. Calculate flow direction in each cell
3. Calculate flow accumulation in each cell
4. Apply a threshold for a "stream"
5. Add additional points where an outlet is desired.







DEM Pit removal – Pit Filling

Original DEM

7	7	6	7	7	7	7	5	7	7
9	9	8	9	9	9	9	7	9	9
11	11	10	11	11	11	11	9	11	11
12	12	8	12	12	12	12	10	12	12
13	12	7	12	13	13	13	11	13	13
14	7	6	11	14	14	14	12	14	14
15	7	7	8	9	15	15	13	15	15
15	8	8	8	7	16	16	14	16	16
15	11	11	11	11	17	17	6	17	17
15	15	15	15	15	18	18	15	18	18

Pits

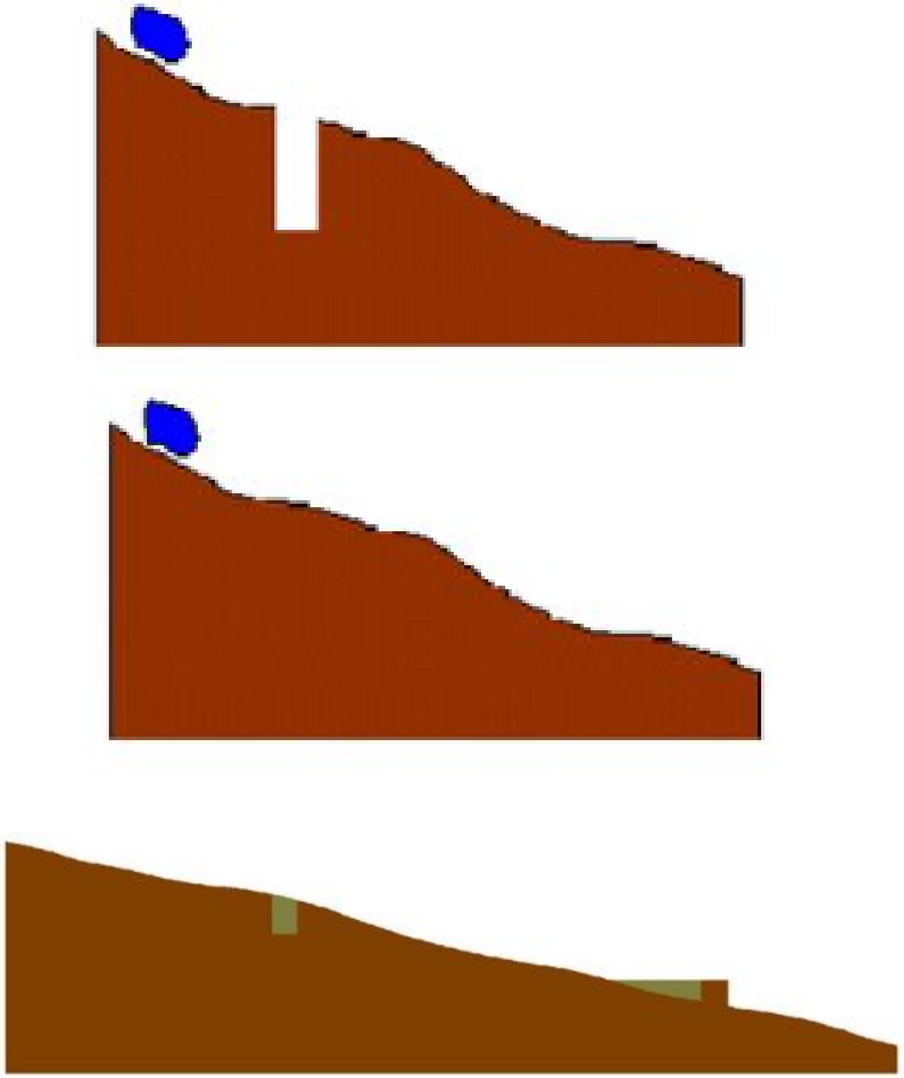
Grid cells or zones completely surrounded by higher terrain

Pits Filled

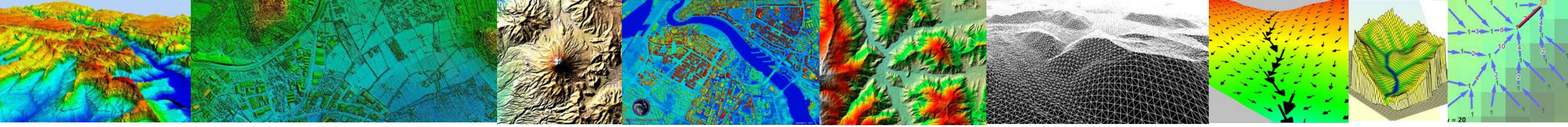
7	7	6	7	7	7	7	5	7	7
9	9	8	9	9	9	9	7	9	9
11	11	10	11	11	11	11	9	11	11
12	12	10	12	12	12	12	10	12	12
13	12	10	12	13	13	13	11	13	13
14	10	10	11	14	14	14	12	14	14
15	10	10	10	10	15	15	13	15	15
15	10	10	10	10	16	16	14	16	16
15	11	11	11	11	17	17	14	17	17
15	15	15	15	15	18	18	15	18	18

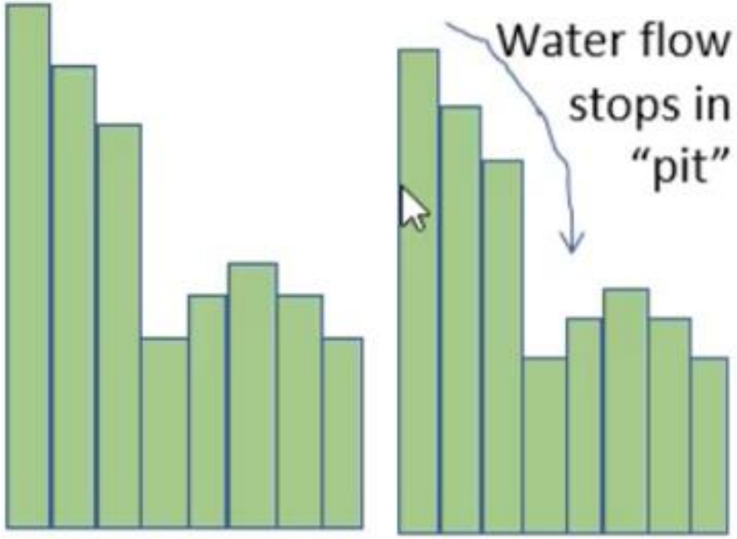
Pour Points

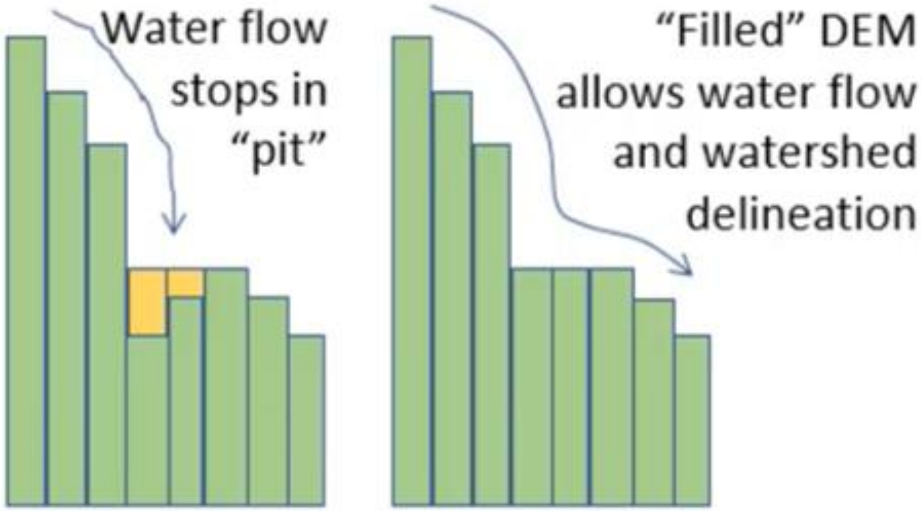
The lowest grid cell adjacent to a pit

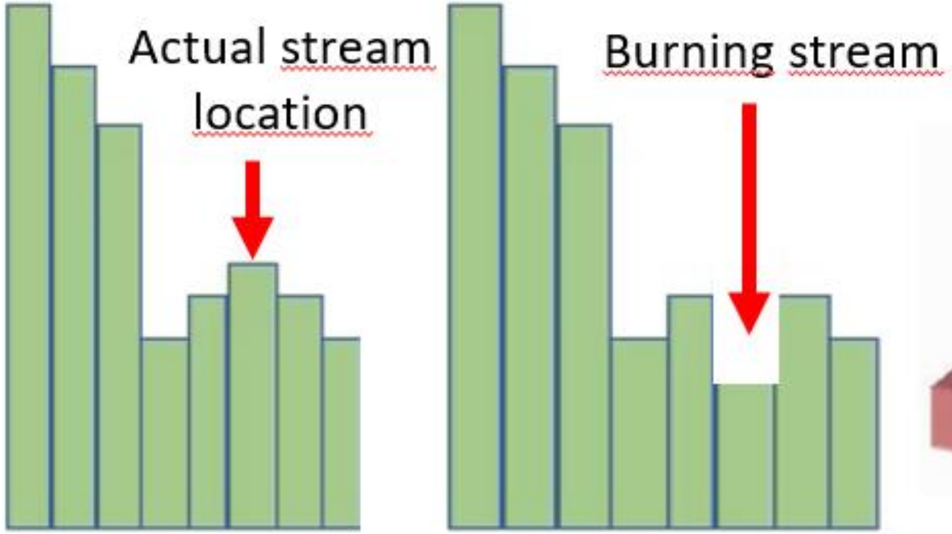


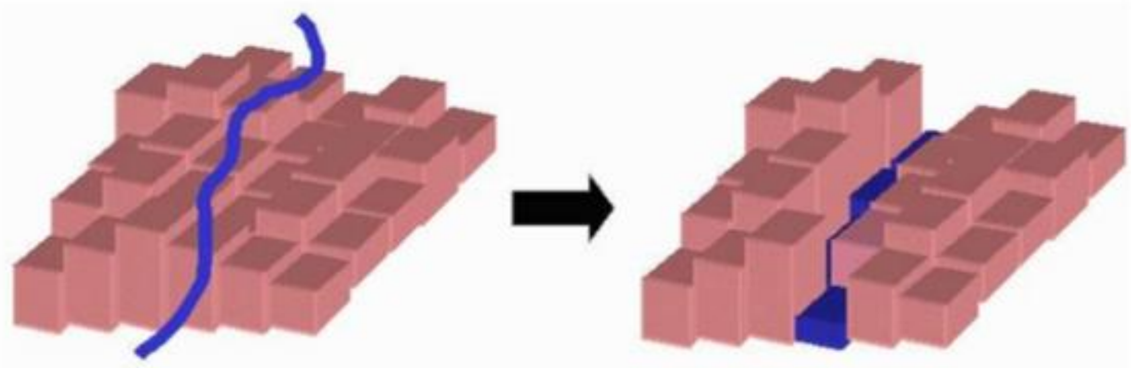
Lane 2020

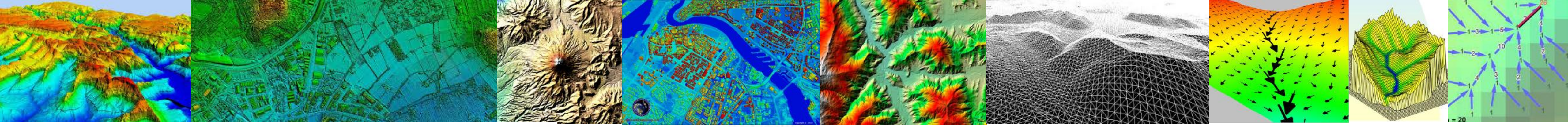


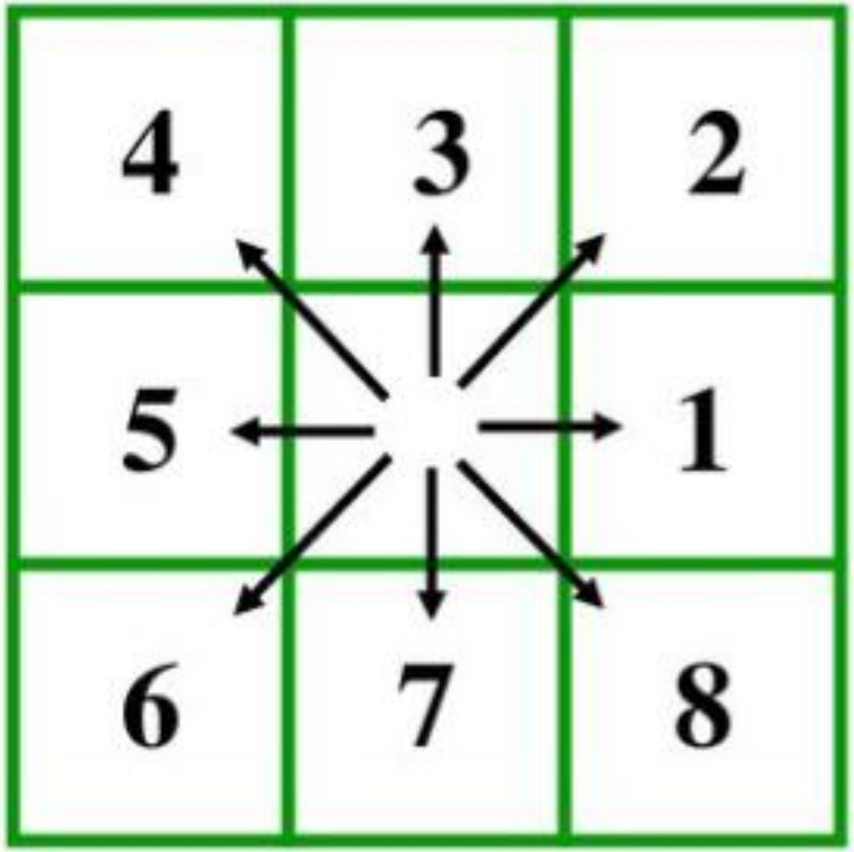

Water flow
stops in
"pit"

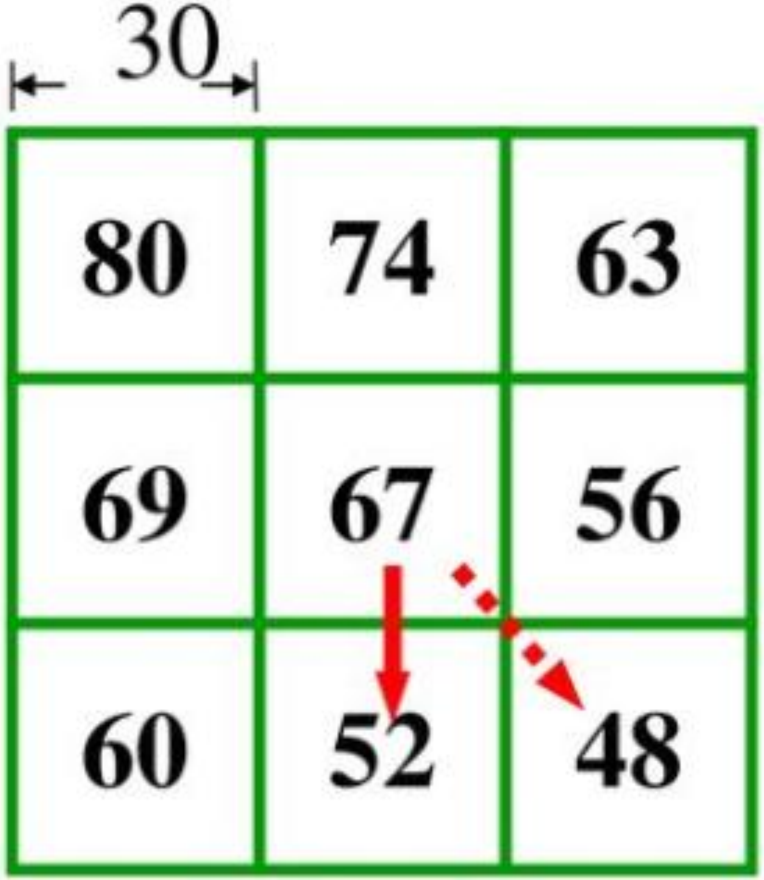

Water flow
stops in
"pit"
"Filled" DEM
allows water flow
and watershed
delineation


Actual stream
location
Burning stream



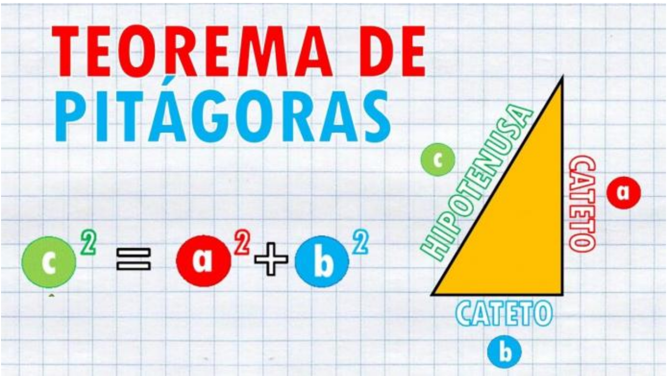


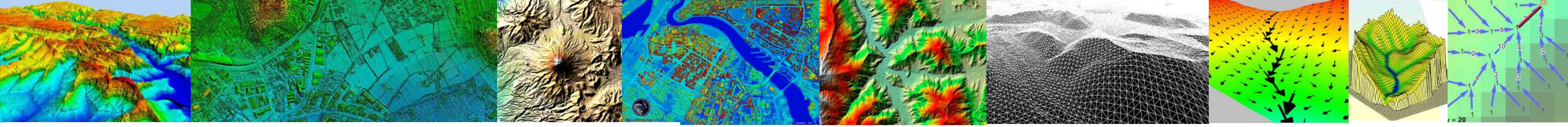


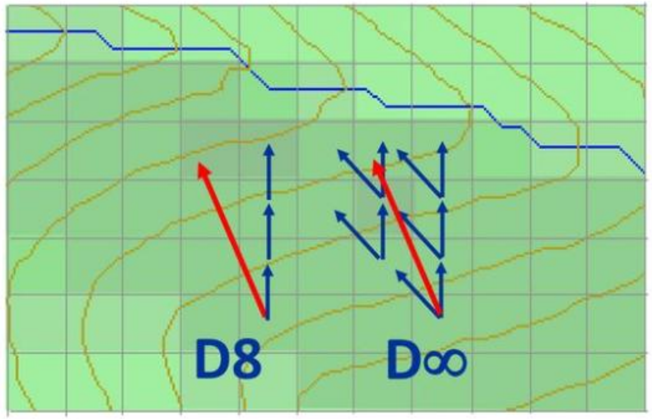


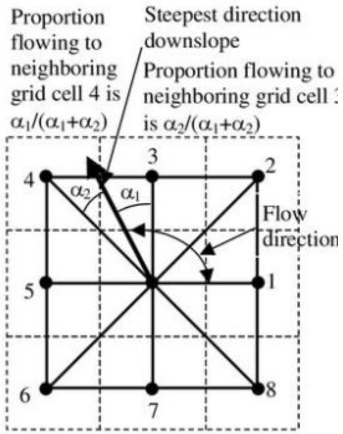
$$\frac{67-52}{30} = 0.50$$

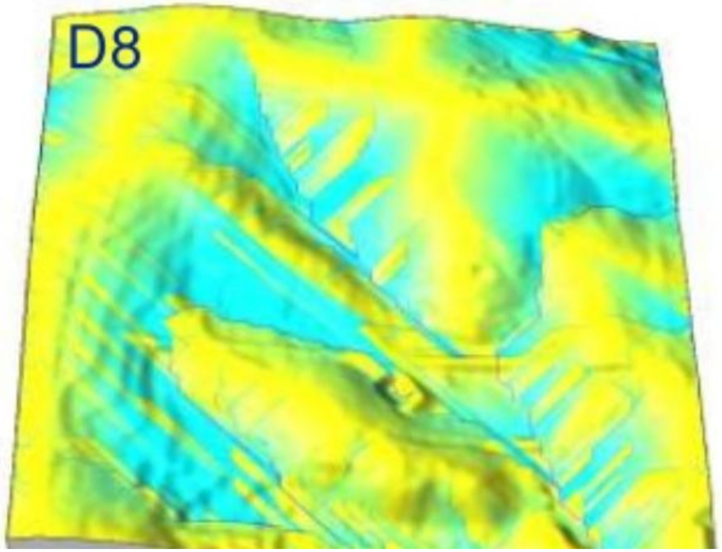
$$\frac{67-48}{30\sqrt{2}} = 0.45$$

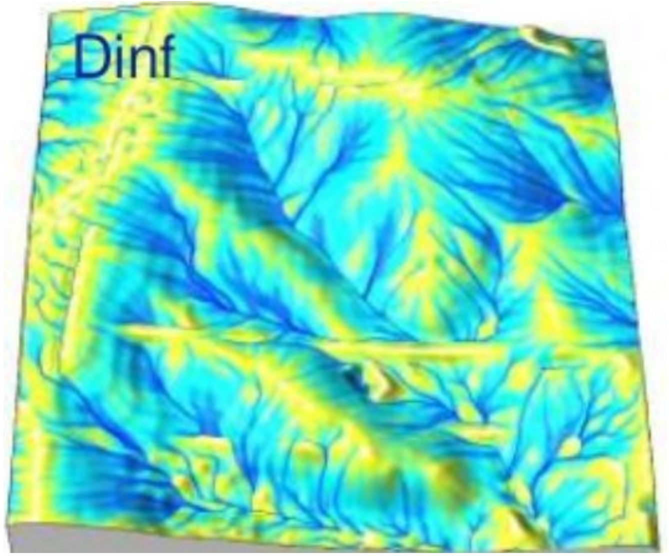


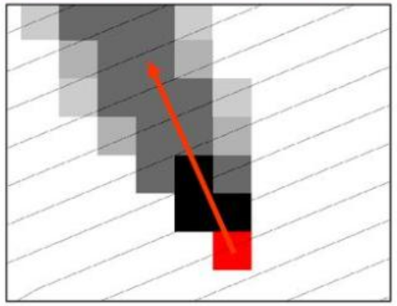


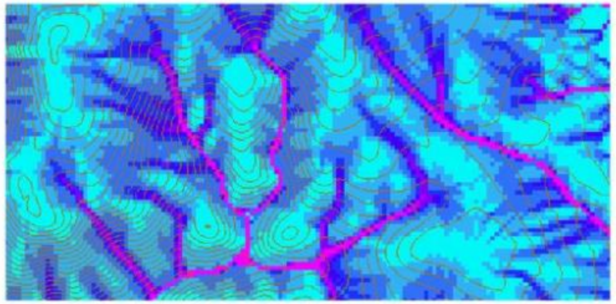




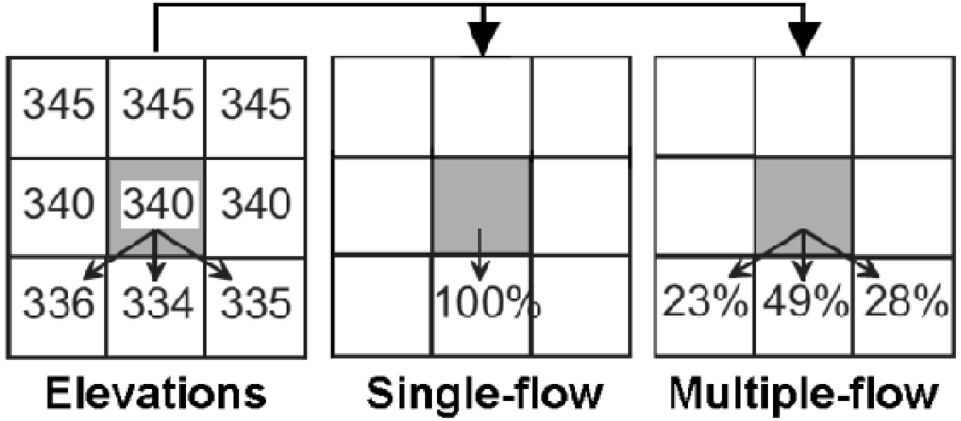


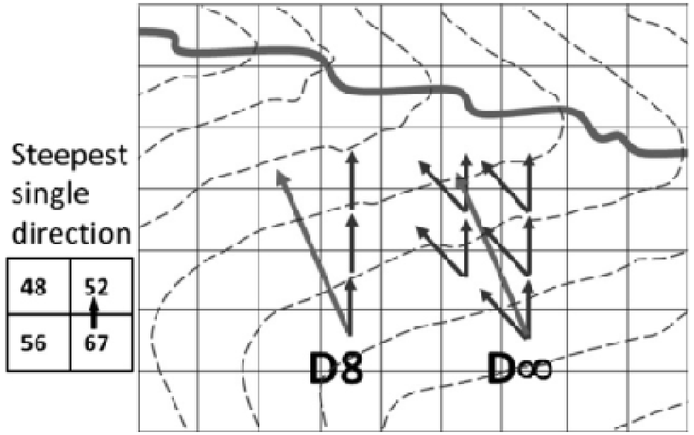


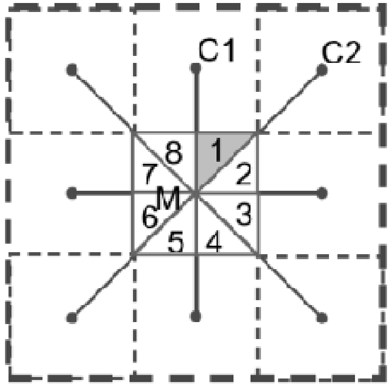




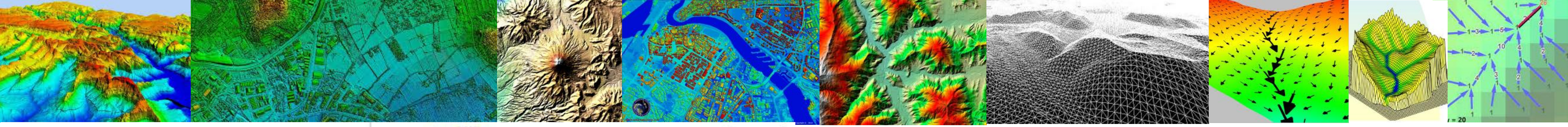
algorithm.

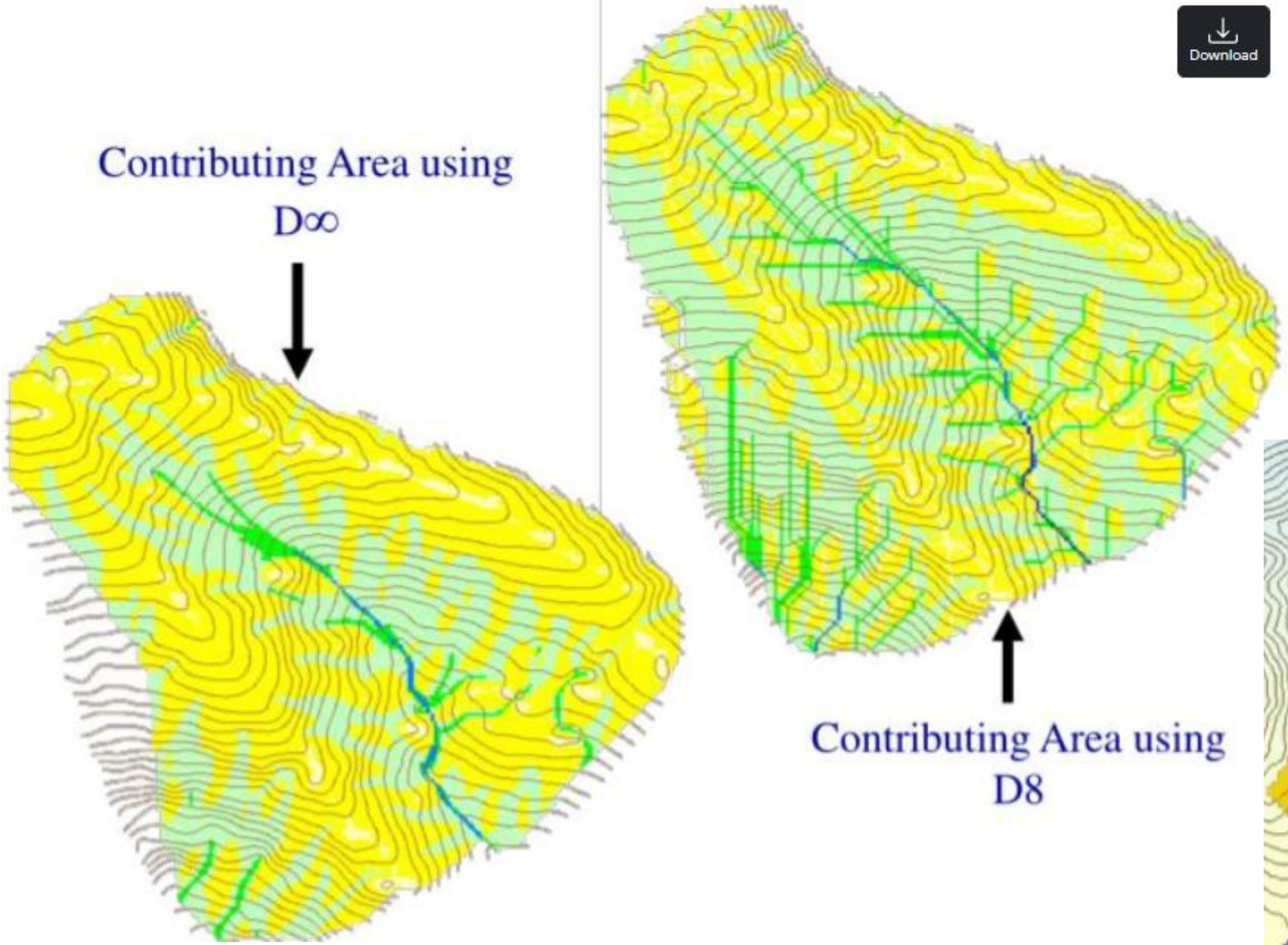


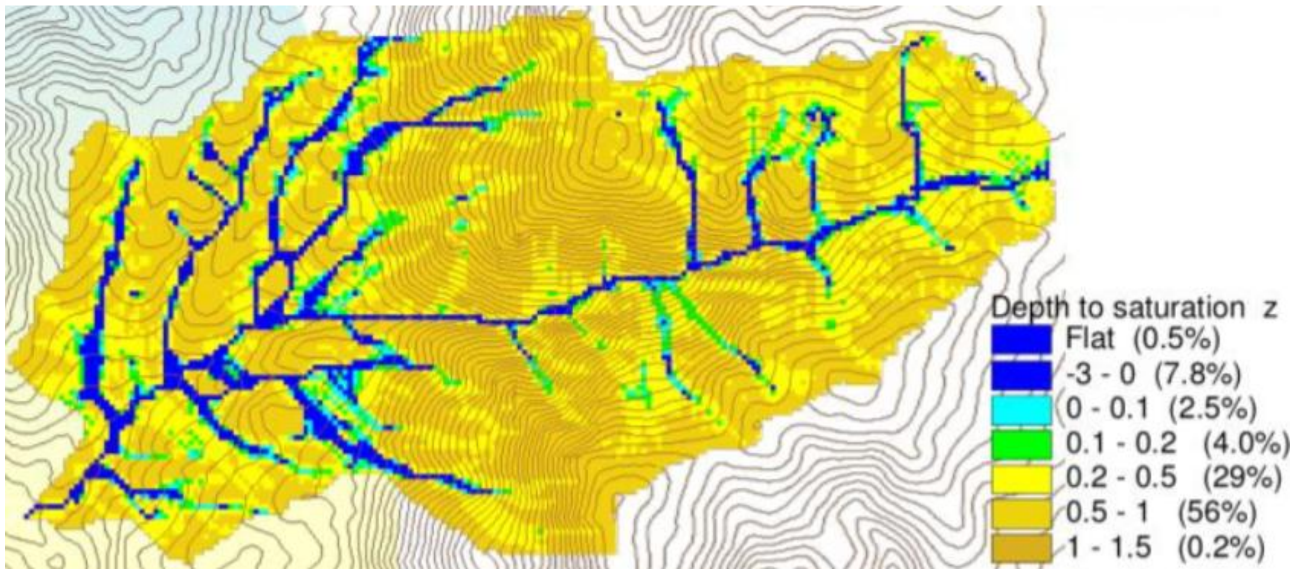


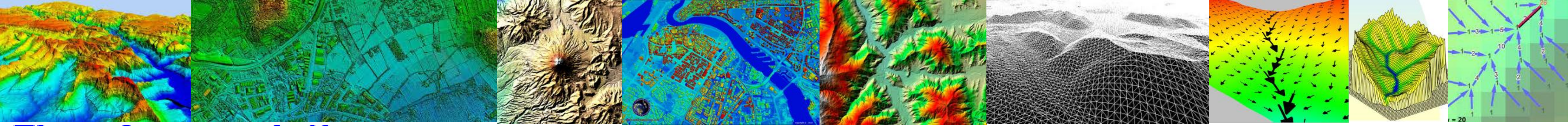


Triangular Multiple Flow



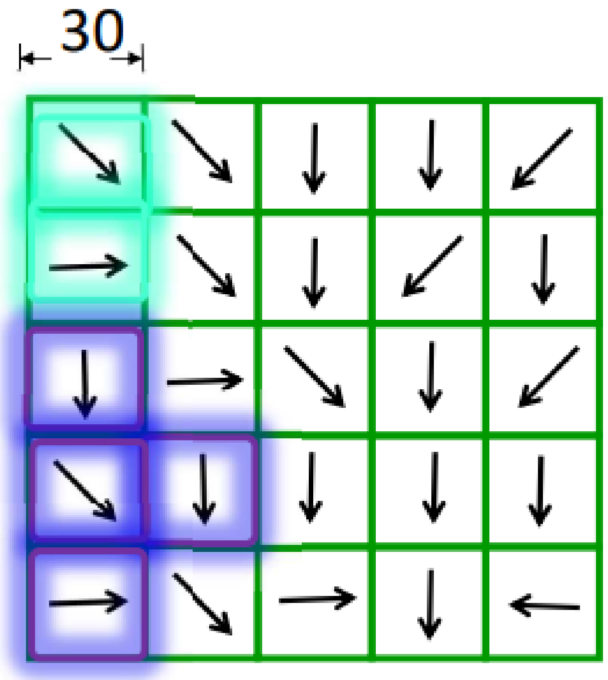

Contributing Area using
D∞
Contributing Area using
D8


Depth to saturation z
Flat (0.5%)
-3 - 0 (7.8%)
0 - 0.1 (2.5%)
0.1 - 0.2 (4.0%)
0.2 - 0.5 (29%)
0.5 - 1 (56%)
1 - 1.5 (0.2%)



Flow Accumulation

Area draining **in** to a grid cell



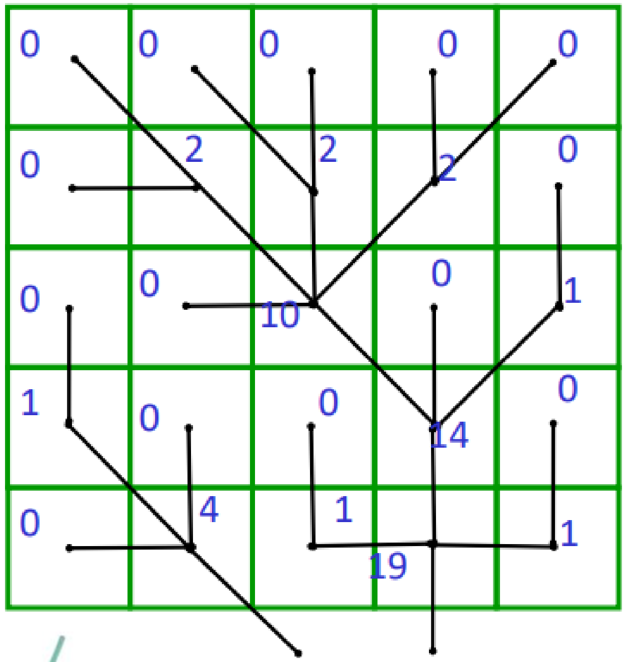
0	0	0	0	0
0	2	2	2	0
0	0	10	0	1
1	0	0	14	0
0	4	1	19	1

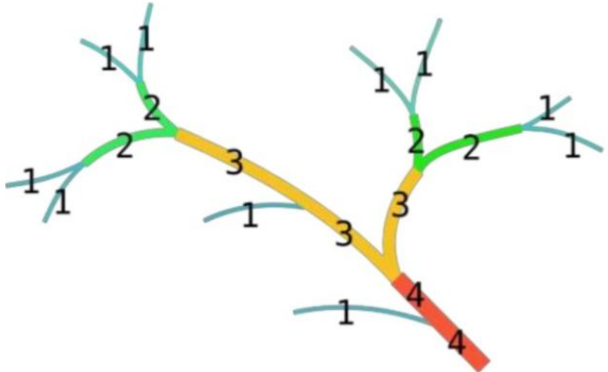
$2*30*30 = 1800\ m^2$

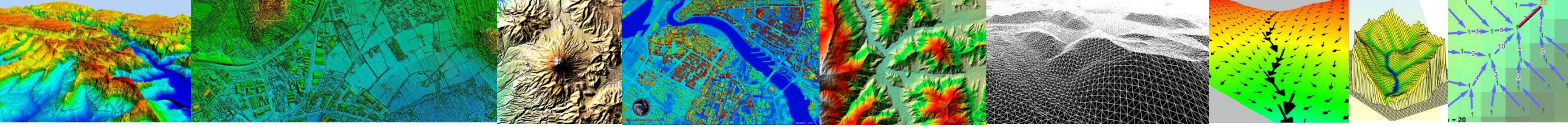
$4*30*30 = ?\ m^2$

Flow Accumulation

Area draining **in** to a grid cell







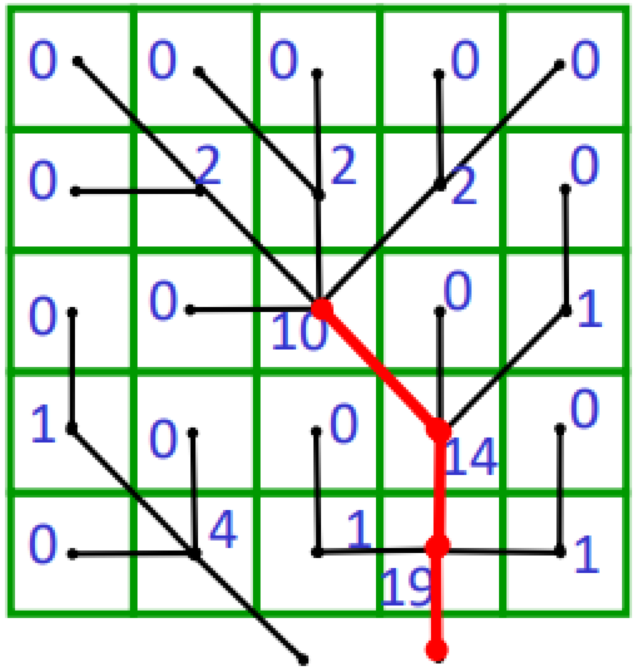
Channel and Watersheds – Stream Definition

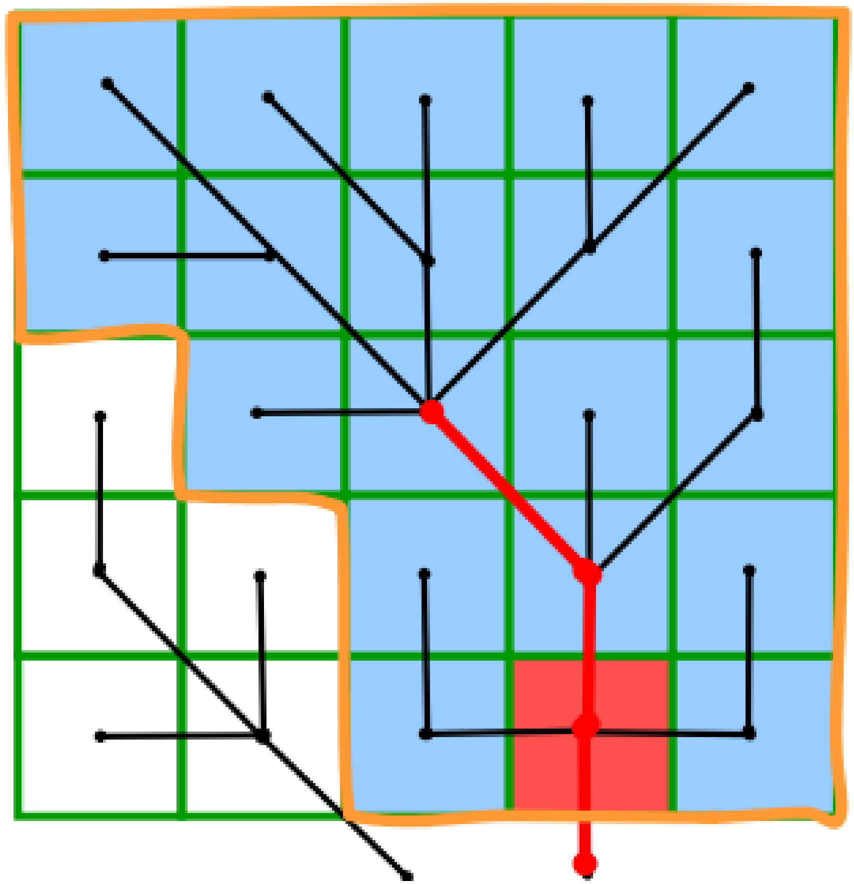
Flow Accumulation
> 10 Cell Threshold

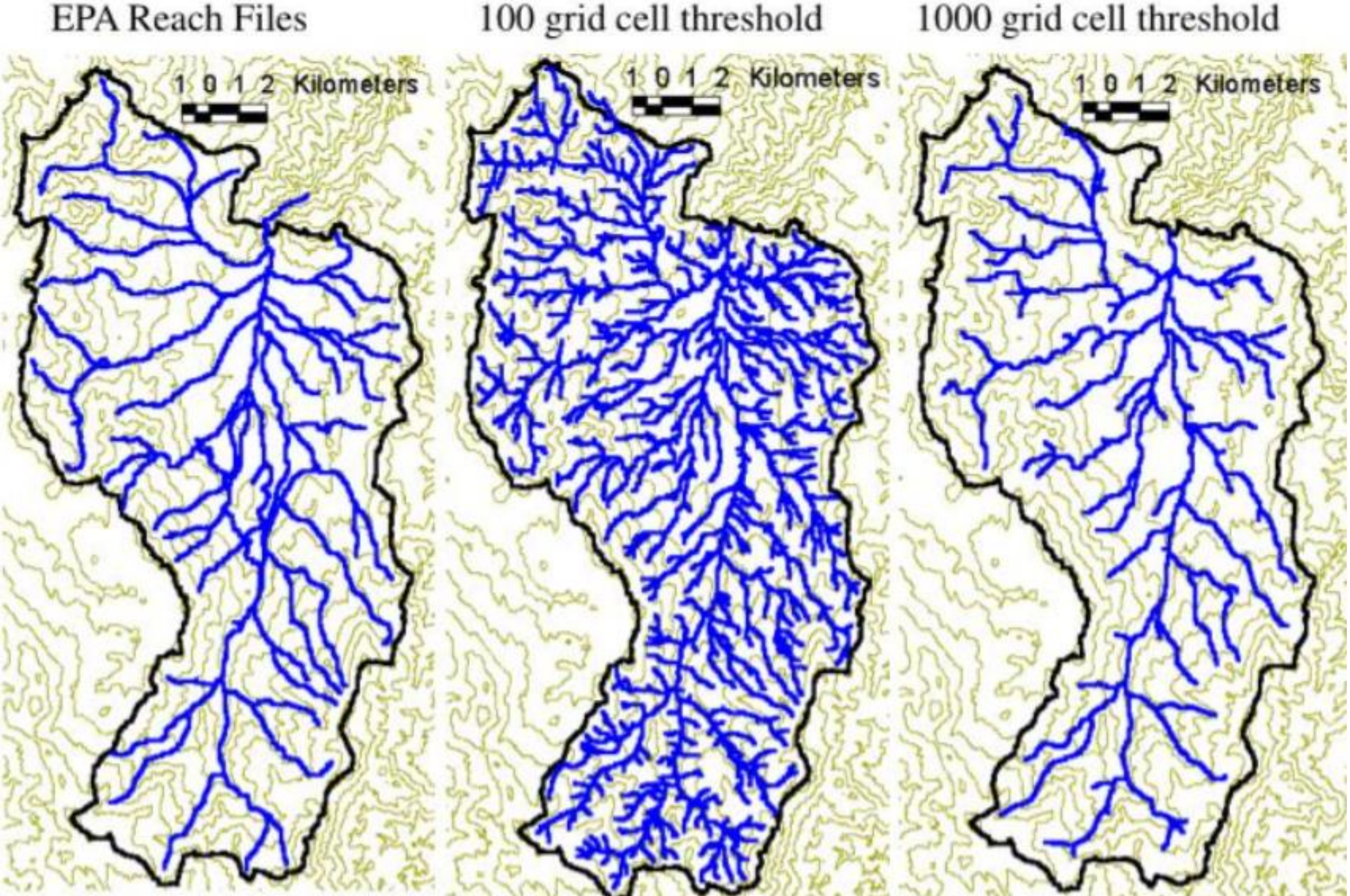
0	0	0	0	0
0	2	2	2	0
0	0	10	0	1
1	0	0	14	0
0	4	1	19	1

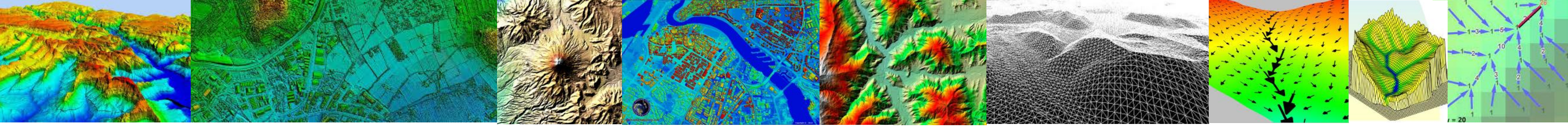
ArcGIS

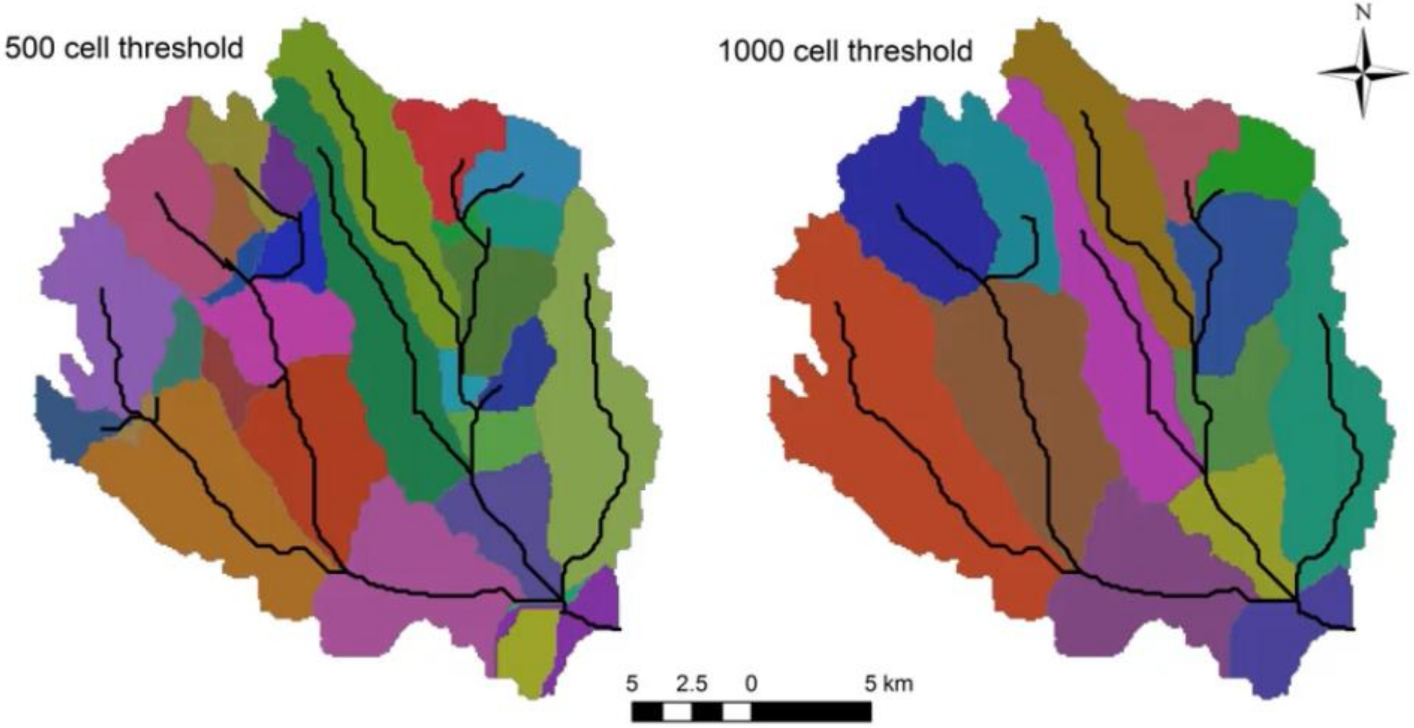
Stream Network for
10 cell Threshold
Drainage Area

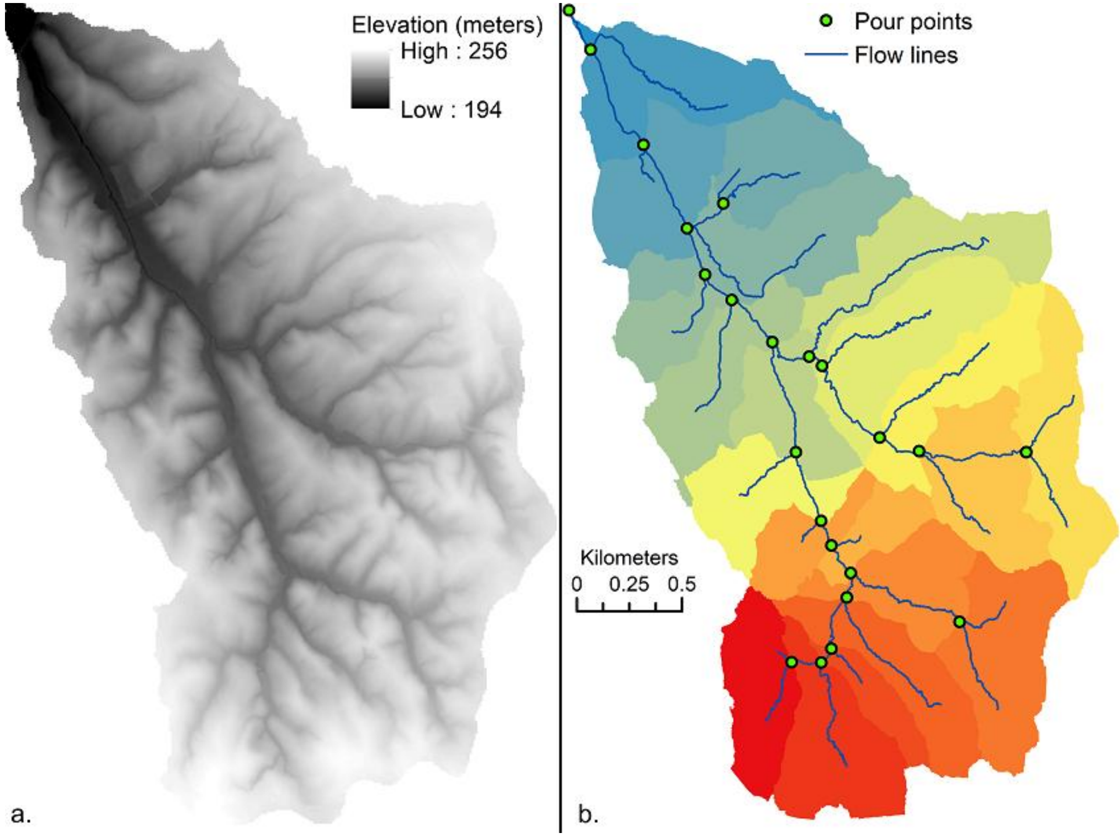


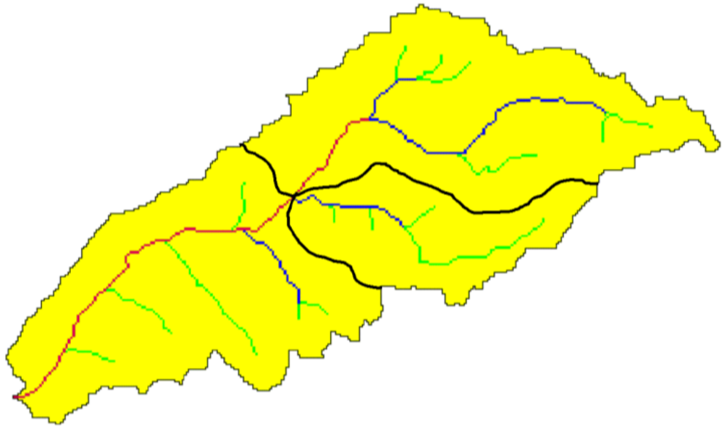


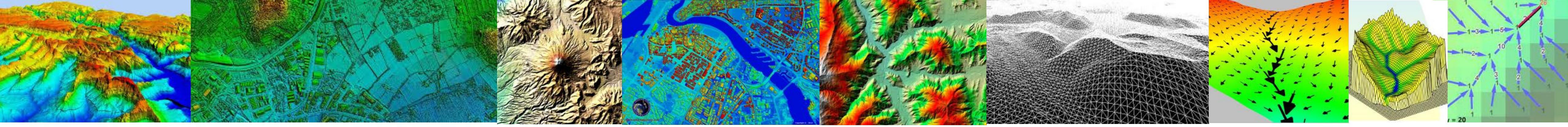

EPA Reach Files
100 grid cell threshold
1000 grid cell threshold
1 0 1 2 Kilometers
1 0 1 2 Kilometers
1 0 1 2 Kilometers

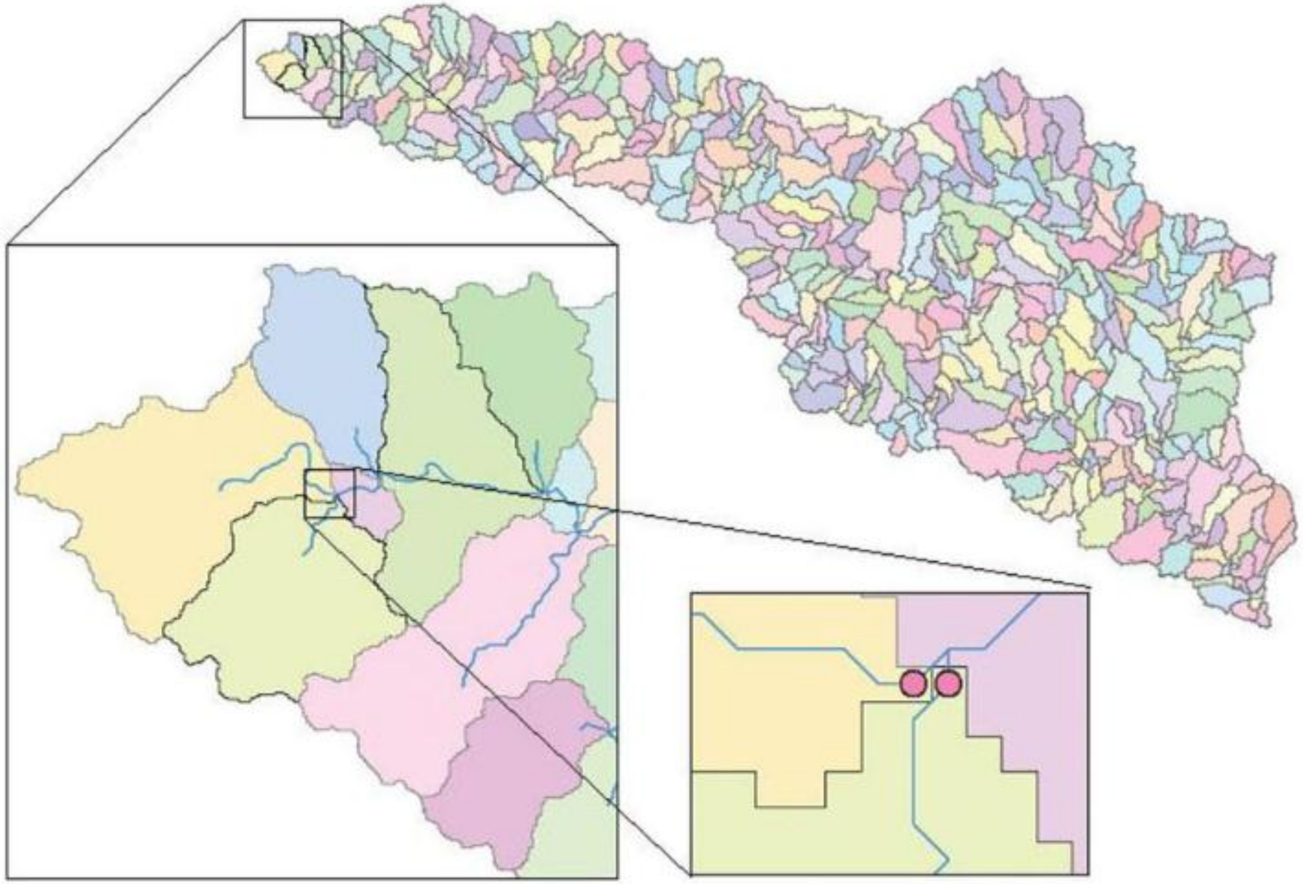



500 cell threshold
1000 cell threshold
N
5 2.5 0 5 km

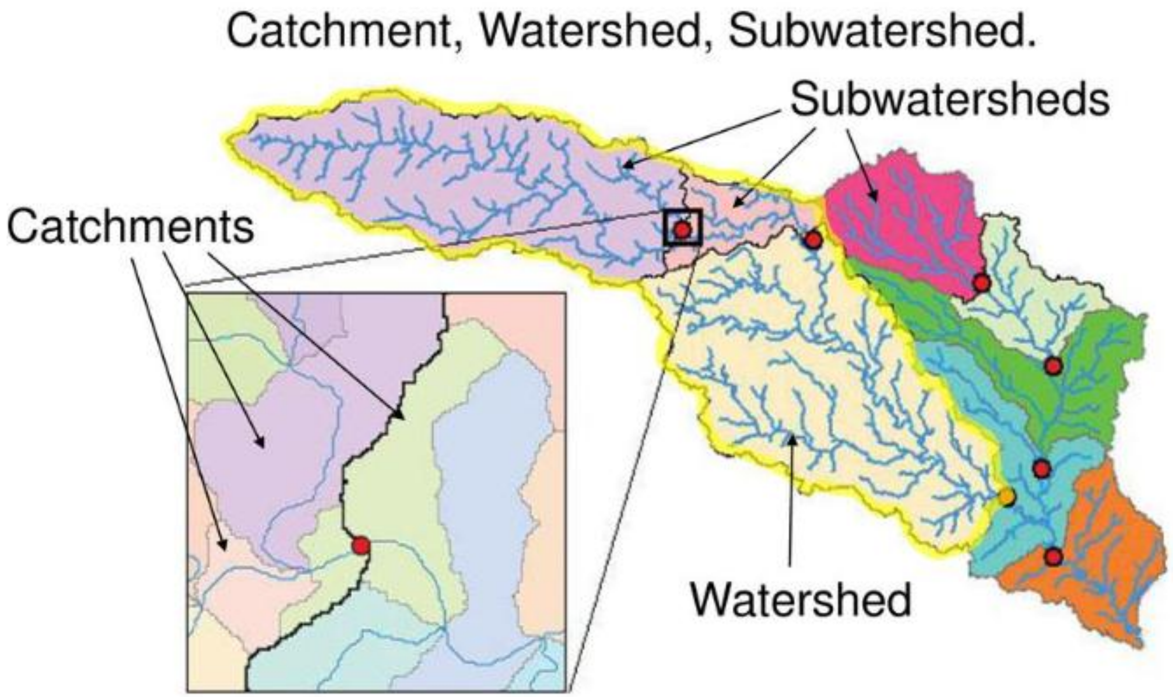

Elevation (meters)
High : 256
Low : 194
Pour points
Flow lines
Kilometers
0 0.25 0.5
a.
b.



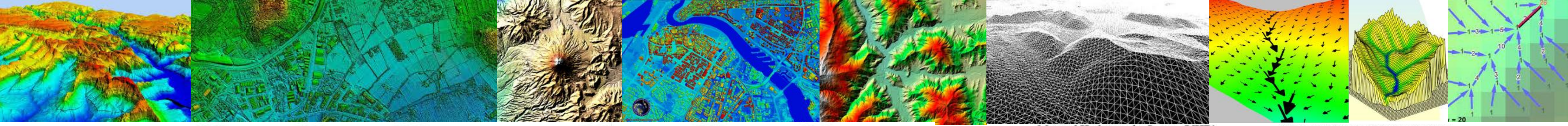


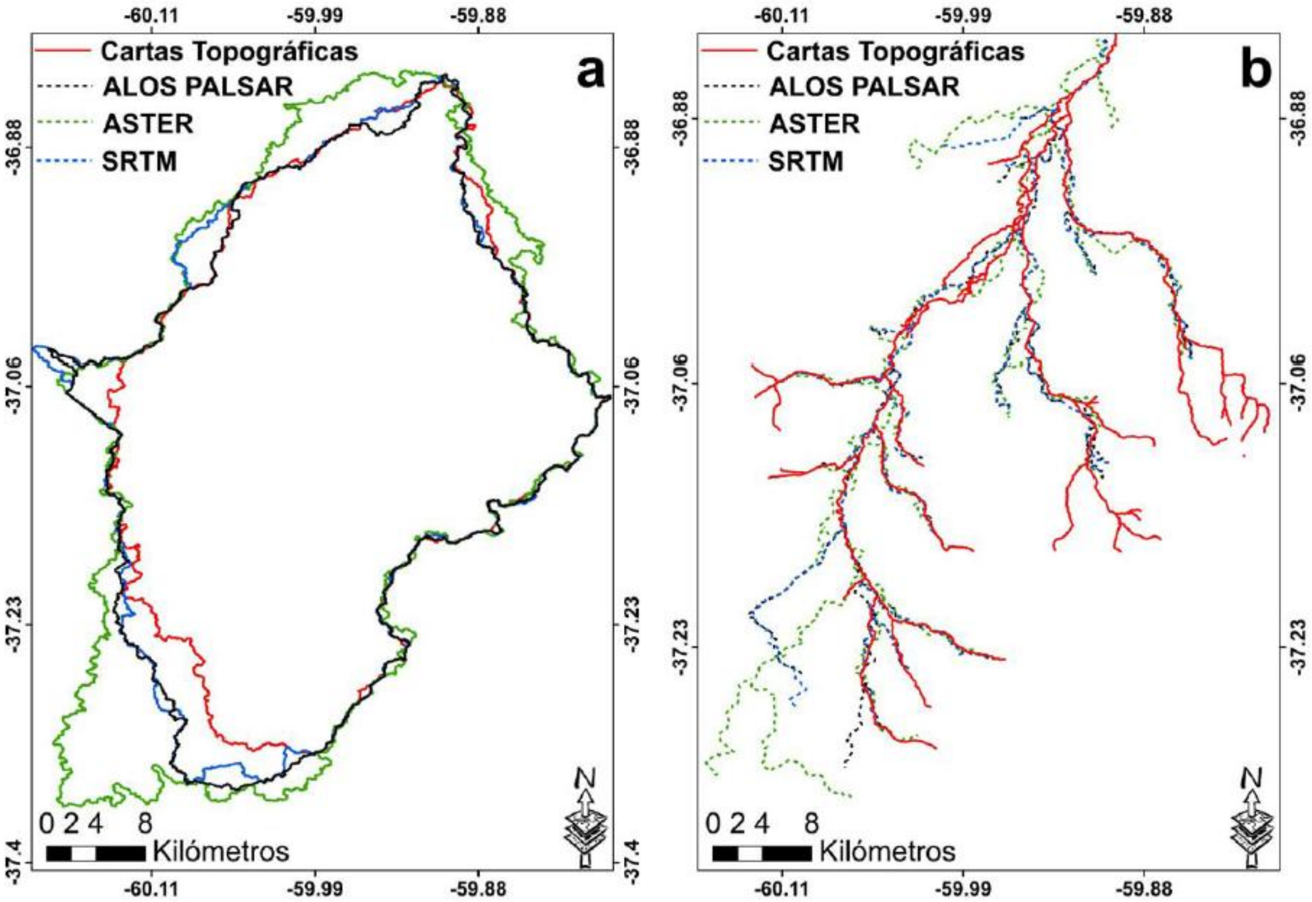


Catchments, DrainageLines and DrainagePoints of the San Marcos basin

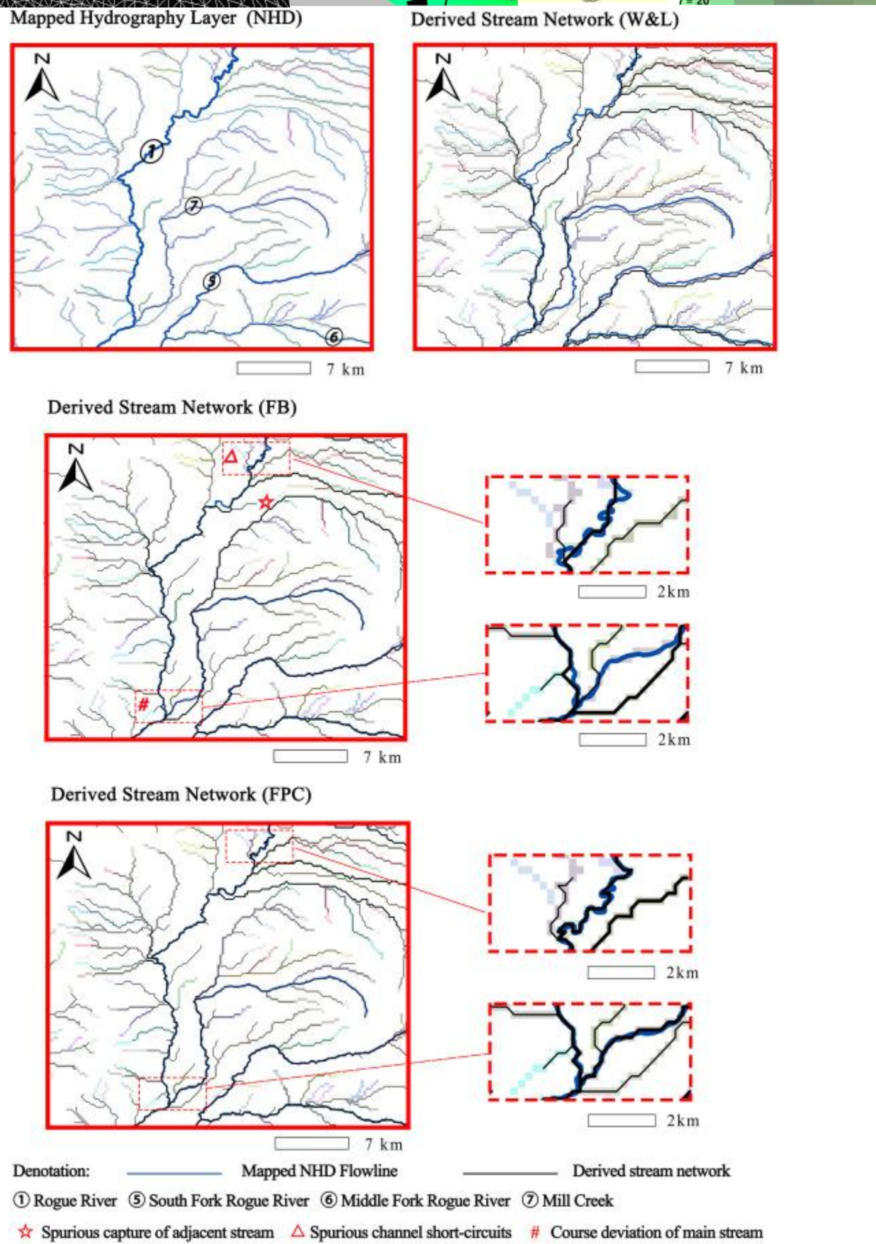


Watershed outlet points may lie within the interior of a catchment, e.g. at a USGS stream-gaging site.





Fuente	Área de la cuenca (km²)	Longitud del drenaje (Km)	Densidad de drenaje (km/km²)
ASTER	1213	252	4.81
SRTM	1050	214	4.9
ALOS PALSAR	1046	220	4.75
CARTA TOPOGRÁFICA	982	312	3.2



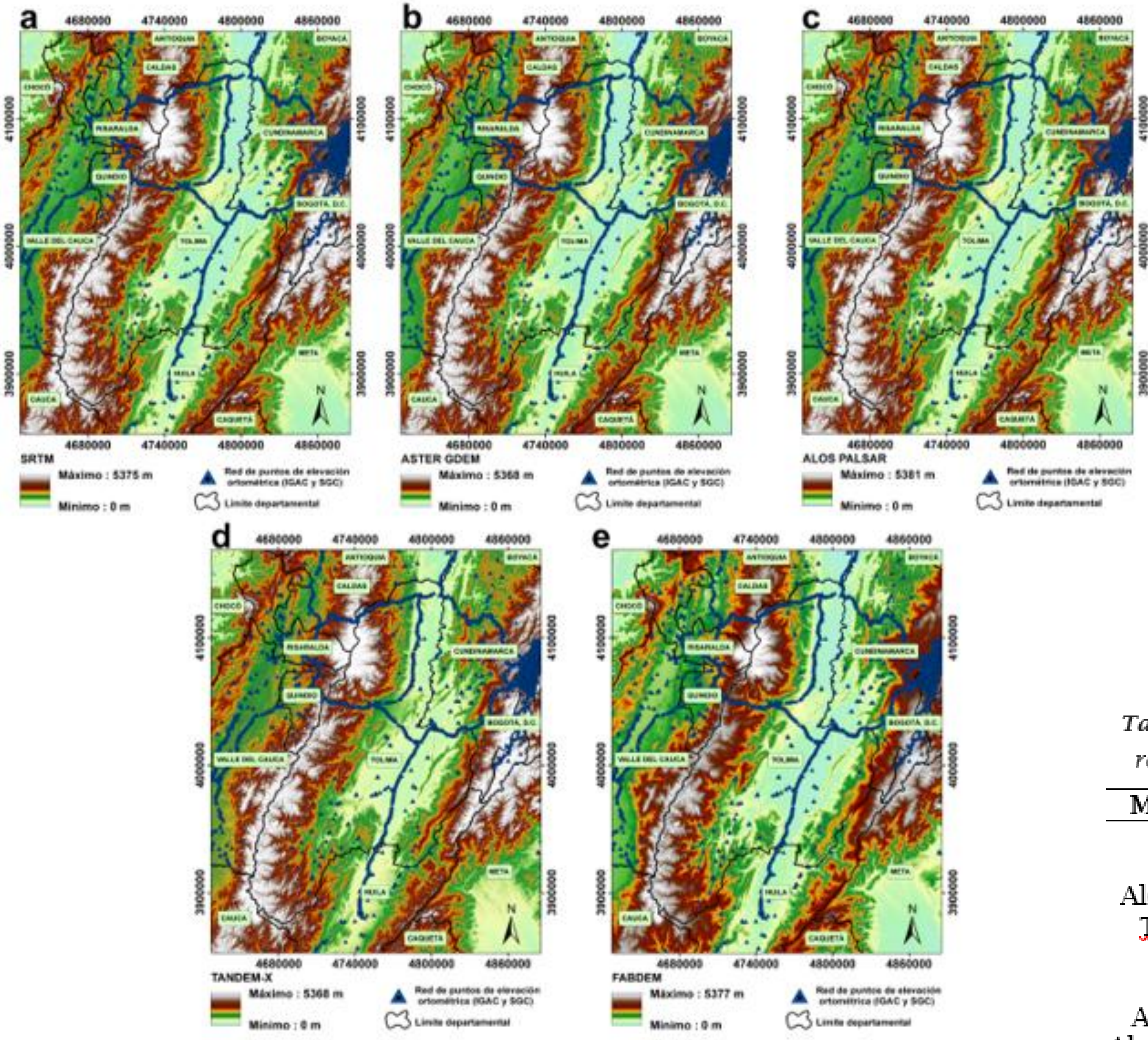


Figura 1. Modelos digitales de elevación analizados en el departamento del Tolima y ubicación de la red de puntos de elevación ortométrica. (a) SRTM, (b) Aster GDEM V3, (c) Alos Palsar, (d) TanDEM-X (e) FABDEM.

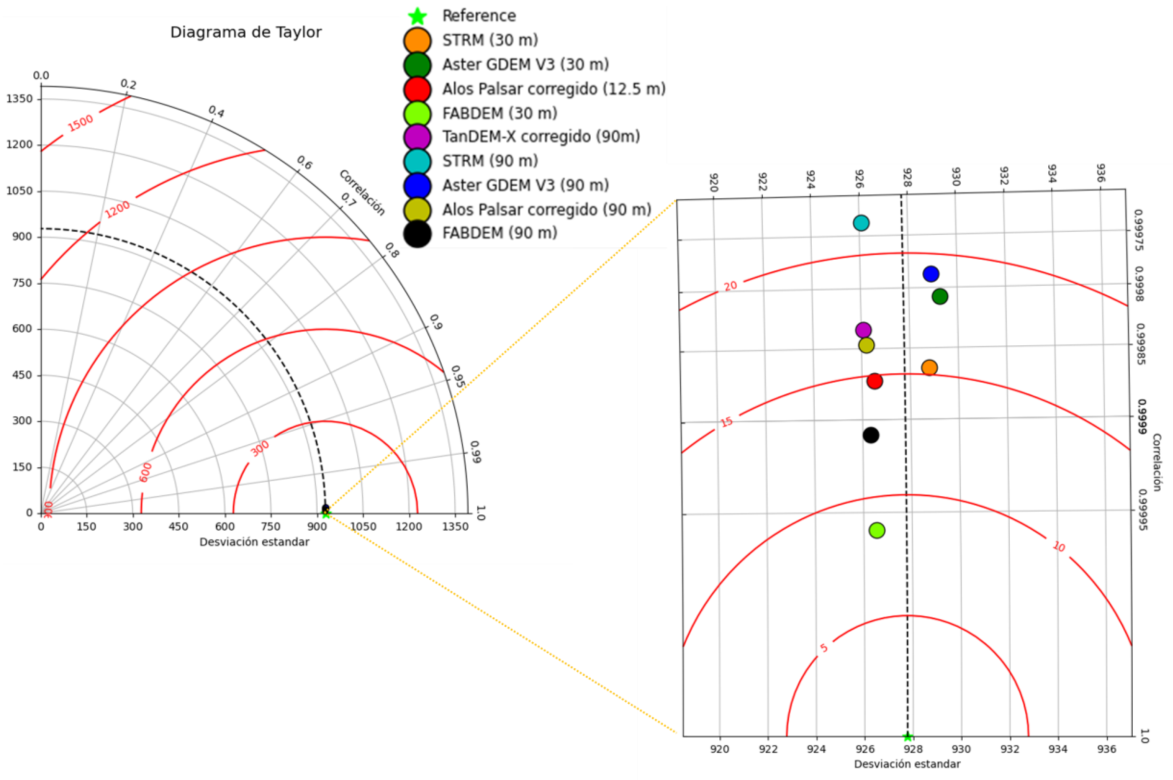
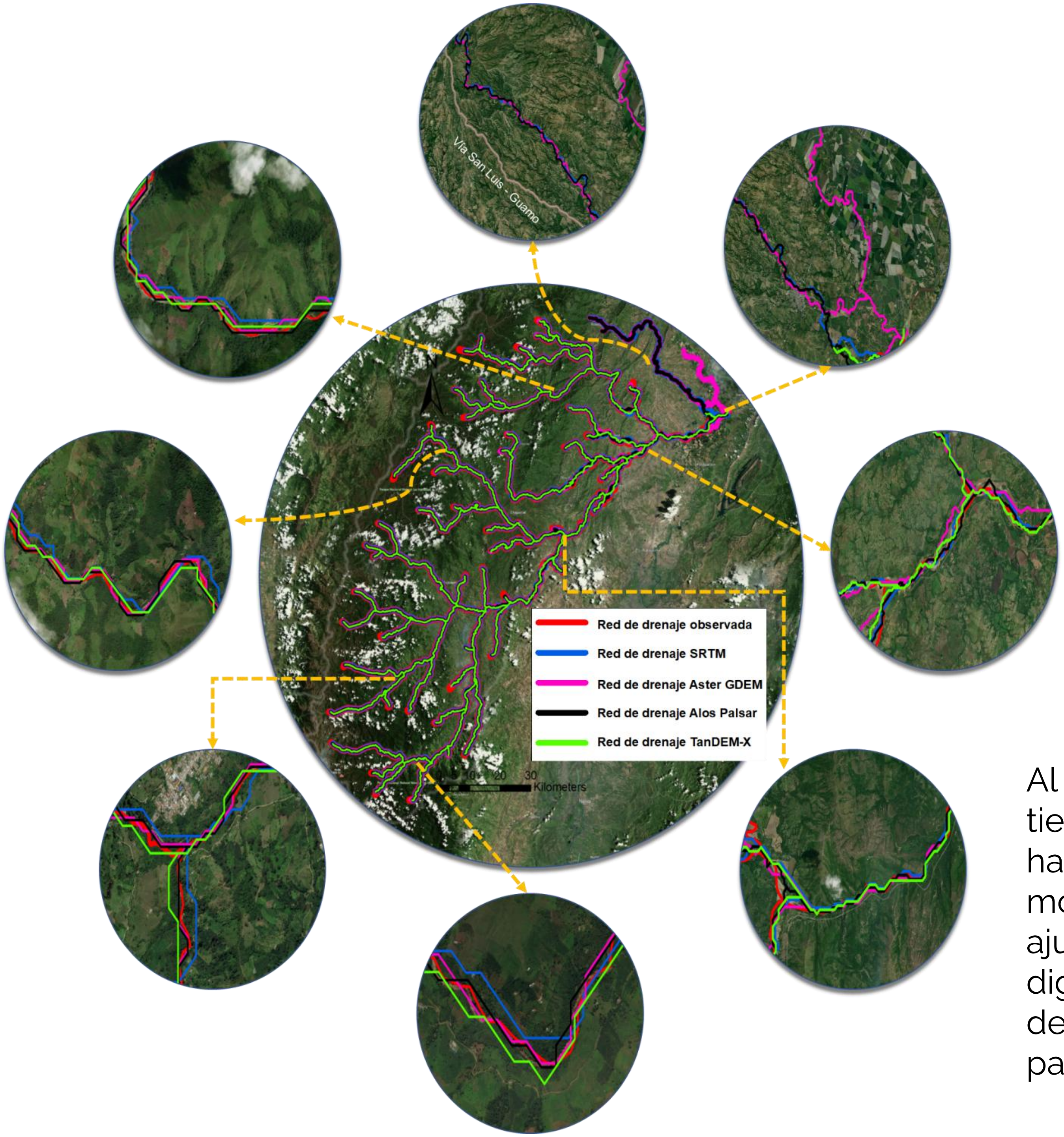
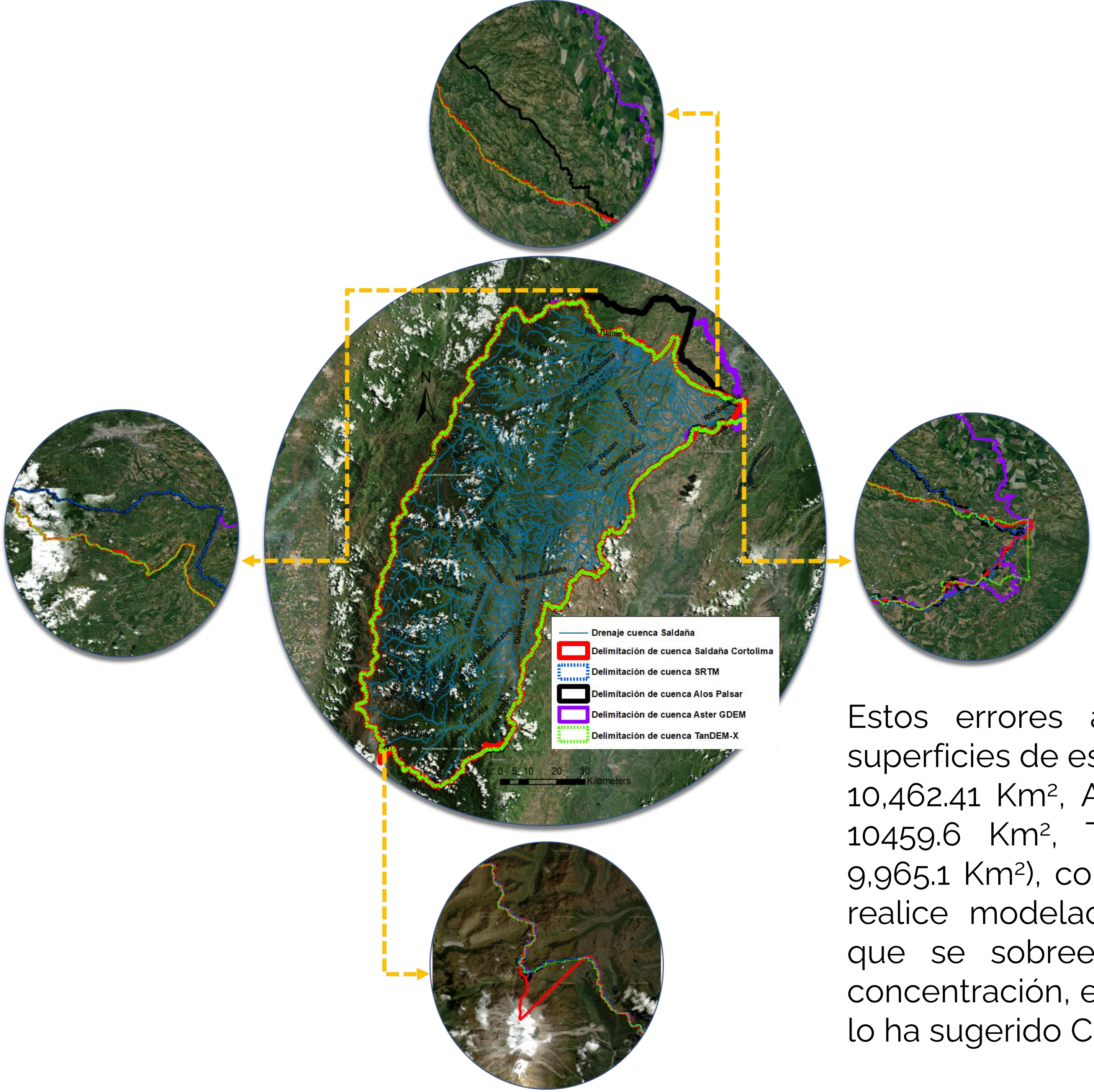


Tabla 2. Estadísticos calculados para medir la incertidumbre topográfica de los MDEs, con respecto a los puntos observación de elevación ortométrica medidos por IGAC y SGC.

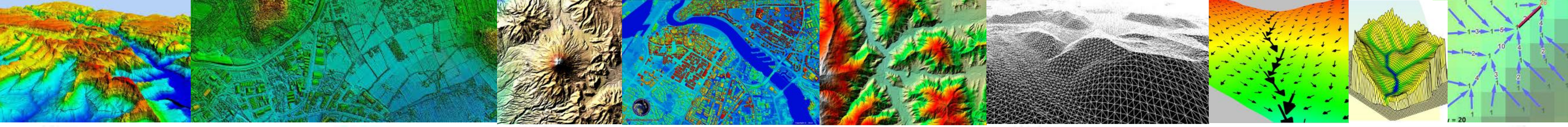
Modelo digital de elevación	STD (m)	MSE (m)	RMSE (m)	R^2
STRM (30 m)	928.53	64.82	8.04	0.9993
Aster GDEM V3 (30 m)	929.04	165.53	12.85	0.9992
Alos Palsar corregido (12.5 m)	926.28	44.93	6.70	0.9993
TanDEM-X corregido (90m)	926.27	115.31	10.73	0.9993
STRM (90 m)	925.84	280.77	16.74	0.9991
Aster GDEM V3 (90 m)	928.69	194.34	13.93	0.9992
Alos Palsar corregido (90 m)	925.95	87.01	9.32	0.9993
Alos Palsar sin corregir (12.5 m)	927.44	659.18	25.65	0.9993
TanDEM-X sin corregir (90m)	927.49	768.15	27.69	0.9993
FABDEM (30m)	926.53	39.48	6.28	0.9999
FABDEM (90m)	926.33	79.70	8.93	0.9999
GPS diferencial	928.04	----------	----------	----------

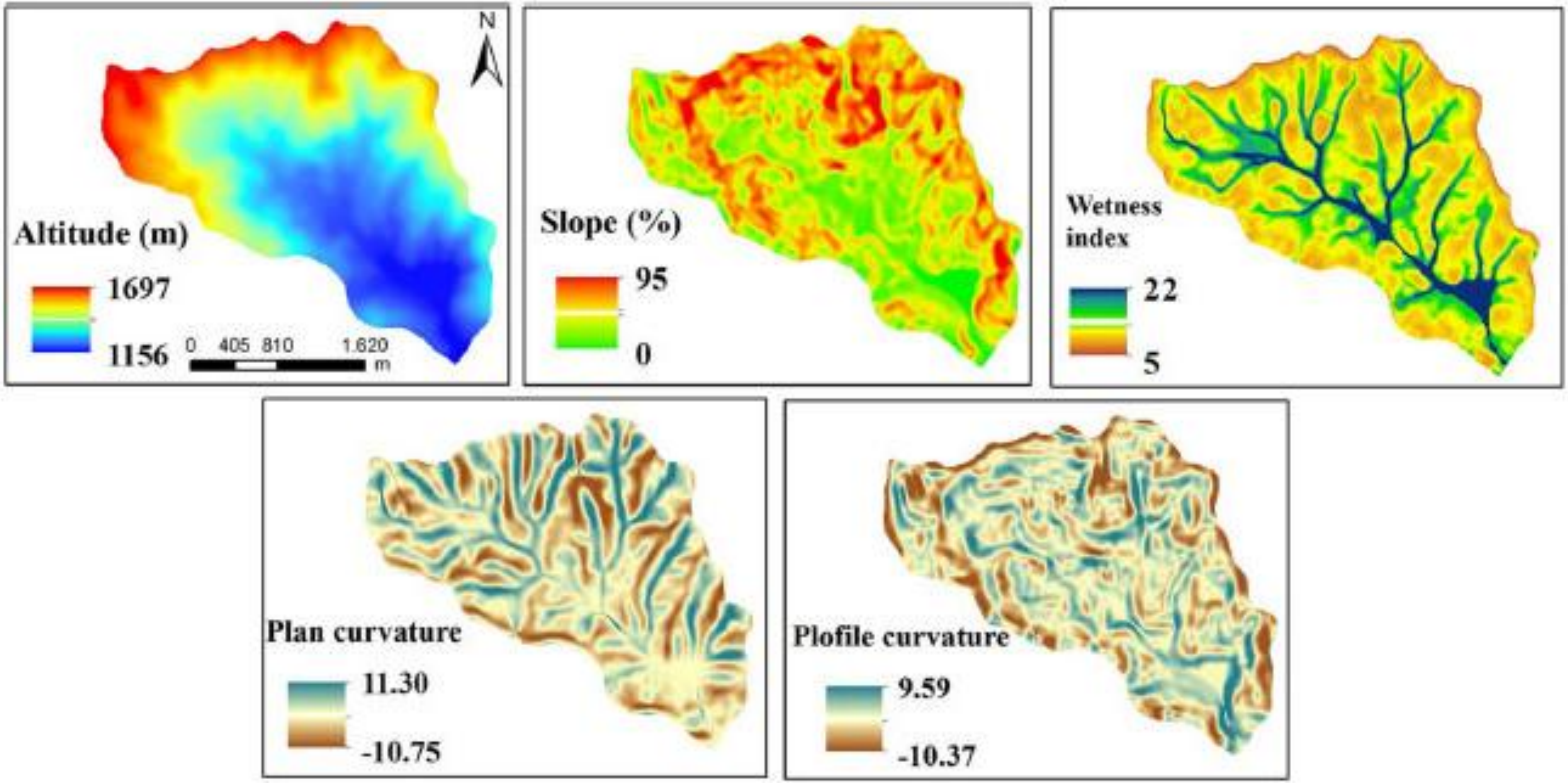


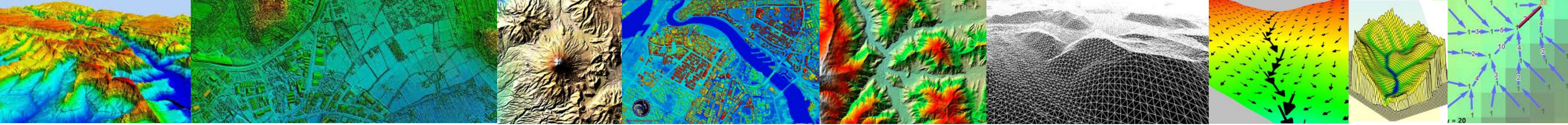
Al analizar la red de drenaje (Figura 6), la cual tiene un área de influencia específica 5000 ha. Se encontró que la red delimitada por los modelos ALOS PALSAR, y TanDEM-X se ajusta mejor con la red de drenaje digitalizada a través de imágenes satelitales de alta resolución espacial horizontal, tanto para la parte alta y baja de la cuenca.

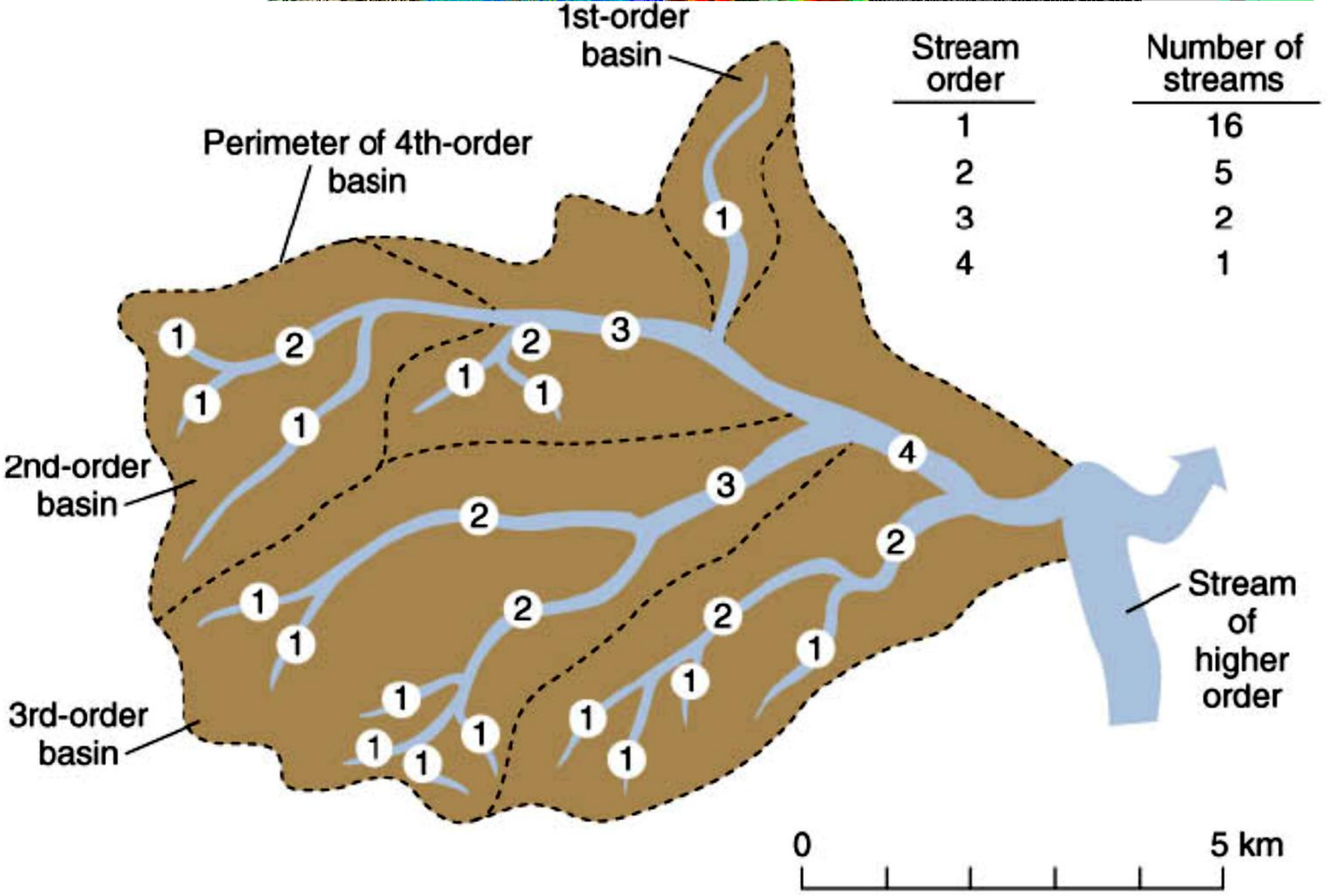


Estos errores anteriormente mencionados generan superficies de escurrimiento de diferentes áreas (SRTM: 10,462.41 Km², Aster GDEM: 10,718.6 Km², Alos Palsar: 10459.6 Km², TanDEM-X: 9983.7 Km² y Cortolima: 9,965.1 Km²), conllevando a futuros errores cuando se realice modelación hidrológica distribuida, debido a que se sobreestima y/o subestima el tiempo de concentración, el flujo superficial y subsuperficial como lo ha sugerido Callow et al. (2007).

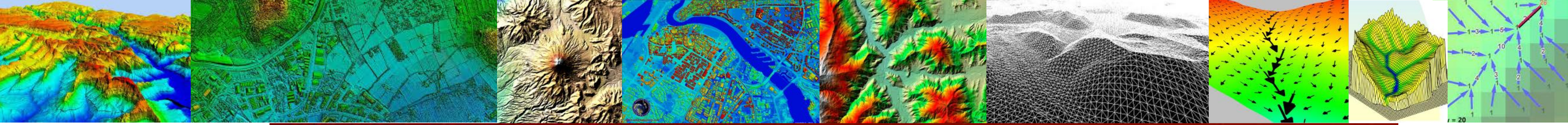



N
Altitude (m)
1697
1156
0 405 810 1.620
m
Slope (%)
95
0
Wetness
index
22
5
Plan curvature
11.30
-10.75
Plofile curvature
9.59
-10.37





Stream order	Number of streams
1	16
2	5
3	2
4	1

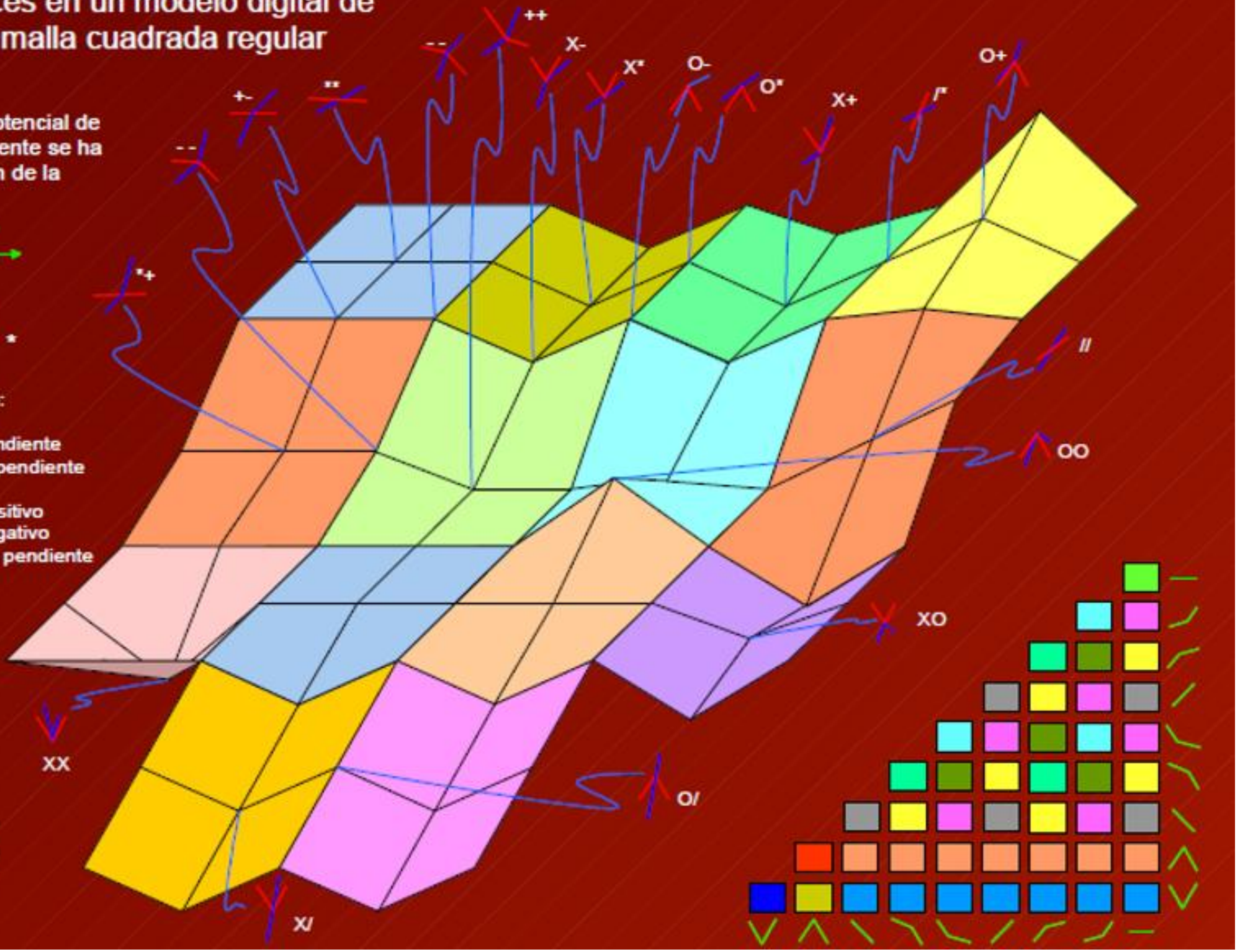


Posibles cruces en un modelo digital de elevación de malla cuadrada regular

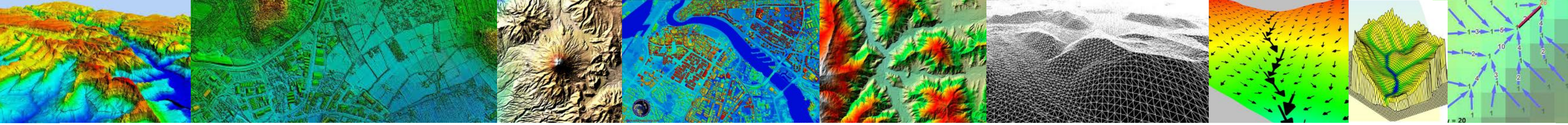
Los símbolos de potencial de cambio en la pendiente se ha valorado en función de la siguiente figura.

Las variaciones son:

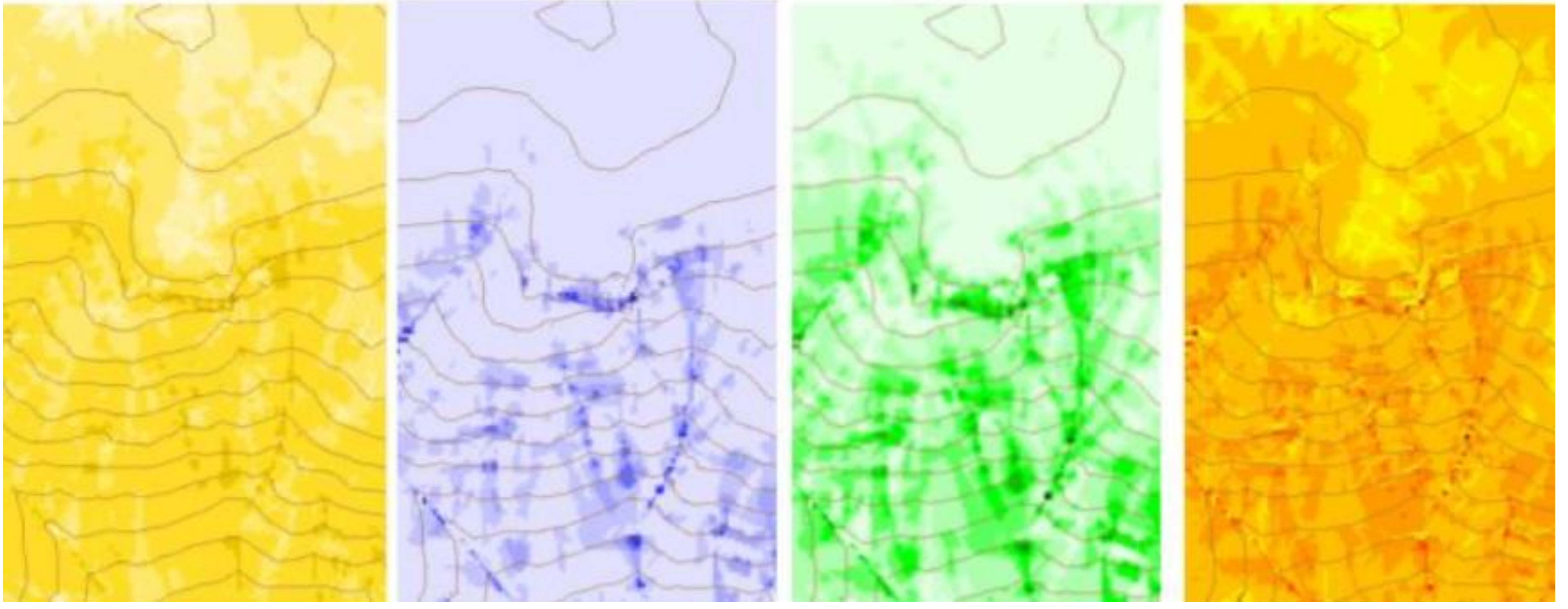
+ aumento de la pendiente
- disminución de la pendiente
* neutro
O punto singular positivo
X punto singular negativo
/ permanencia de la pendiente

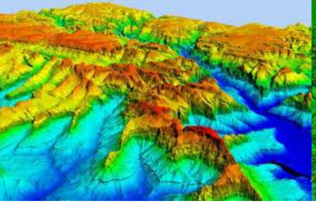


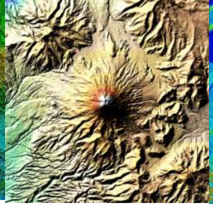
- Llano
- Divisoria
- Valle
- Cima
- Collado
- Sima
- Rectilíneo
- Cóncavo rectilíneo
- Convexo rectilíneo
- Cóncavo-convexo
- Cóncavo
- Convexo

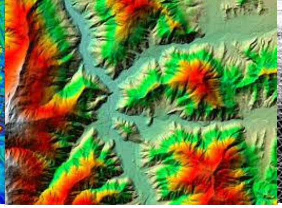
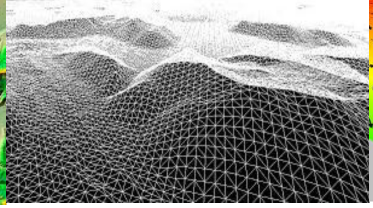
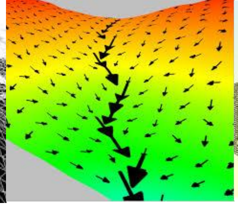
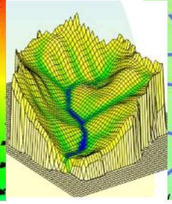
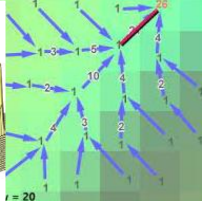


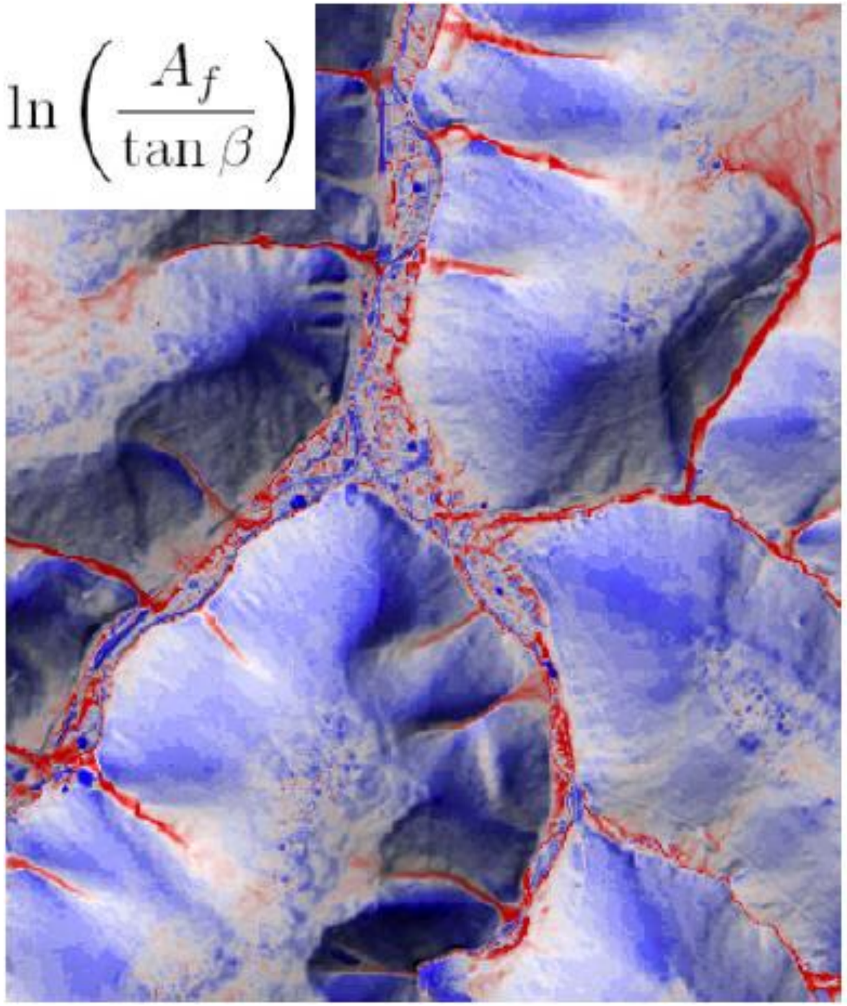
Erodability	Transport Capacity	Transport Flux,	Deposition,
e.g. $E = \beta\, a^{0.7}\, S^{0.6}$	e.g. $T_{cap} = \chi\, a^{2}\, S^{2}$	T	D



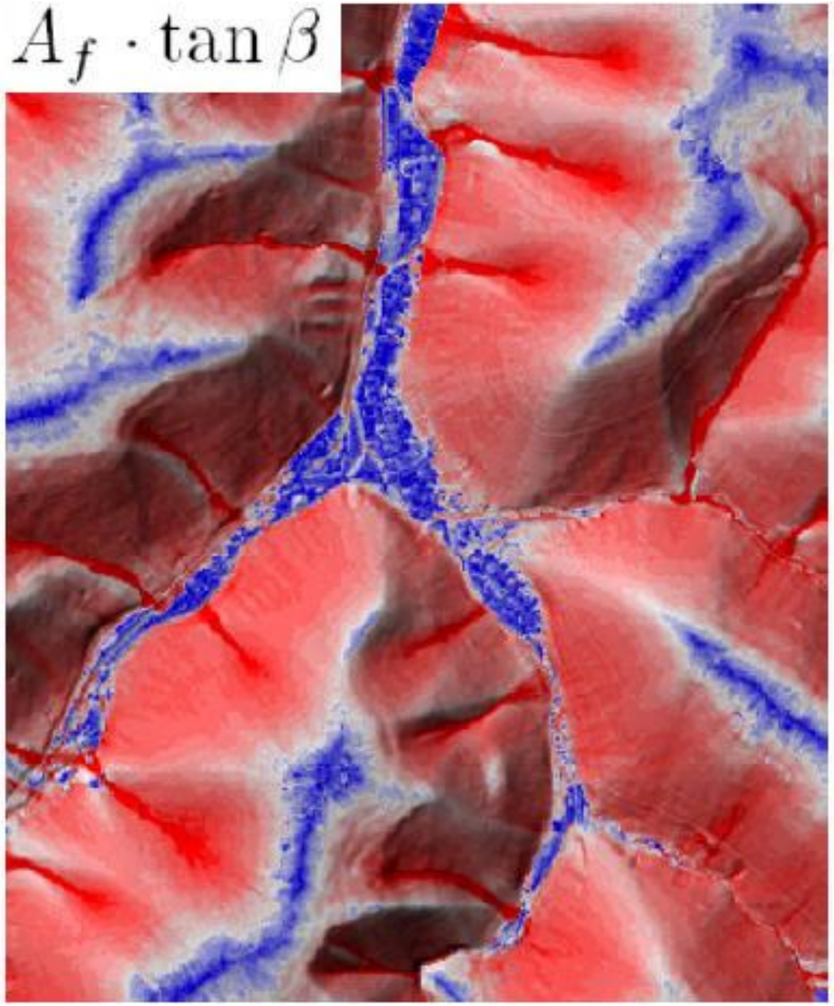




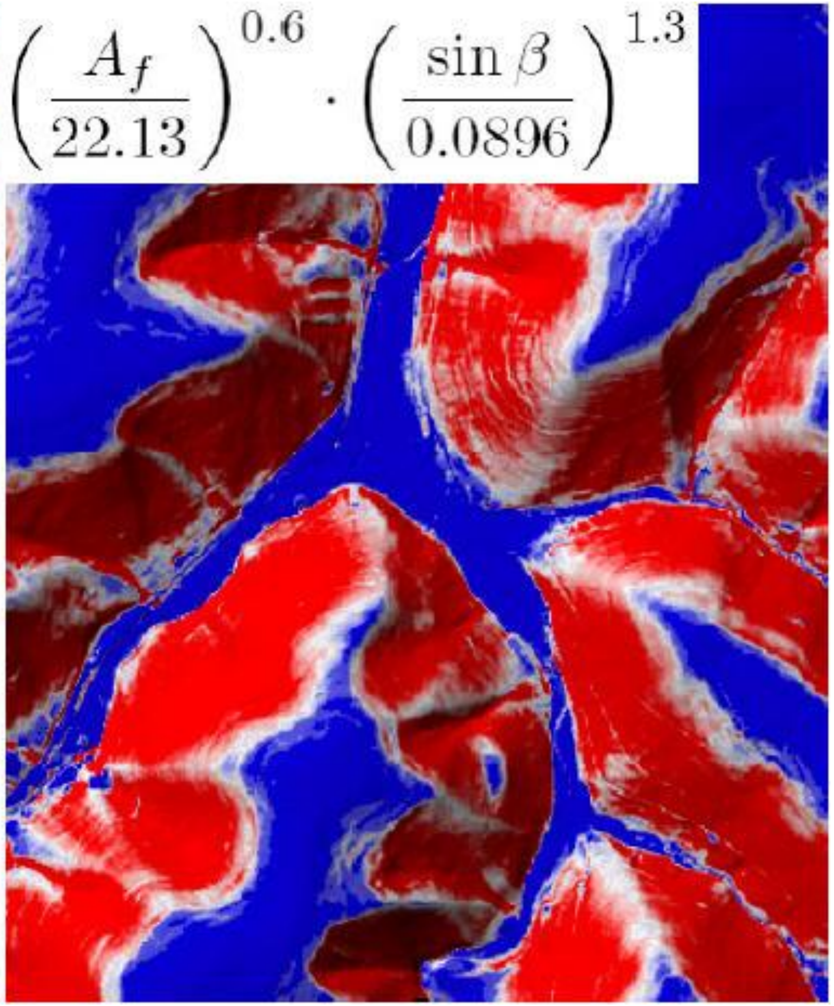








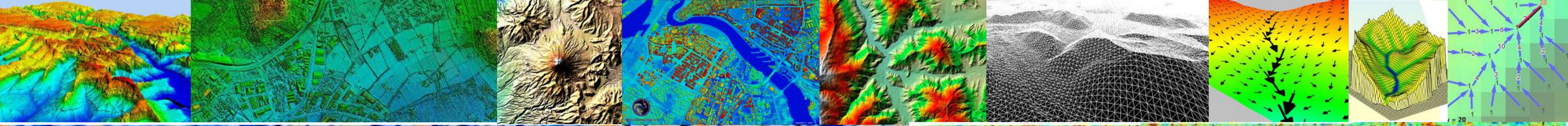
Topographic Wetness Index TWI

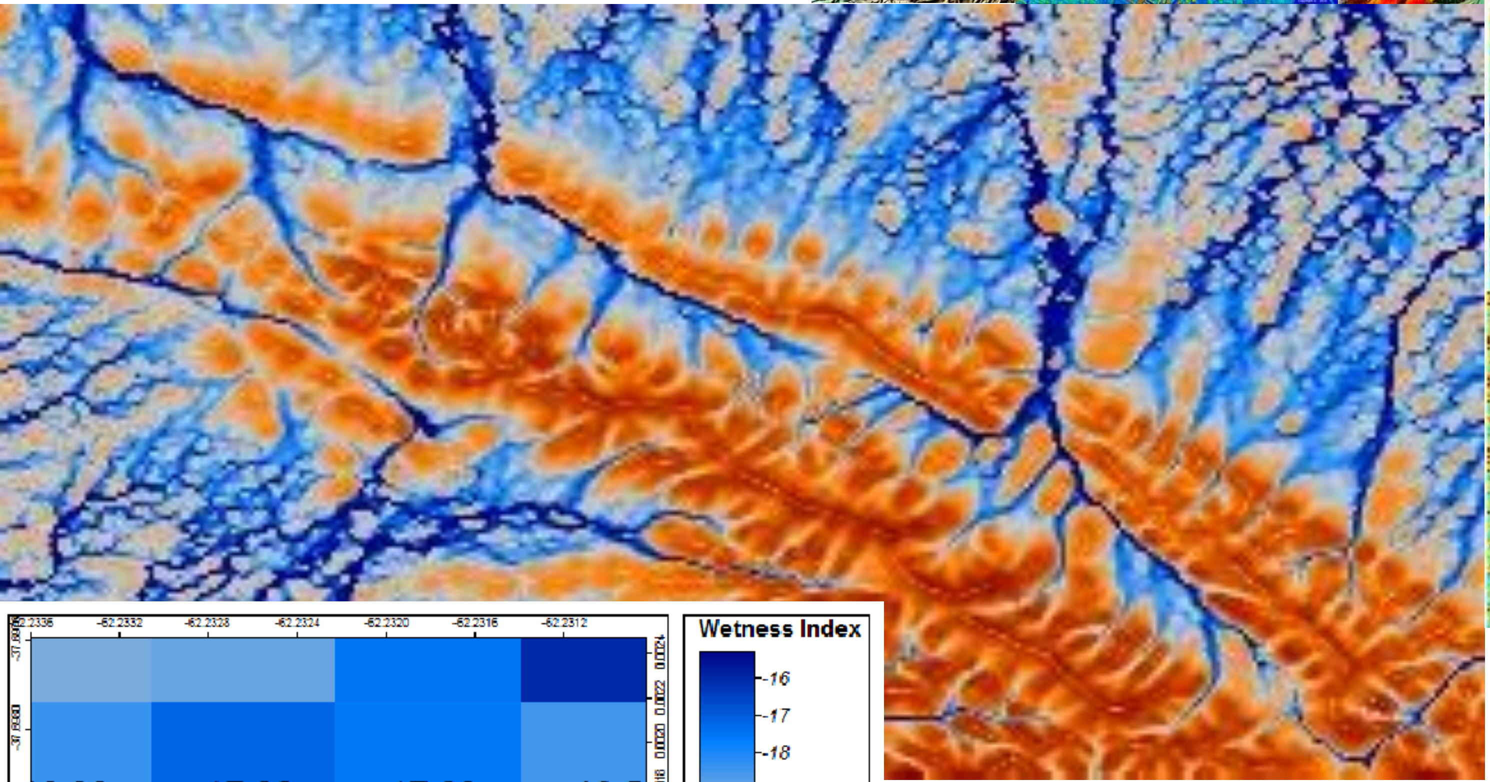


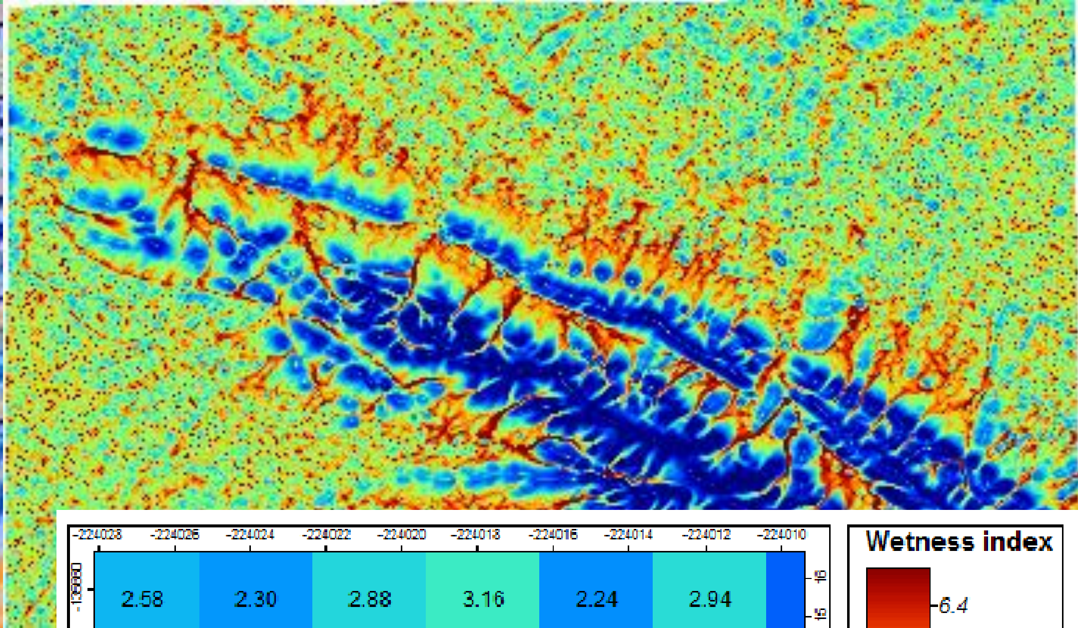
Stream Power Index

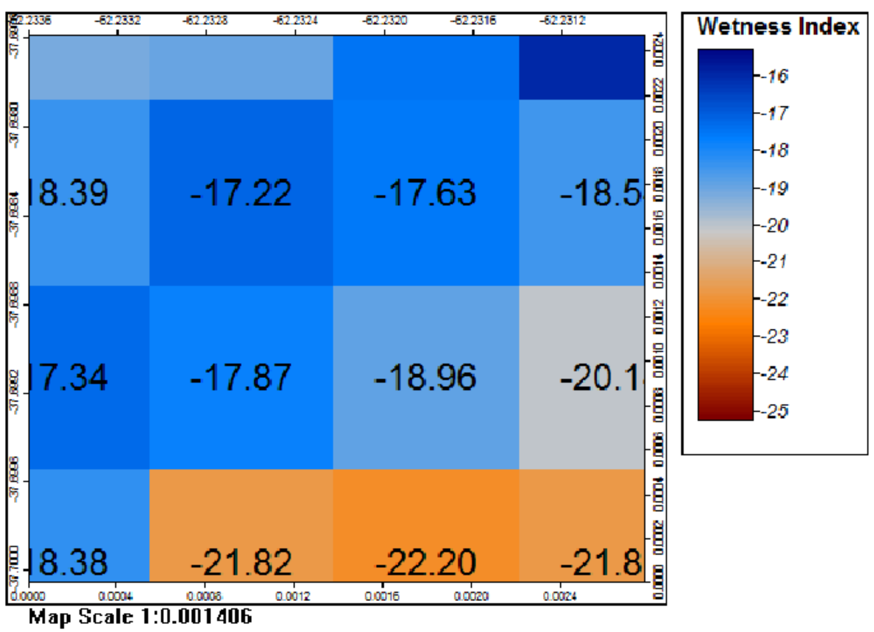


Slope Length Index

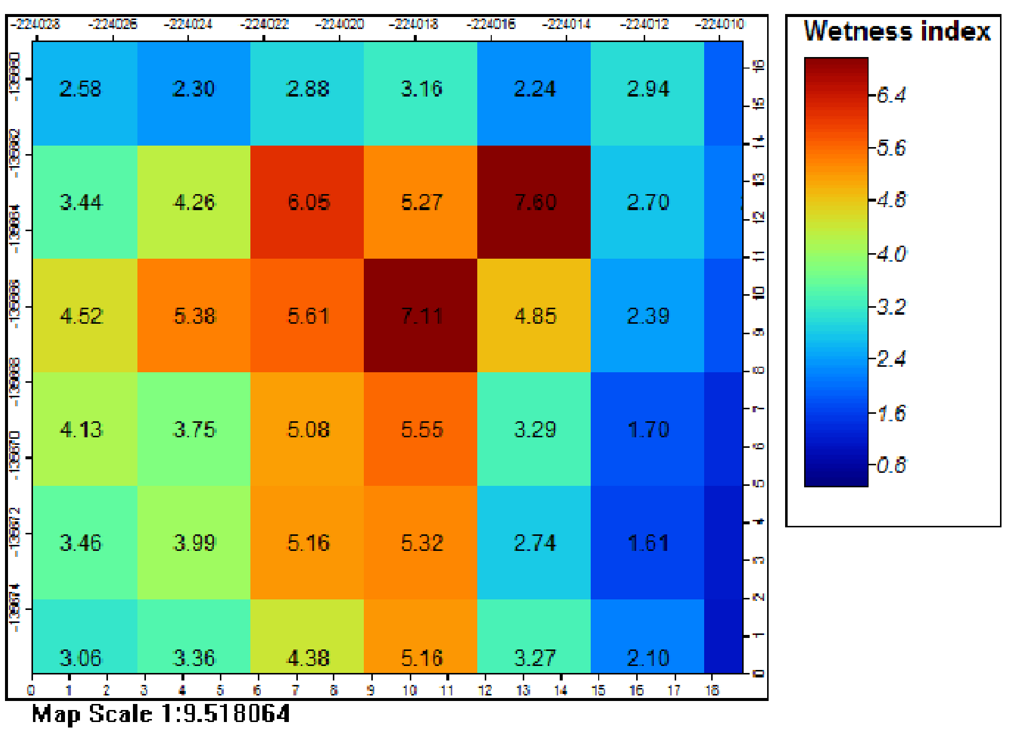


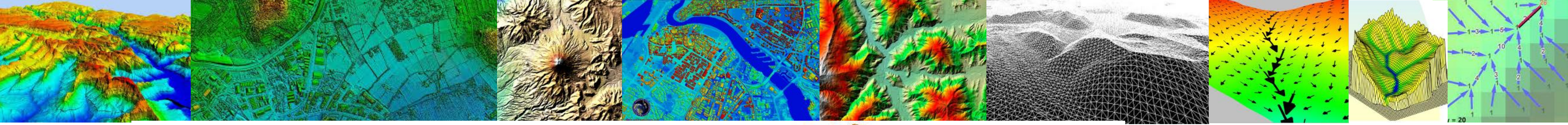


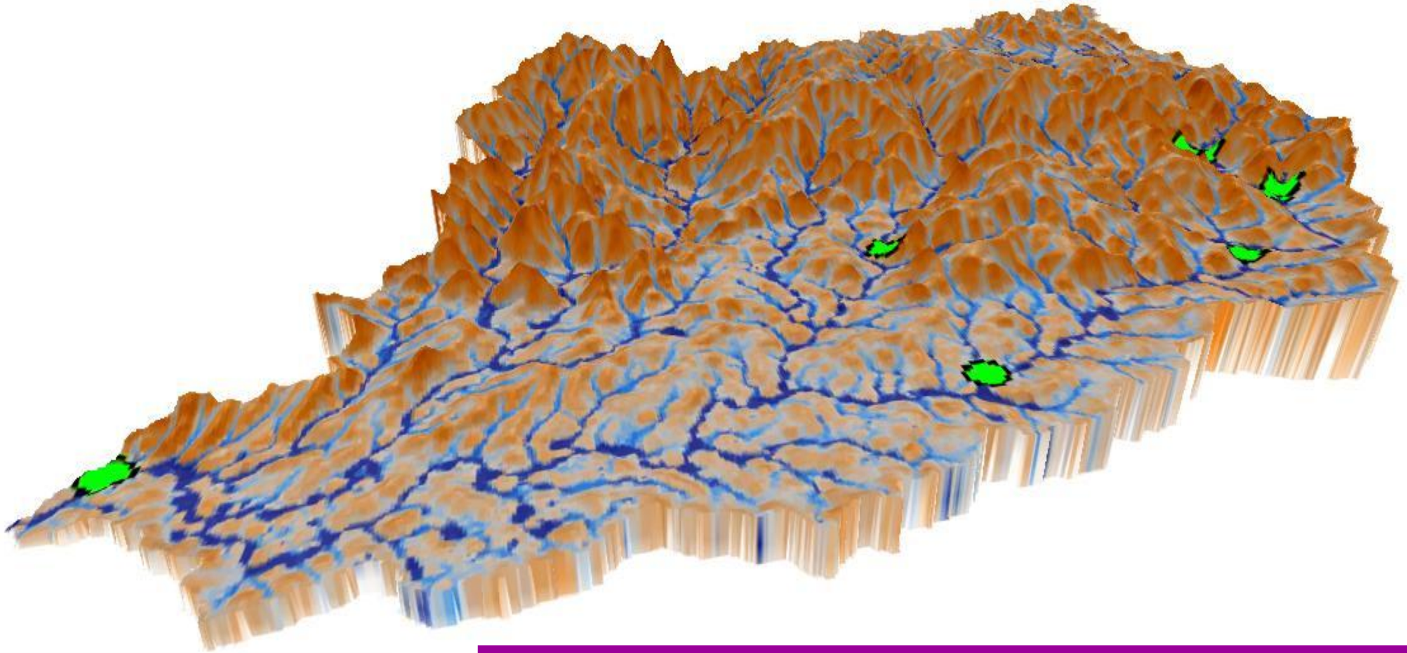


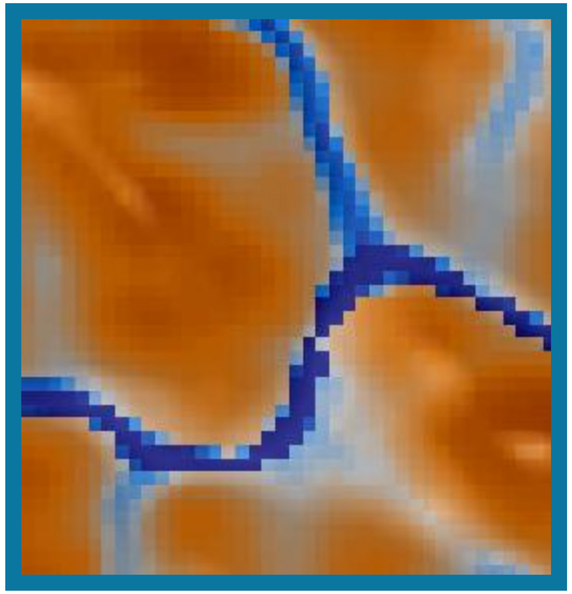


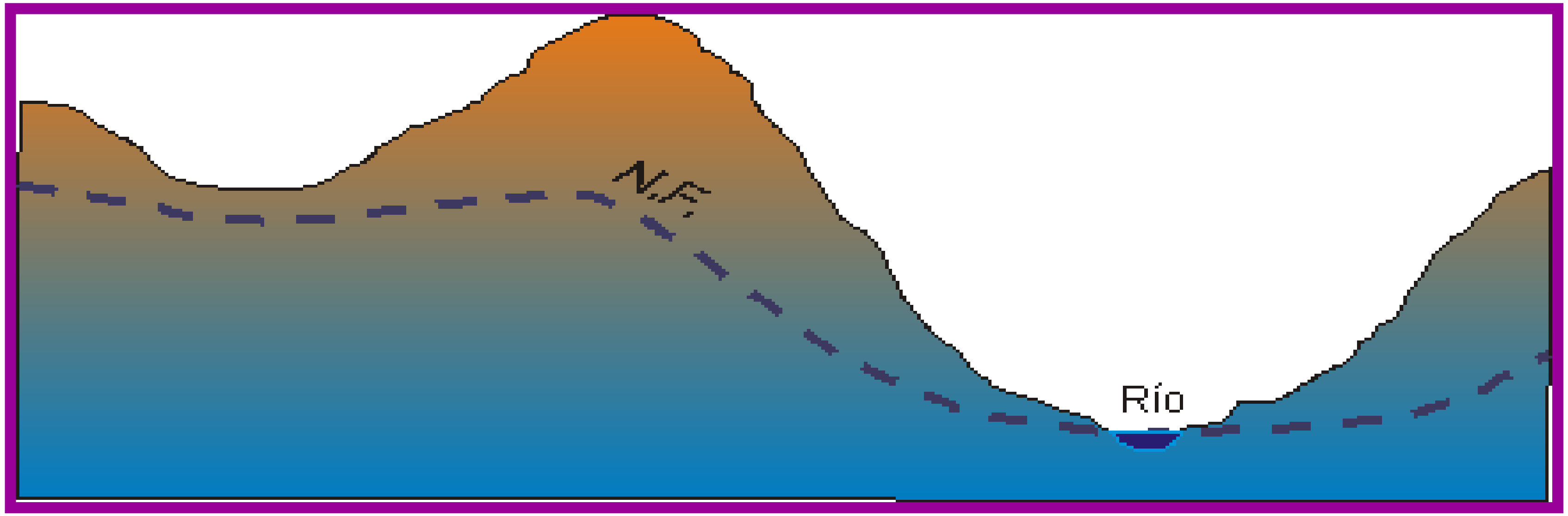
Wetness Index

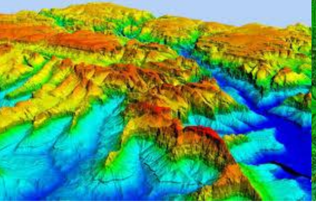


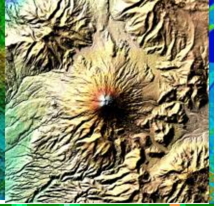


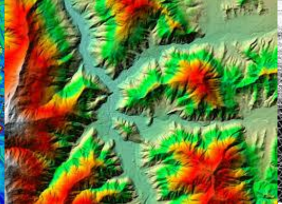


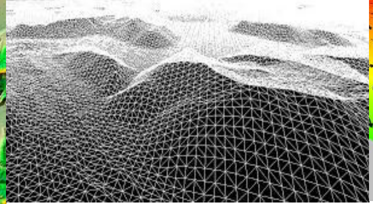


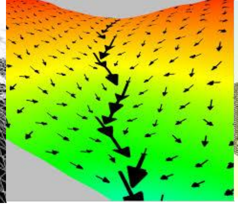

N.F.
Río

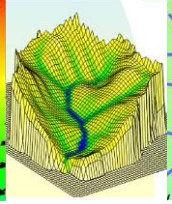


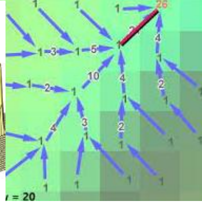


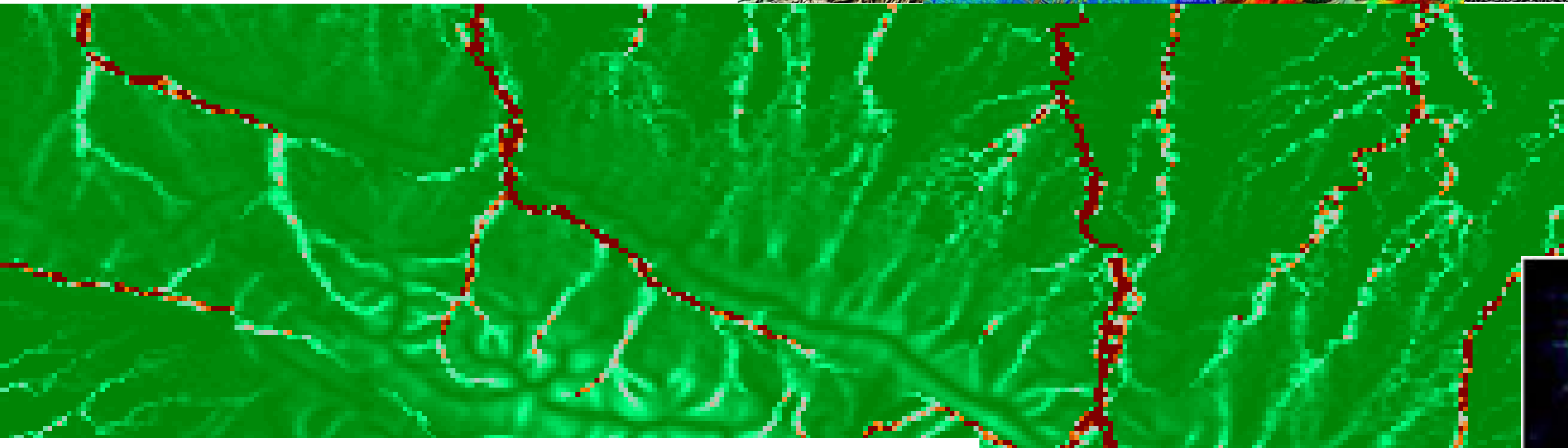


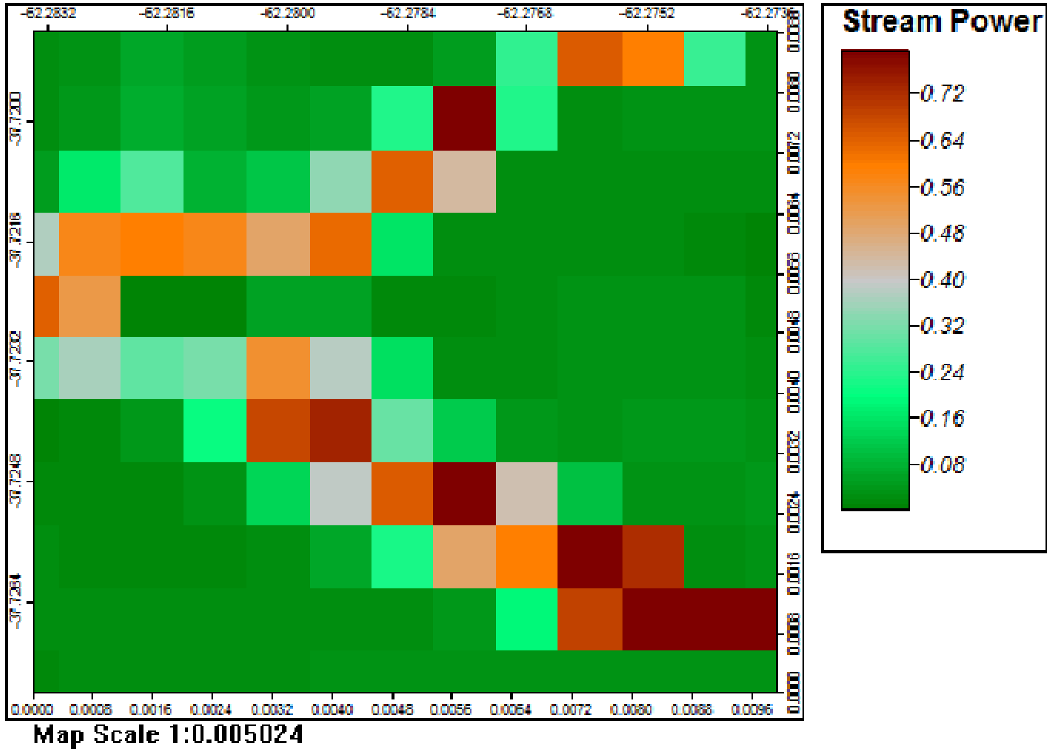


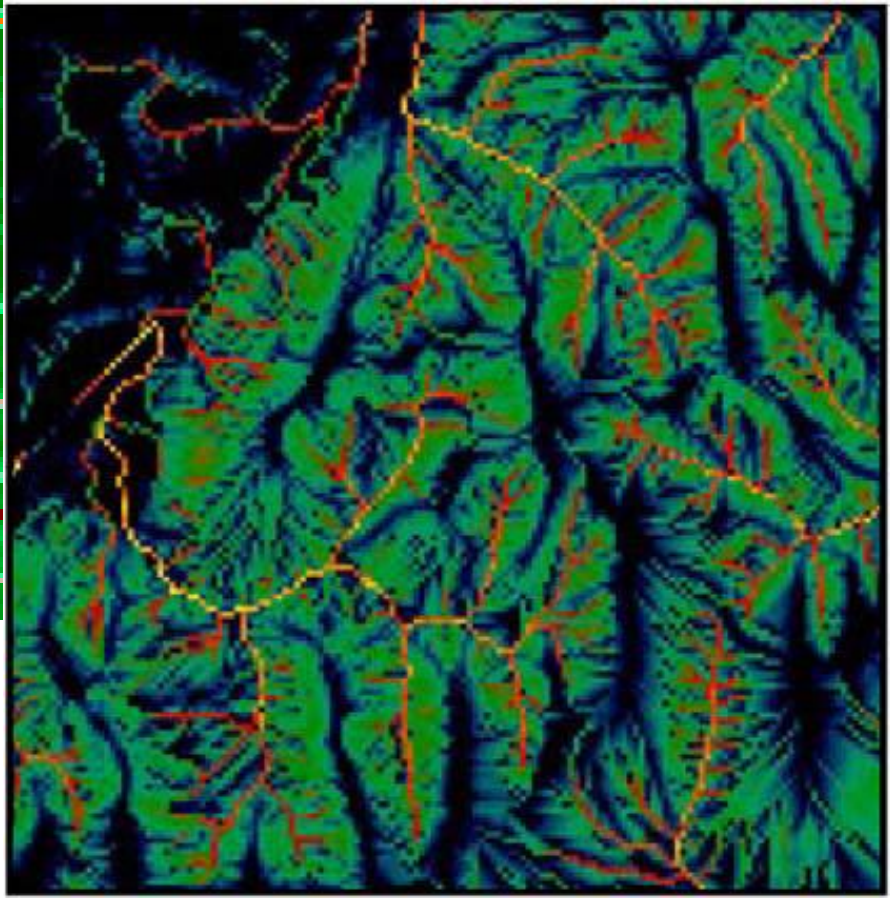


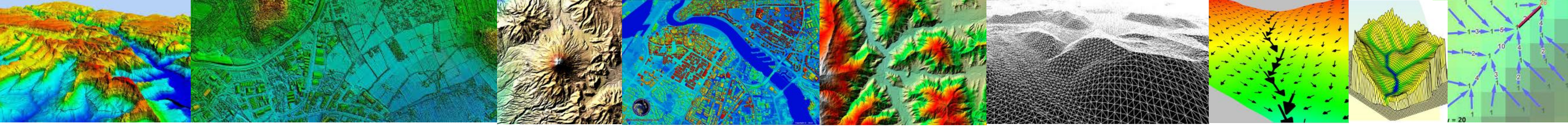




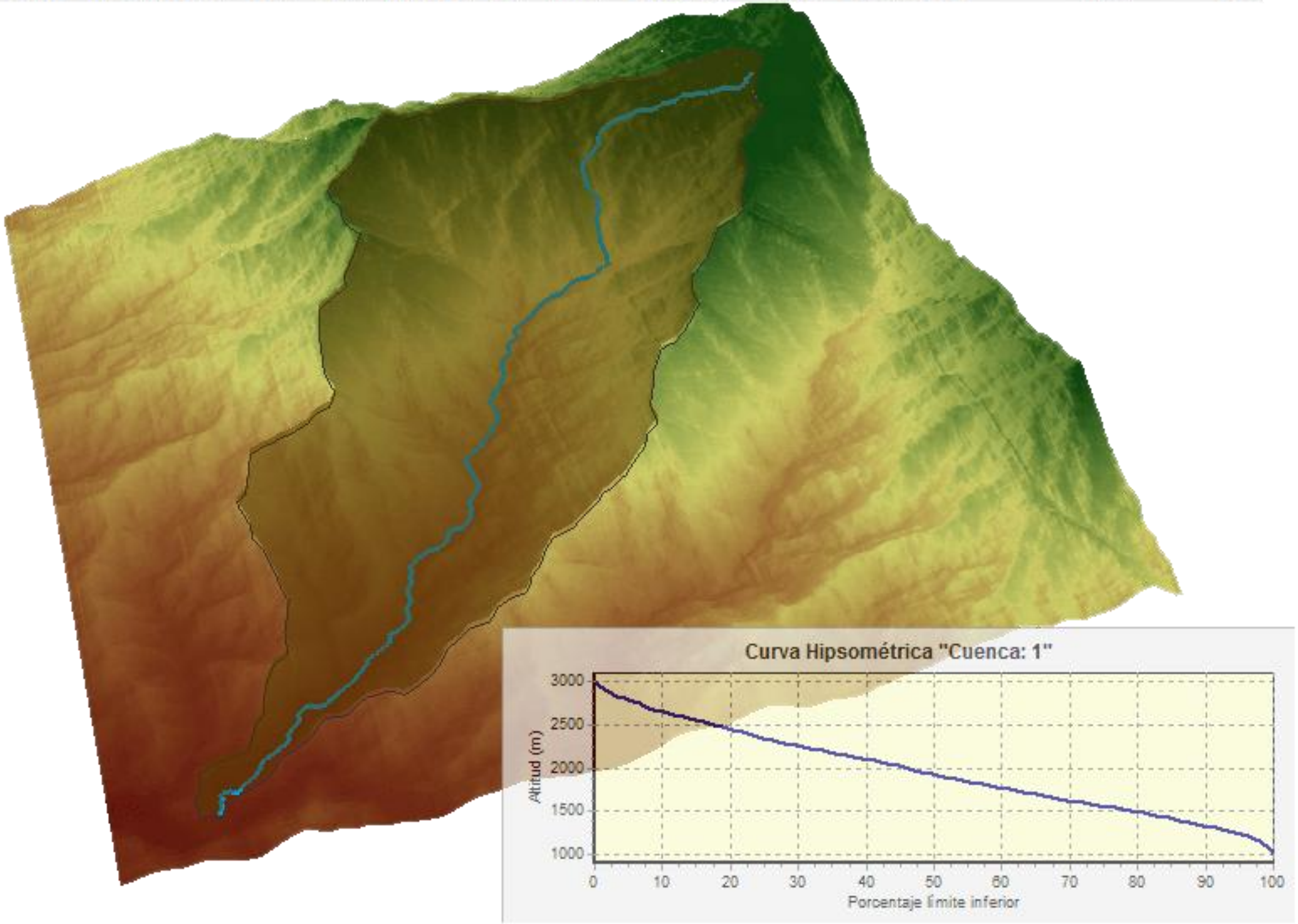


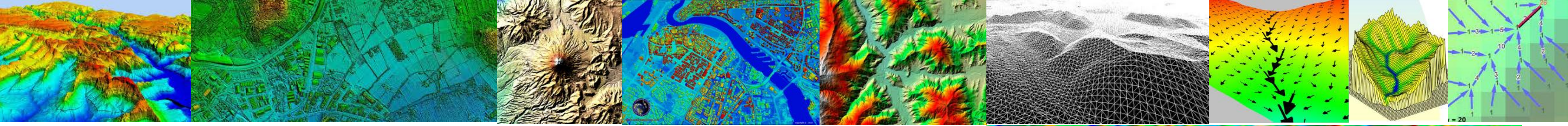

Stream Power
0.72
0.64
0.56
0.48
0.40
0.32
0.24
0.16
0.08
-62.2832
-62.2816
-62.2800
-62.2784
-62.2768
-62.2752
0.0000
0.0008
0.0016
0.0024
0.0032
0.0040
0.0048
0.0056
0.0064
0.0072
0.0080
0.0088
0.0096
Map Scale 1:0.005024

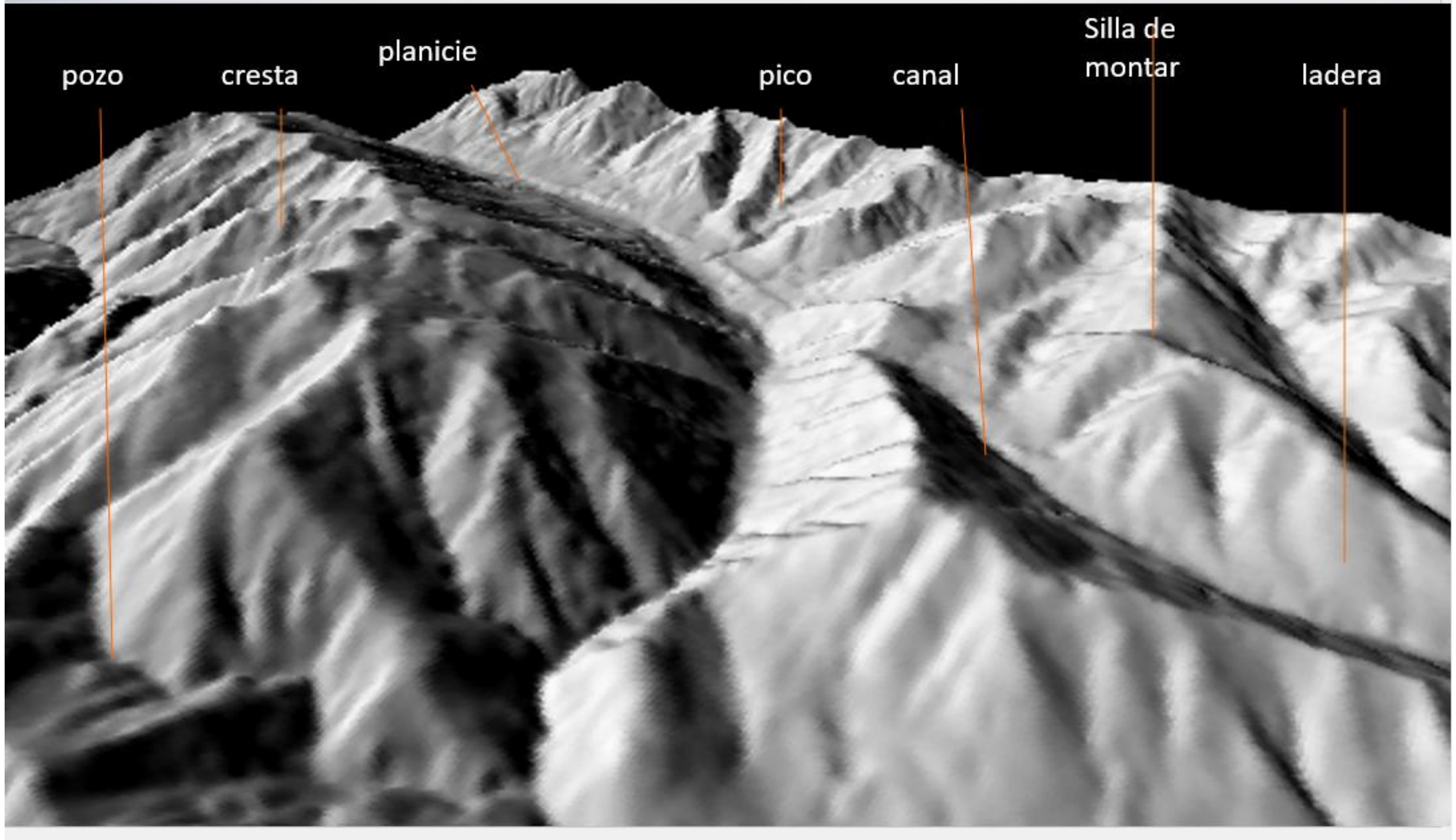


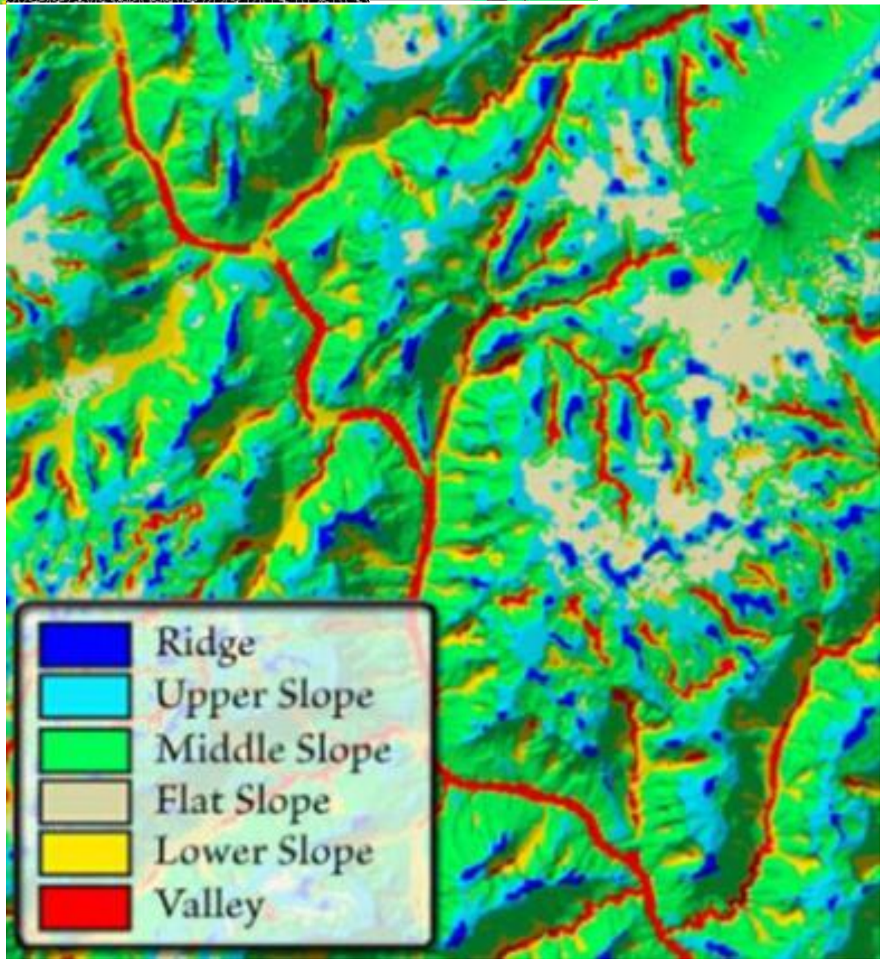


	ClvRgn	A_km2	P_km	Em_m	Pm_g	Pm_p	Kc	Rci	Rh	Lc_km	La_km	Sh	Emx_m	Emn_m	Sc_p	Tc_Kirpich_h	Tc_CHPW_h
▸	1	23.76	31.86	1920.72	25.7	50.1	1.8	0.29	1.0	12.23	9.72	1.2	2914	976	22.28	0.92	0.93

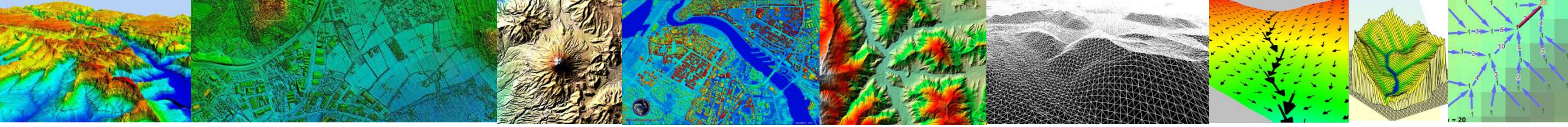








El Indice de posción topográfica (TPI) proporciona un método simple y repetible para clasificar el paisaje con la posición de la pendiente



Earth Surface Processes and Landforms
BSG British Society for Geomorphology

RESEARCH ARTICLE

Assessing open-access digital elevation models for hydrological applications in a large scale plain: Drainage networks, shallow water bodies and vertical accuracy

Ailé Selenne Golin, Hugo Ramiro Páez Campos, Cristian Guevara Ochoa, Claudia Fernanda Dávila, Luis Sebastián Vives

First published: 20 November 2024 | https://doi.org/10.1002/esp.6035

Funding information: This study was funded by CONICET Project UE N22920200100035CO.

Read the full text > PDF TOOLS SHARE

Abstract

This study evaluated six open-access digital elevation models (DEMs) for the Del Azul Creek Basin in the Argentine Chaco-Pampean Plain: Shuttle Radar Topography Mission, Advanced Land Observing Satellite Phased Array L-Band Synthetic Aperture Radar, TerraSAR-X Add-On for Digital Elevation Measurements (TanDEM-X), NASADEM Global

Home / Photogrammetric Engineering & Remote Sensing, Volume 85, Number 3

Analysis and Correction of Digital Elevation Models for Plain Areas

Download Article:
Download
(PDF 2.072,4 kb)

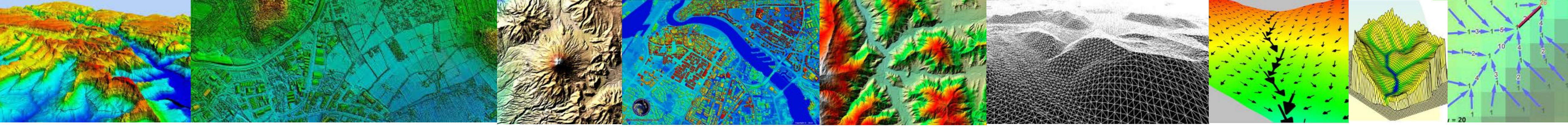
Authors: Ochoa, Cristian Guevara; Vives, Luis; Zimmermann, Erik; Masson, Ignacio; Fajardo, Luisa; Scioli, Carlos

Source: Photogrammetric Engineering & Remote Sensing, Volume 85, Number 3, March 2019, pp. 209-219(11)
Publisher: American Society for Photogrammetry and Remote Sensing
DOI: https://doi.org/10.14358/PERS.85.3.209

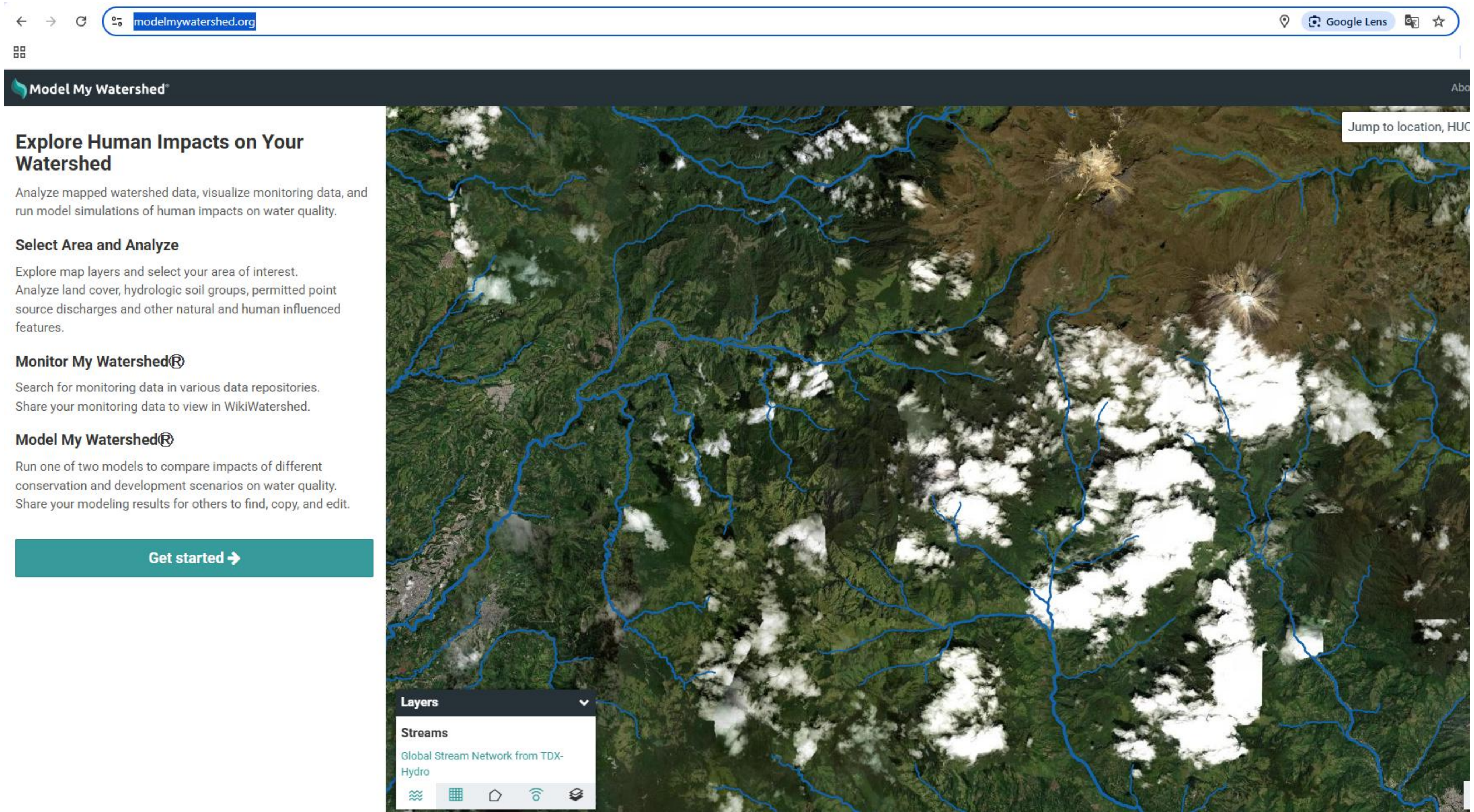
< previous article view table of contents next article > ADD TO FAVOURITES

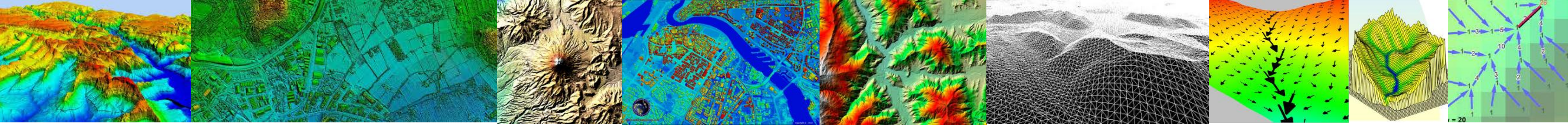
Abstract References Citations Supplementary Data Suggestions

Water movement modeling in plain areas requires digital elevation models (DEMs) adequately representing the morphological and geomorphological land patterns including the presence of civil structures that could affect water flow patterns. This has a direct effect on water accumulation and water flow direction. The objectives of this work were to analyze, compare and improve DEMs so surface water



https://modelmywatershed.org/





https://mghydro.com/watersheds/